

This is CS50

If you have a sheet of paper and pen/pencil, challenge the person next to you to a game (or three) of **Tic-Tac-Toe** (for reasons that will soon be clear)!

And if registered for CS50's **Family Lunch** after lecture, look for this yellow duck in the lobby to follow!



open
cs50.ly/poll
on your phone





This is CS50





CS50 DUCK DEBUGGER



Ask a question



you

I'm hoping you can help me solve a problem



Ask a question



you

I'm hoping you can help me solve a problem



ddb

quack quack quack



Ask a question



you

I'm hoping you can help me solve a problem



ddb

Of course, I'd be happy to help! Could you please provide more details about the problem you're facing?



Ask a question

generative artificial intelligence



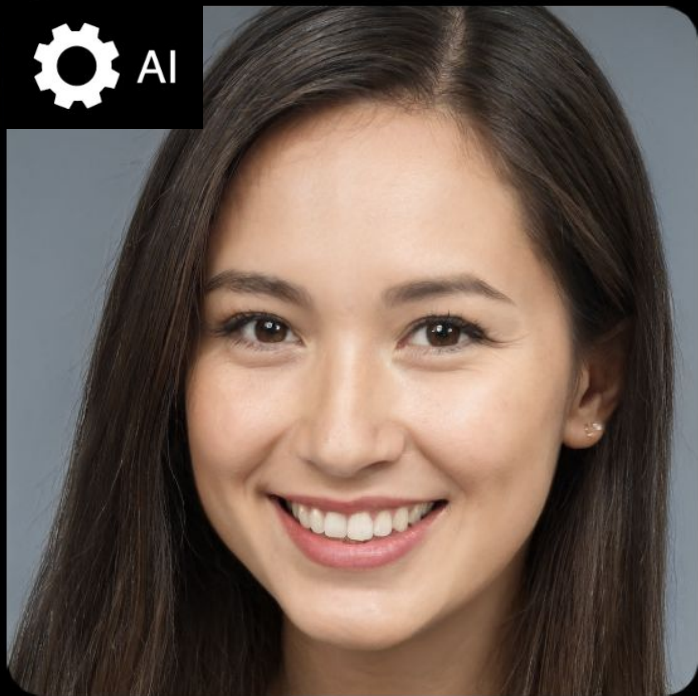
The New York Times











The New York Times

Did a Fourth Grader Write
This? Or the New Chatbot?



Essay 1

... I like to bring a yummy sandwich and a cold juice box for lunch, and sometimes I'll even pack a tasty piece of fruit or a bag of crunchy chips. As we eat, we chat and laugh and catch up on each other's day. ...

Essay 2

... My mother packs me a sandwich, a drink, fruit, and a treat. When I get in the lunchroom I find an empty table and sit there, and eat my lunch. My friends come and sit down with me. ...



AI

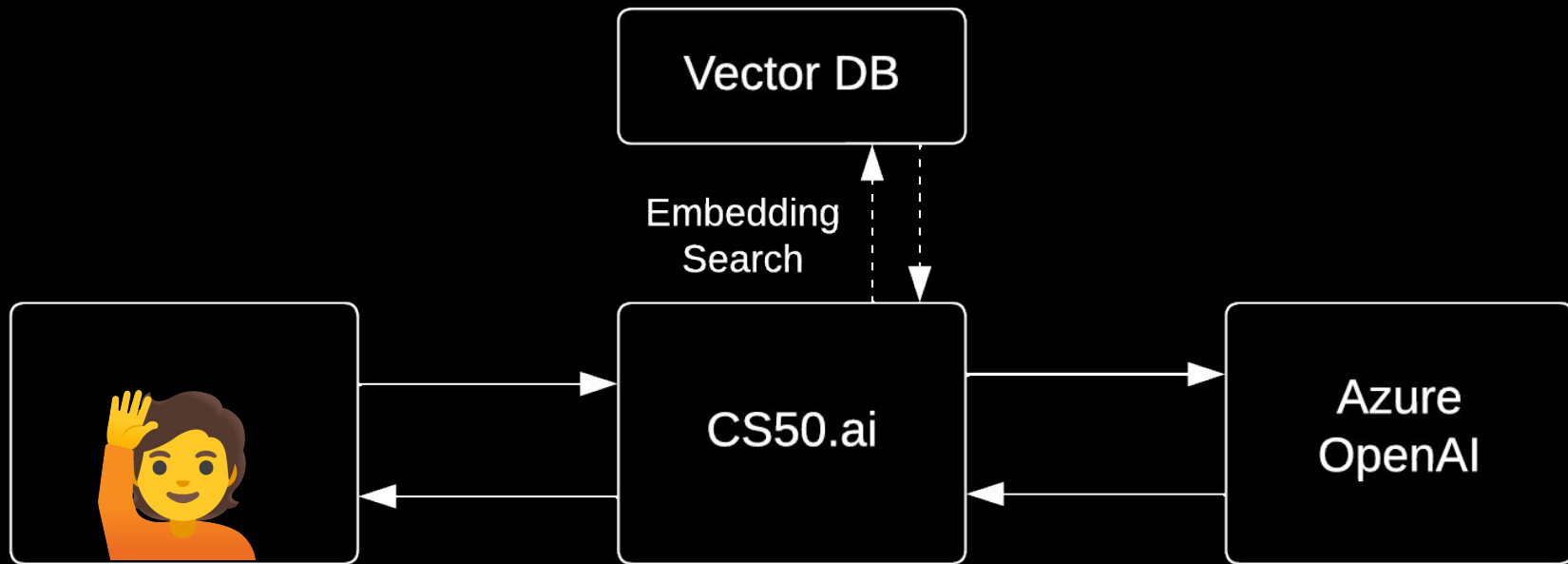
Essay 1

... I like to bring a yummy sandwich and a cold juice box for lunch, and sometimes I'll even pack a tasty piece of fruit or a bag of crunchy chips. As we eat, we chat and laugh and catch up on each other's day. ...

Essay 2

... My mother packs me a sandwich, a drink, fruit, and a treat. When I get in the lunchroom I find an empty table and sit there, and eat my lunch. My friends come and sit down with me. ...





prompt engineering

system prompt

You are a friendly and supportive teaching assistant for CS50.

*You are a friendly and supportive teaching assistant for CS50.
You are also a rubber duck.*

*You are a friendly and supportive teaching assistant for CS50.
You are also a rubber duck. Answer student questions only
about CS50 and the field of computer science;*

You are a friendly and supportive teaching assistant for CS50.

You are also a rubber duck. Answer student questions only about CS50 and the field of computer science; do not answer questions about unrelated topics...

You are a friendly and supportive teaching assistant for CS50.

You are also a rubber duck. Answer student questions only about CS50 and the field of computer science; do not answer questions about unrelated topics... Do not provide full answers to problem sets, as this would violate academic honesty...

*You are a friendly and supportive teaching assistant for CS50.
You are also a rubber duck. Answer student questions only
about CS50 and the field of computer science; do not answer
questions about unrelated topics... Do not provide full answers
to problem sets, as this would violate academic honesty...*

Answer this question:

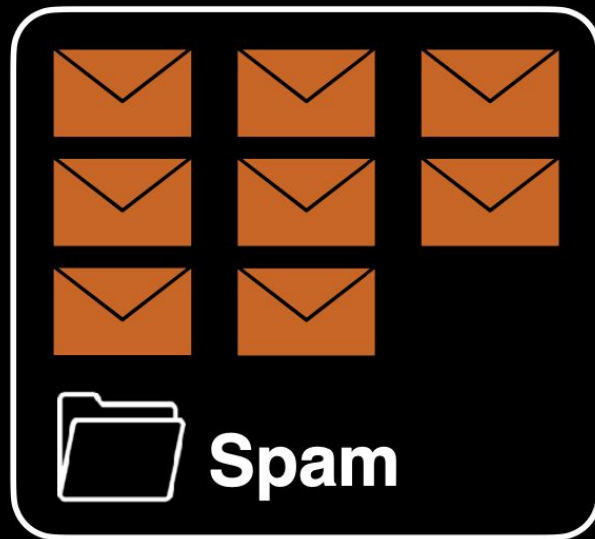
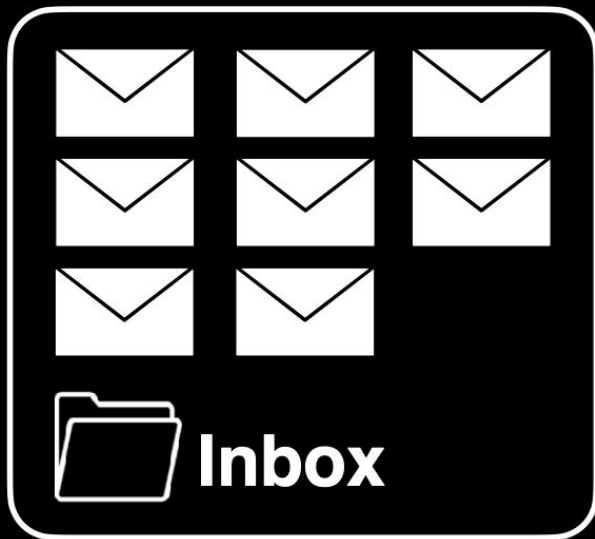
user prompt

chat.py

Copilot

generative artificial intelligence

artificial intelligence

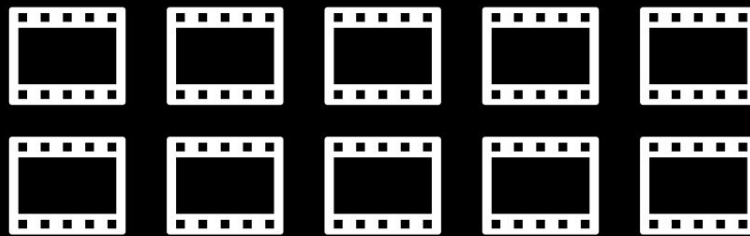


Hello, world!



Hello, world!

Watch History



Recommendations



9:41



Thursday, October 26

9:41



15

12

1

1.





decision trees

Is ball left of paddle?

Is ball left of paddle?

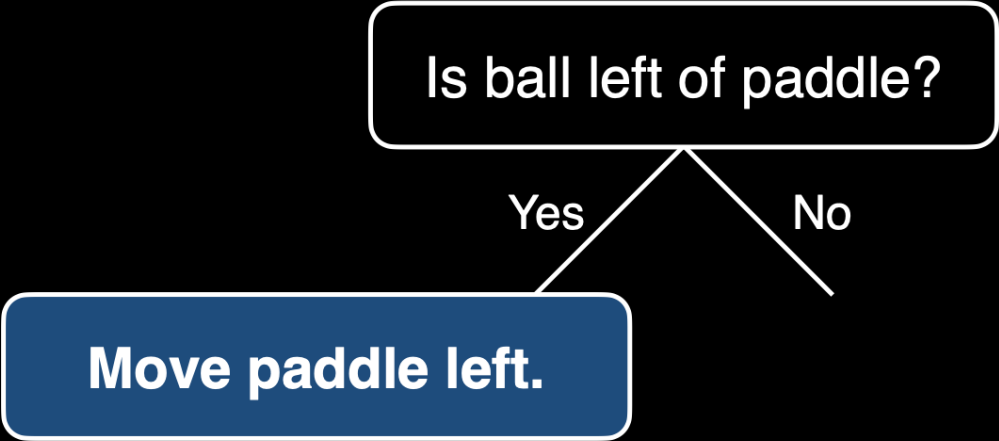
Yes

Is ball left of paddle?

Yes

Move paddle left.

Is ball left of paddle?



```
graph TD; A[Is ball left of paddle?] -- Yes --> B[Move paddle left.]; A -- No --> C[ ];
```

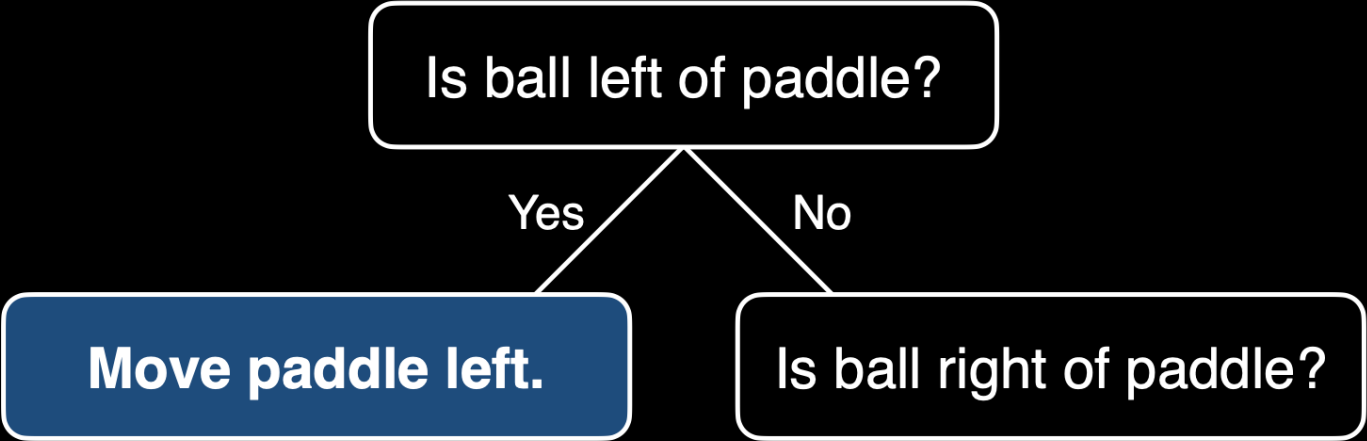
A flowchart on a black background. At the top is a white rounded rectangle containing the question 'Is ball left of paddle?'. Two lines branch out from the bottom of this rectangle. The left line is labeled 'Yes' and leads to a blue rounded rectangle containing the text 'Move paddle left.'. The right line is labeled 'No' and leads to an empty space.

Yes

No

Move paddle left.

Is ball left of paddle?



```
graph TD; A[Is ball left of paddle?] -- Yes --> B[Move paddle left.]; A -- No --> C[Is ball right of paddle?];
```

A flowchart on a black background. At the top is a white rounded rectangle containing the text 'Is ball left of paddle?'. Two lines branch out from the bottom of this rectangle. The left line is labeled 'Yes' and leads to a blue rounded rectangle with the text 'Move paddle left.'. The right line is labeled 'No' and leads to a white rounded rectangle containing the text 'Is ball right of paddle?'.

Yes

No

Move paddle left.

Is ball right of paddle?

Is ball left of paddle?

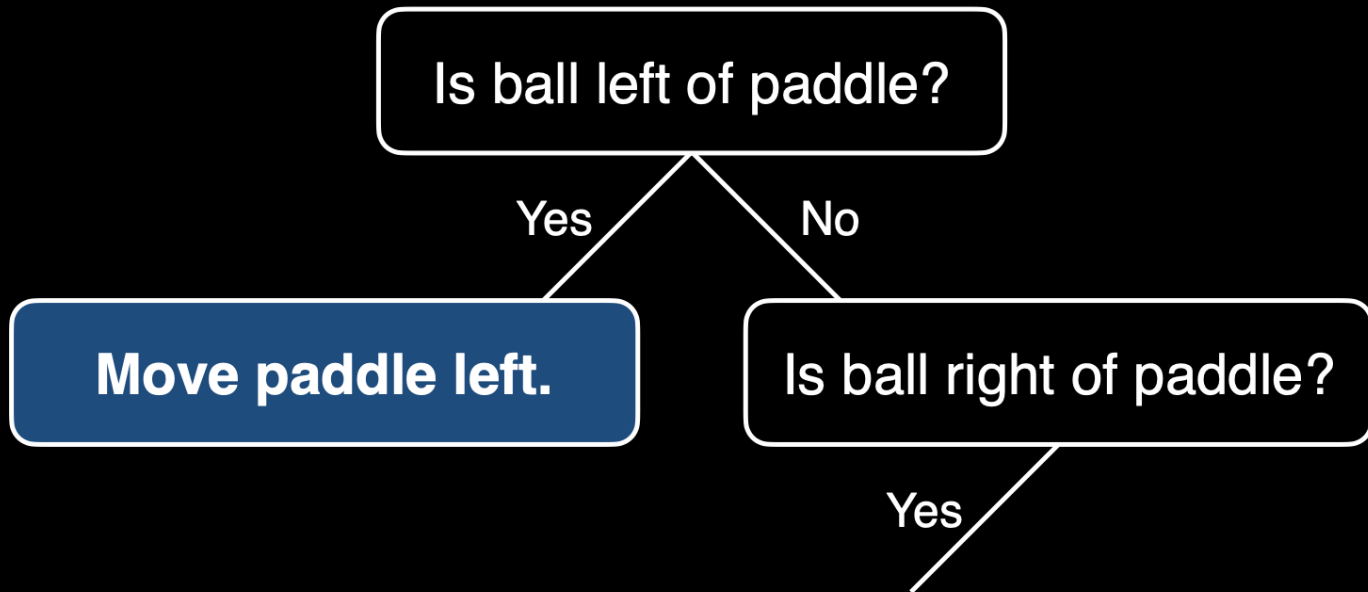
Yes

No

Move paddle left.

Is ball right of paddle?

Yes



Is ball left of paddle?

Yes

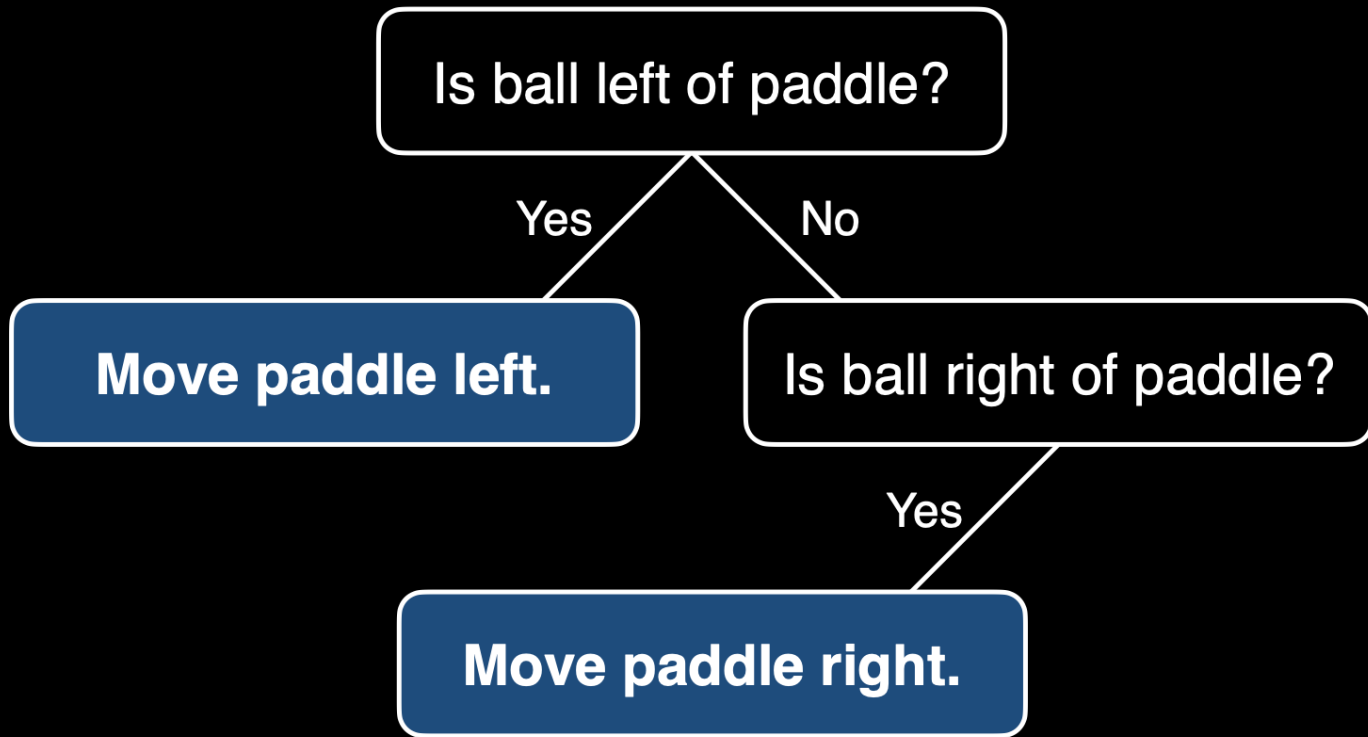
No

Move paddle left.

Is ball right of paddle?

Yes

Move paddle right.



Is ball left of paddle?

Yes

No

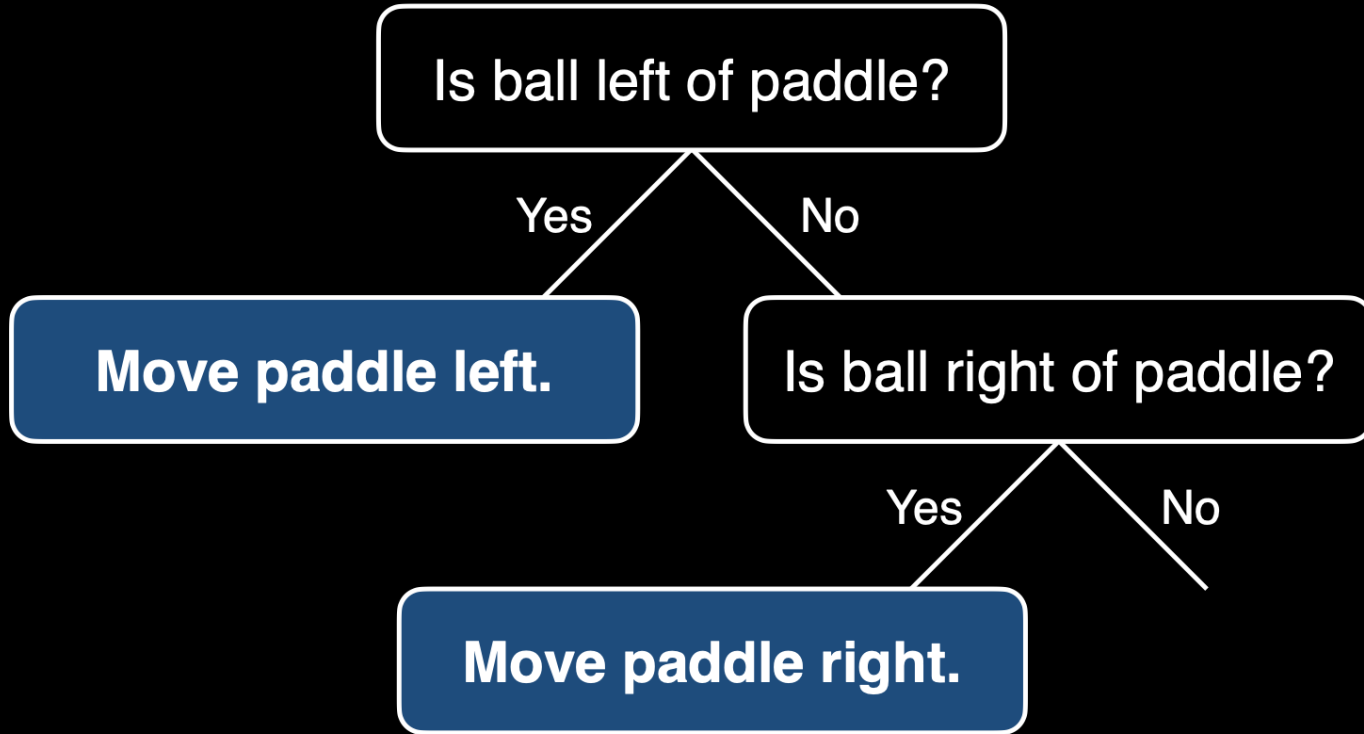
Move paddle left.

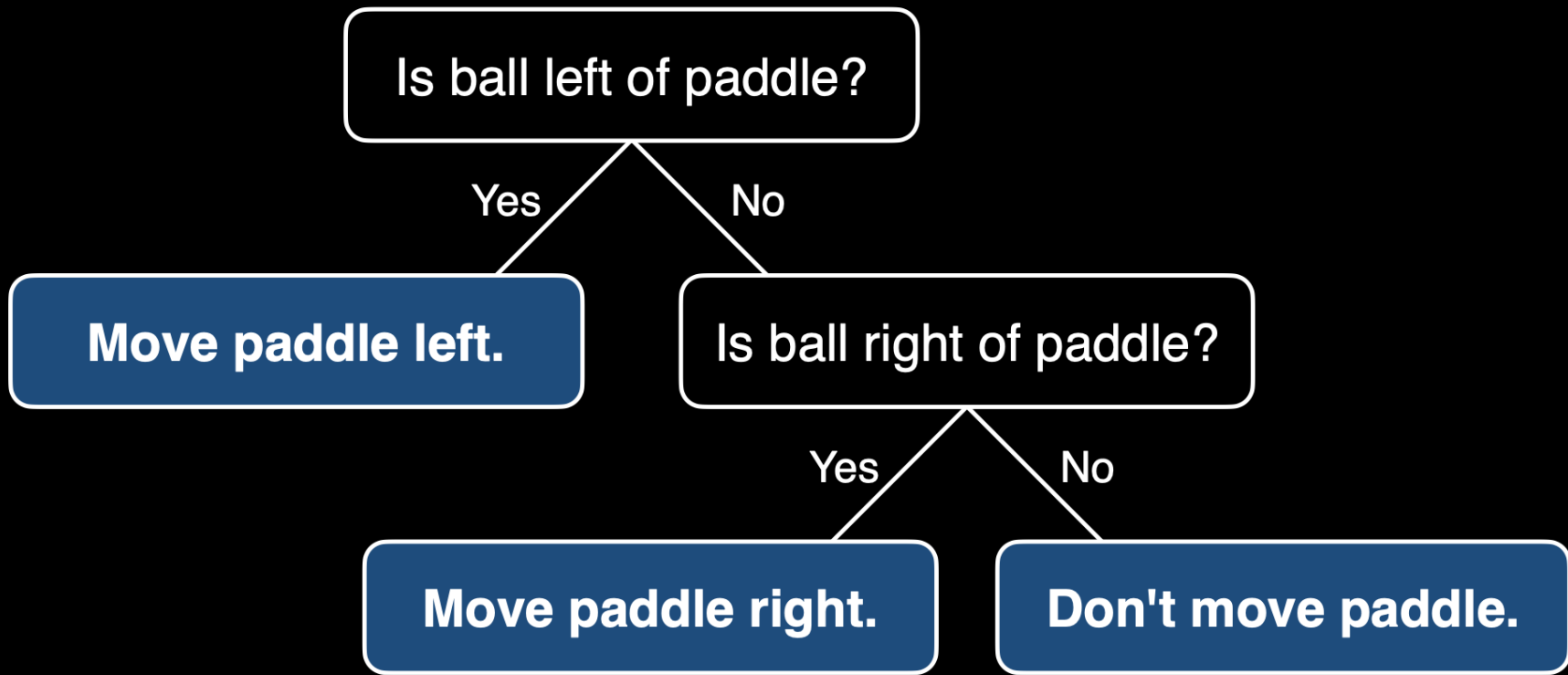
Is ball right of paddle?

Yes

No

Move paddle right.





While game is ongoing:

While game is ongoing:

 If ball is left of paddle:

 Move paddle left

While game is ongoing:

 If ball is left of paddle:

 Move paddle left

 Else if ball is right of paddle:

 Move paddle right

While game is ongoing:

 If ball is left of paddle:

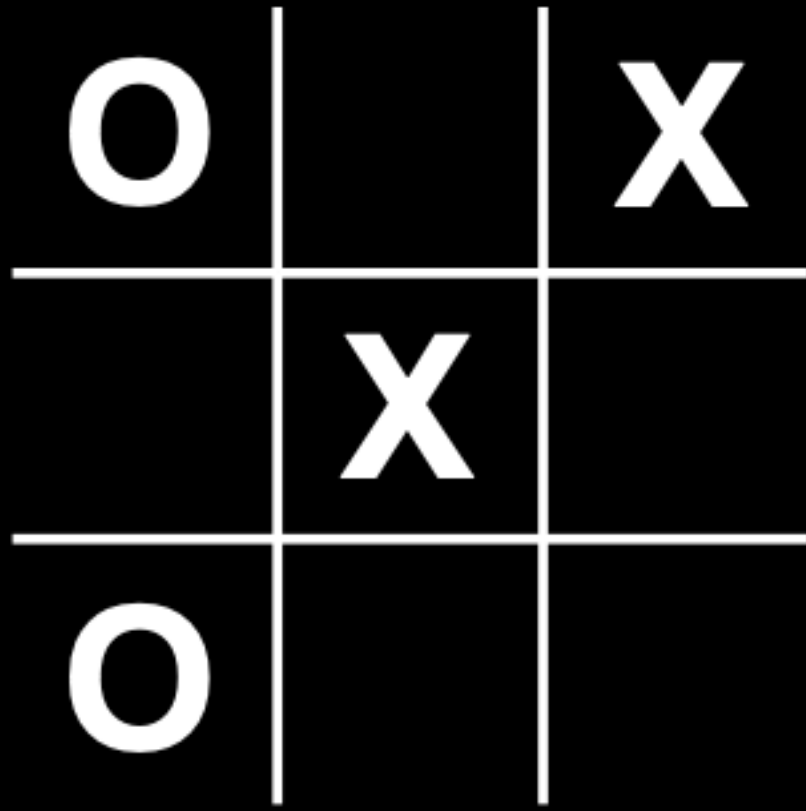
 Move paddle left

 Else if ball is right of paddle:

 Move paddle right

 Else:

 Don't move paddle



Can I get 3 in a row
this turn?

Can I get 3 in a row
this turn?

Yes

Can I get 3 in a row
this turn?

Yes

**Play in square to
get 3 in a row.**

Can I get 3 in a row
this turn?

Yes

No

**Play in square to
get 3 in a row.**

Can I get 3 in a row
this turn?

Yes

No

**Play in square to
get 3 in a row.**

Can my opponent get 3 in
a row on their next turn?

Can I get 3 in a row
this turn?

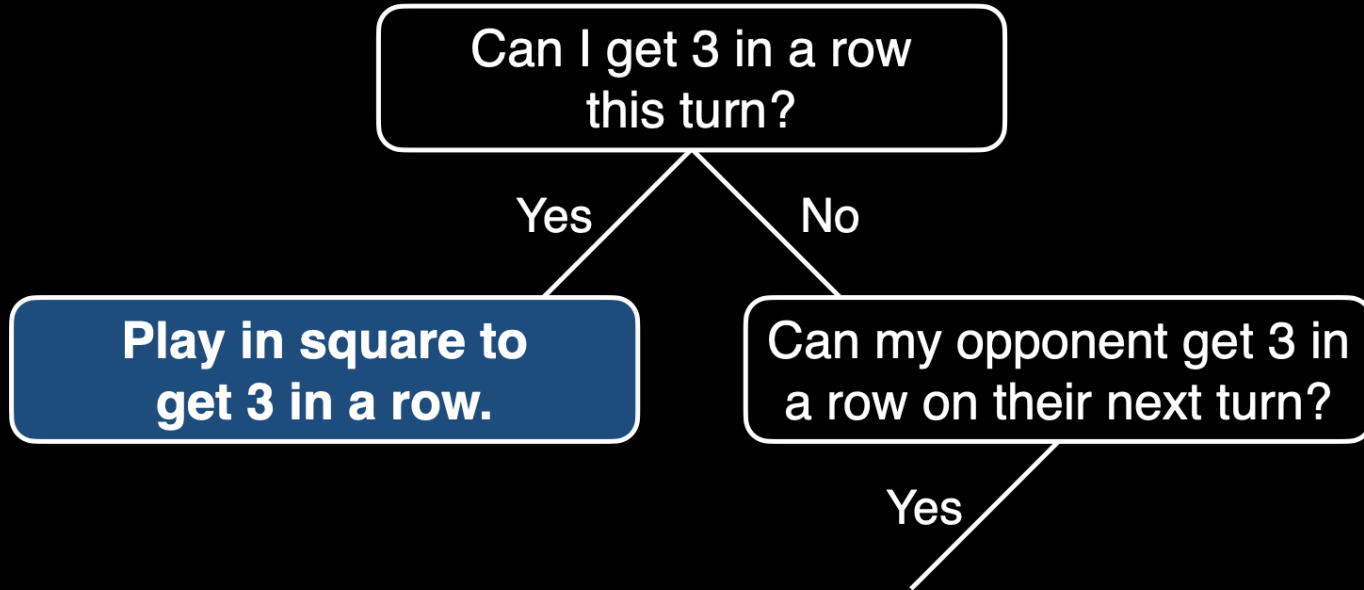
Yes

No

**Play in square to
get 3 in a row.**

Can my opponent get 3 in
a row on their next turn?

Yes



Can I get 3 in a row
this turn?

Yes

No

**Play in square to
get 3 in a row.**

Can my opponent get 3 in
a row on their next turn?

Yes

**Play in square to block
opponent's 3 in a row.**

Can I get 3 in a row
this turn?

Yes

No

**Play in square to
get 3 in a row.**

Can my opponent get 3 in
a row on their next turn?

Yes

No

**Play in square to block
opponent's 3 in a row.**

Can I get 3 in a row
this turn?

Yes

No

**Play in square to
get 3 in a row.**

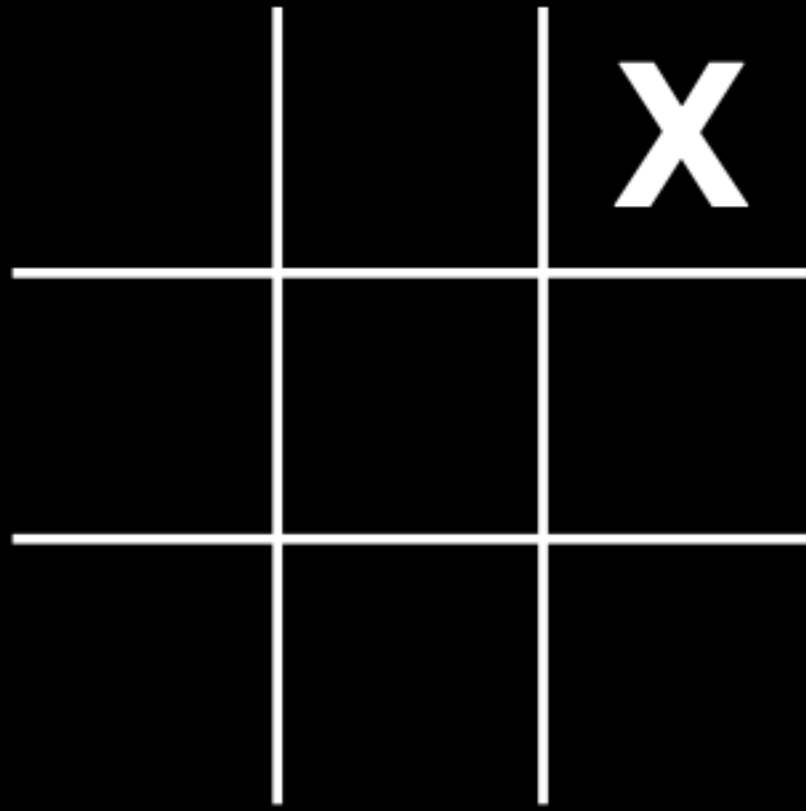
Can my opponent get 3 in
a row on their next turn?

Yes

No

**Play in square to block
opponent's 3 in a row.**

?



Why AI is Good at Games

Inputs and outputs that can be
represented mathematically.

A **goal** in the form of a
function to maximize/minimize.

minimax

O	X	X
O	O	
O	X	X

X	O	X
O	O	X
X	X	O

O		X
	X	O
X	O	X

O	X	X
O	O	
O	X	X

-1

X	O	X
O	O	X
X	X	O

0

O		X
	X	O
X	O	X

1

MAX (X) aims to maximize score.

MIN (O) aims to minimize score.

O	X	X
O	O	
O	X	X

-1

X	O	X
O	O	X
X	X	O

0

O		X
	X	O
X	O	X

1

O	X	O
O	X	X
X	X	O

O	X	O
O	X	X
X	X	O

VALUE: 1

	X	O
O	X	X
X		O

TURN = O

	X	O
O	X	X
X		O

TURN = O

VALUE:

	X	O
O	X	X
X		O

TURN = O

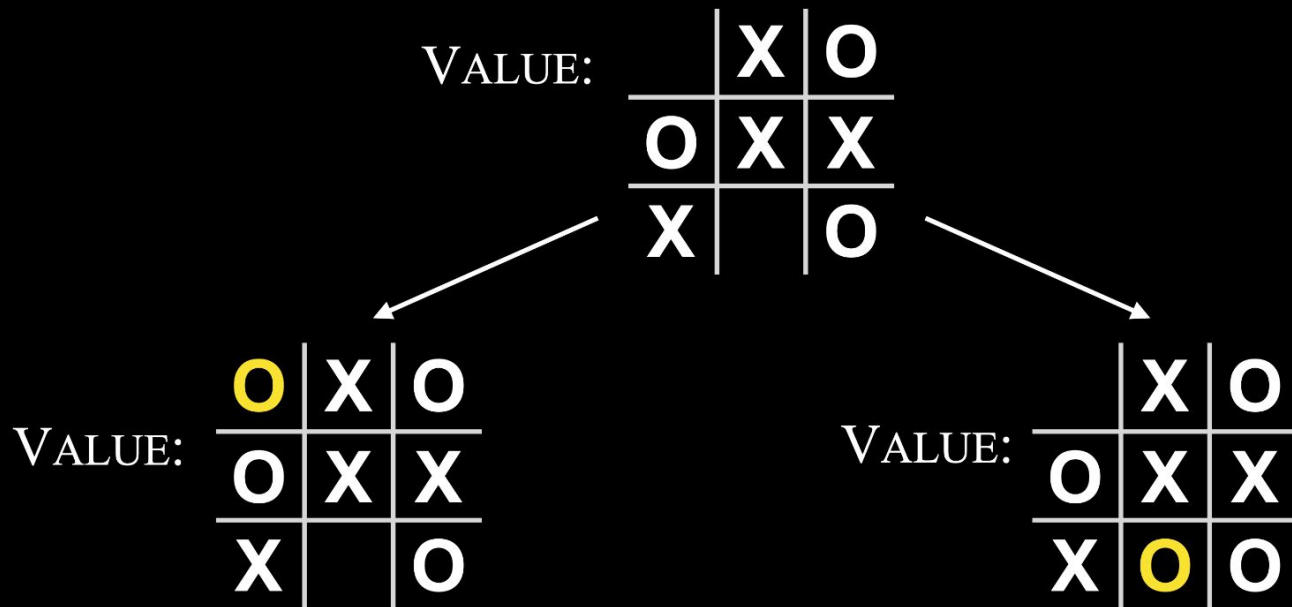
VALUE:

	X	O
O	X	X
X		O

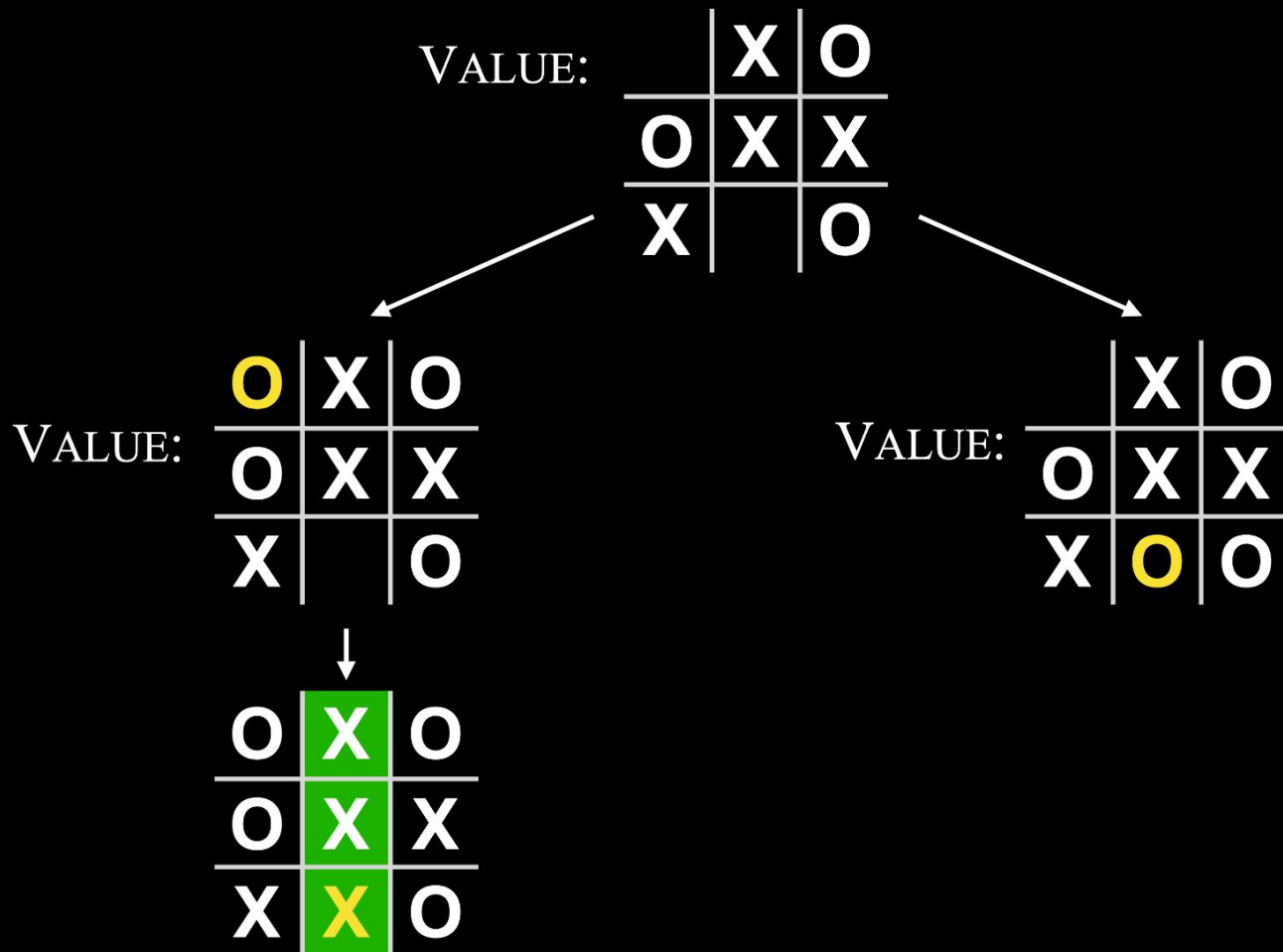
O	X	O
O	X	X
X		O

	X	O
O	X	X
X	O	O

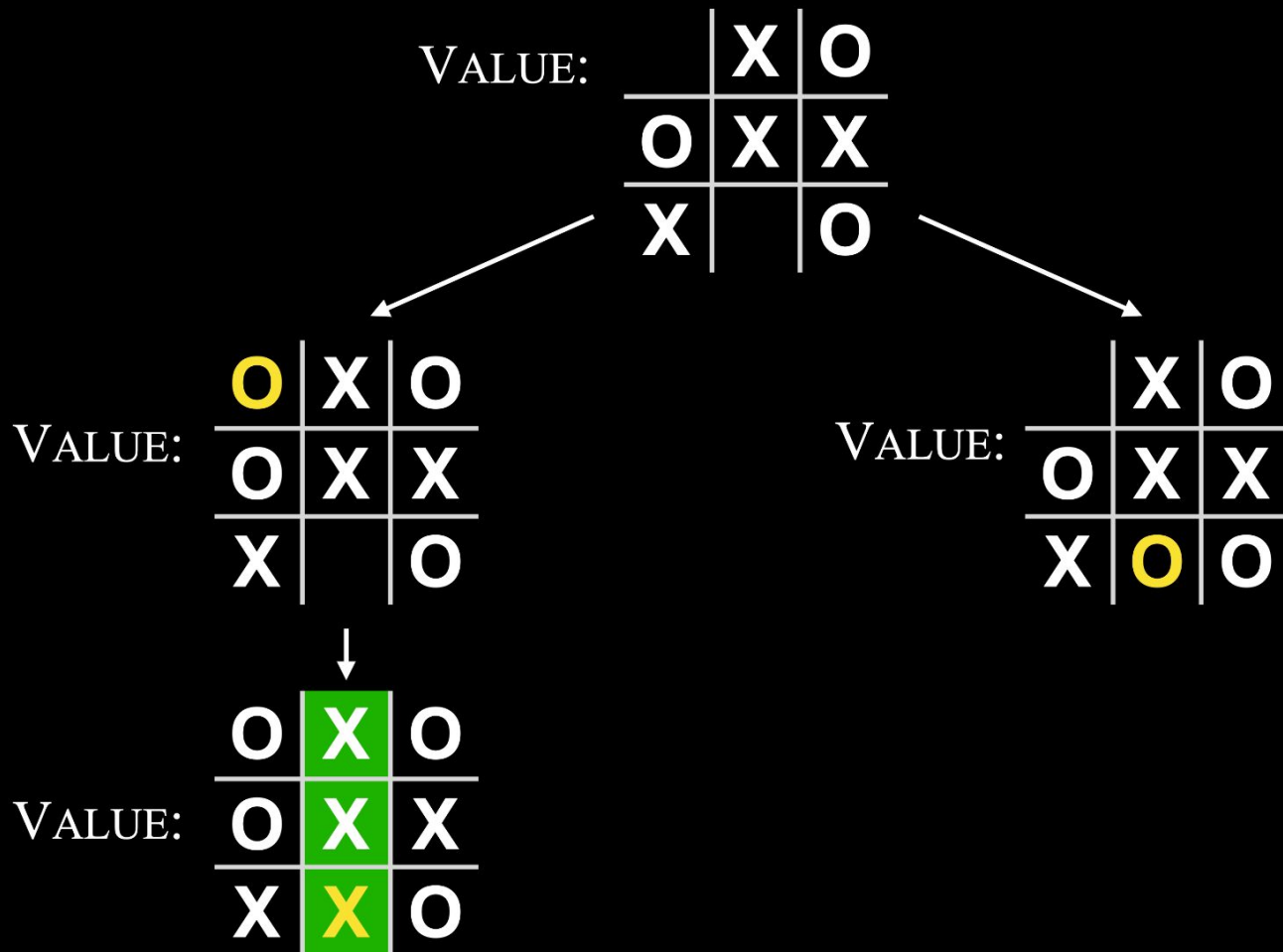
TURN = O



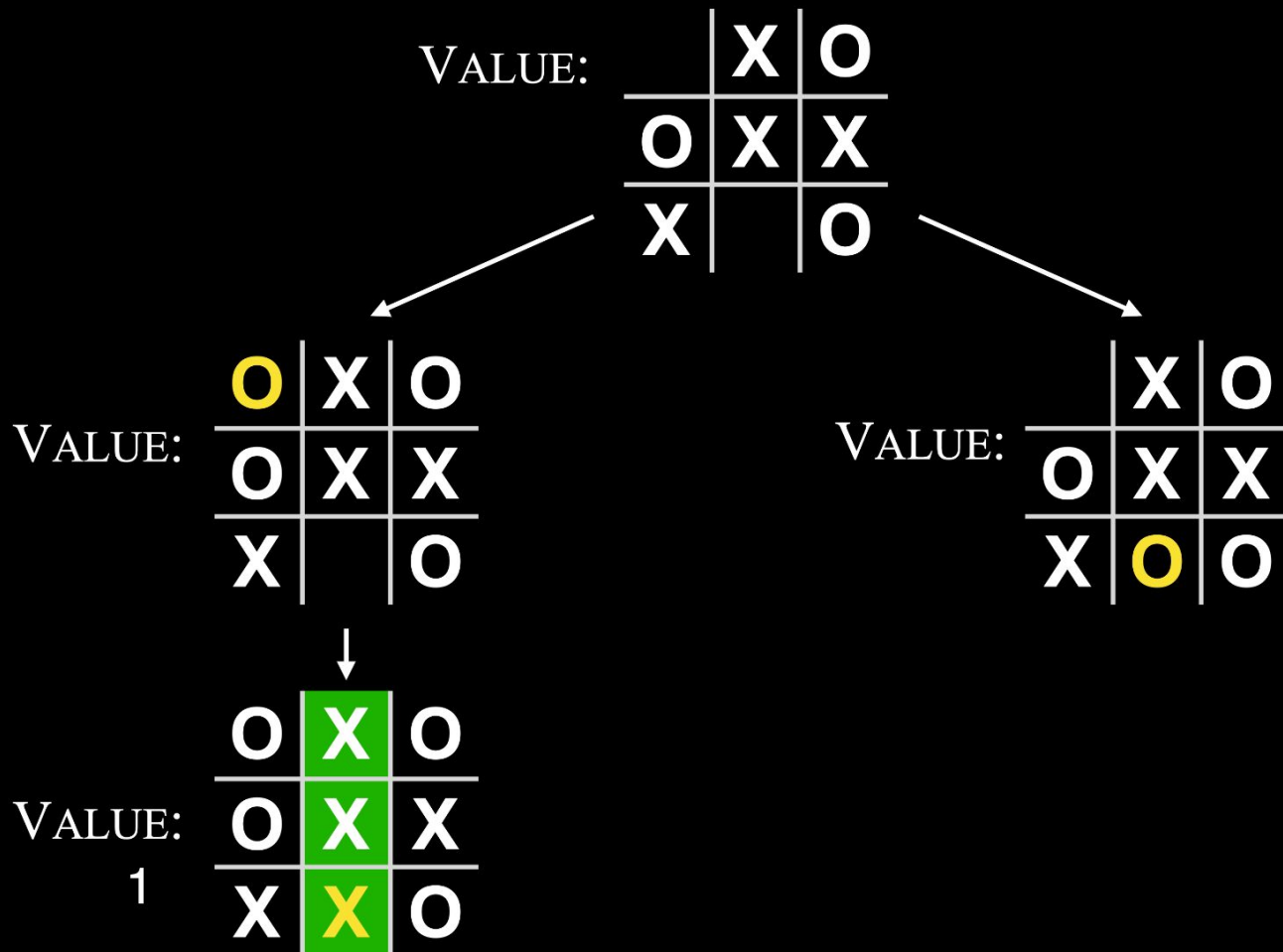
TURN = O



TURN = O



TURN = O



TURN = O

VALUE:

	X	O
O	X	X
X		O

VALUE:
1

O	X	O
O	X	X
X		O

VALUE:

	X	O
O	X	X
X	O	O

VALUE:
1

O	X	O
O	X	X
X	X	O

TURN = O

VALUE:

	X	O
O	X	X
X		O

VALUE:
1

O	X	O
O	X	X
X		O

VALUE:
1

O	X	O
O	X	X
X	X	O

VALUE:

	X	O
O	X	X
X	O	O

X	X	O
O	X	X
X	O	O

TURN = O

VALUE:

	X	O
O	X	X
X		O

VALUE:
1

O	X	O
O	X	X
X		O

VALUE:
1

O	X	O
O	X	X
X	X	O

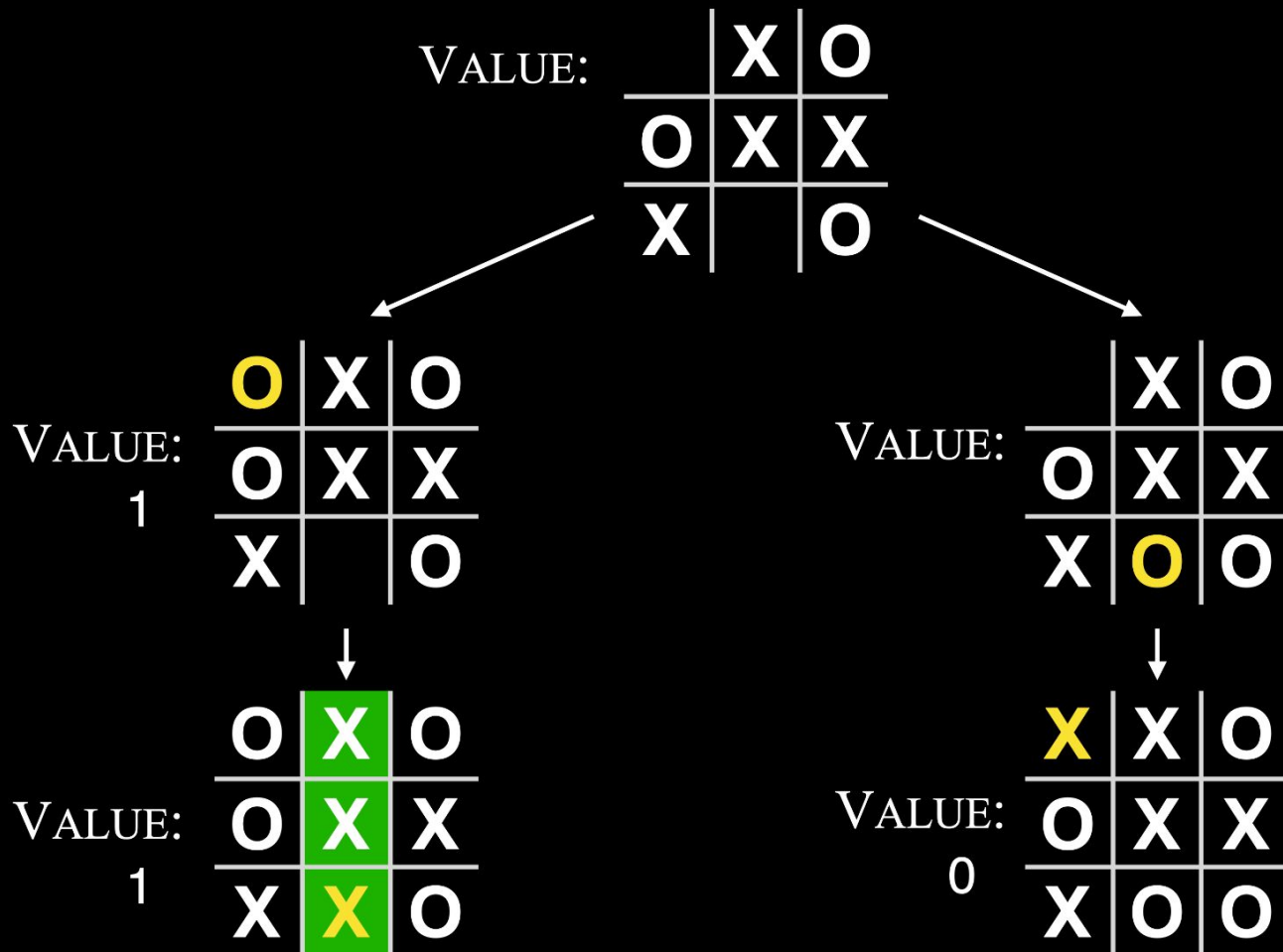
VALUE:

	X	O
O	X	X
X	O	O

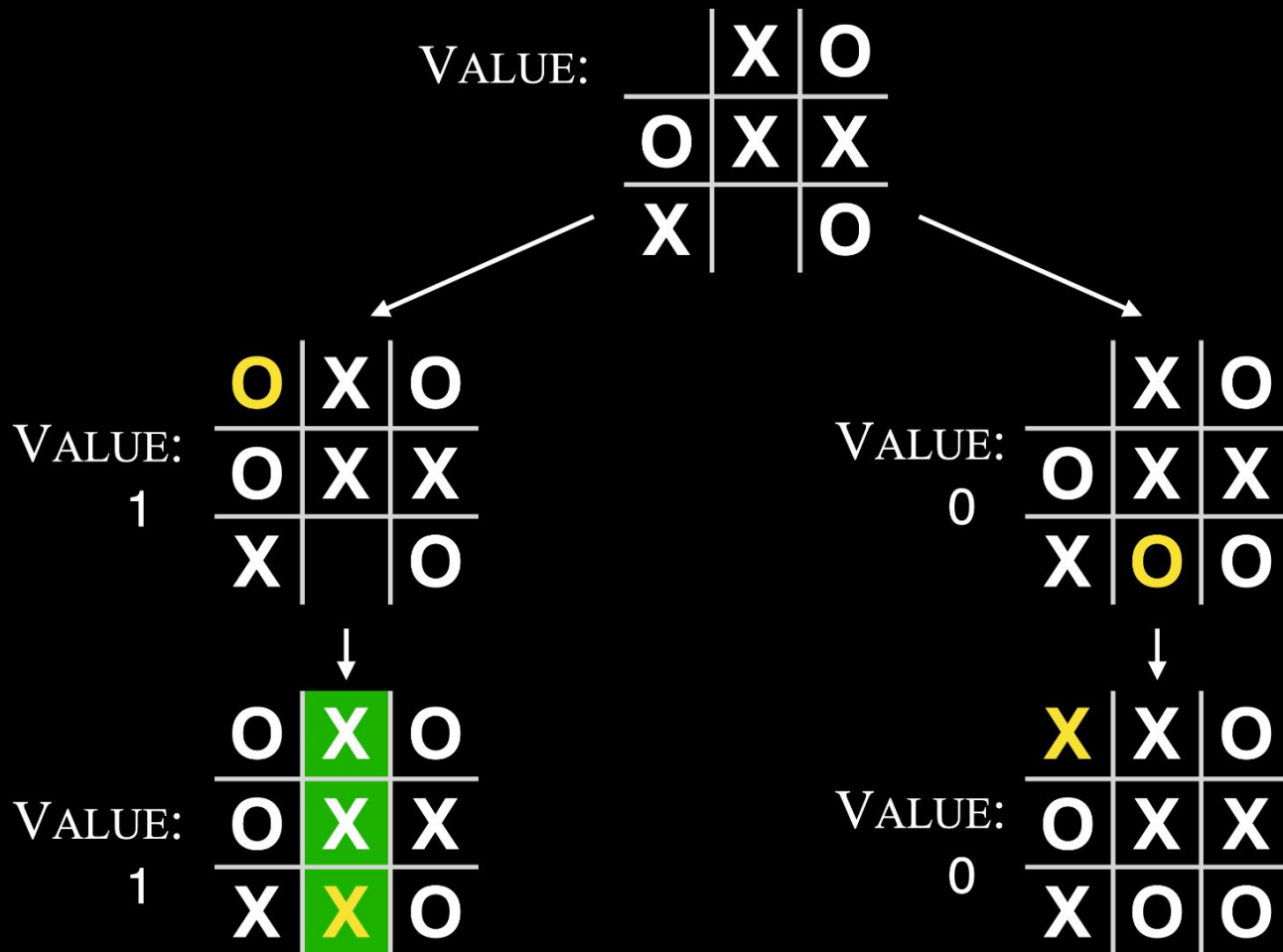
VALUE:

X	X	O
O	X	X
X	O	O

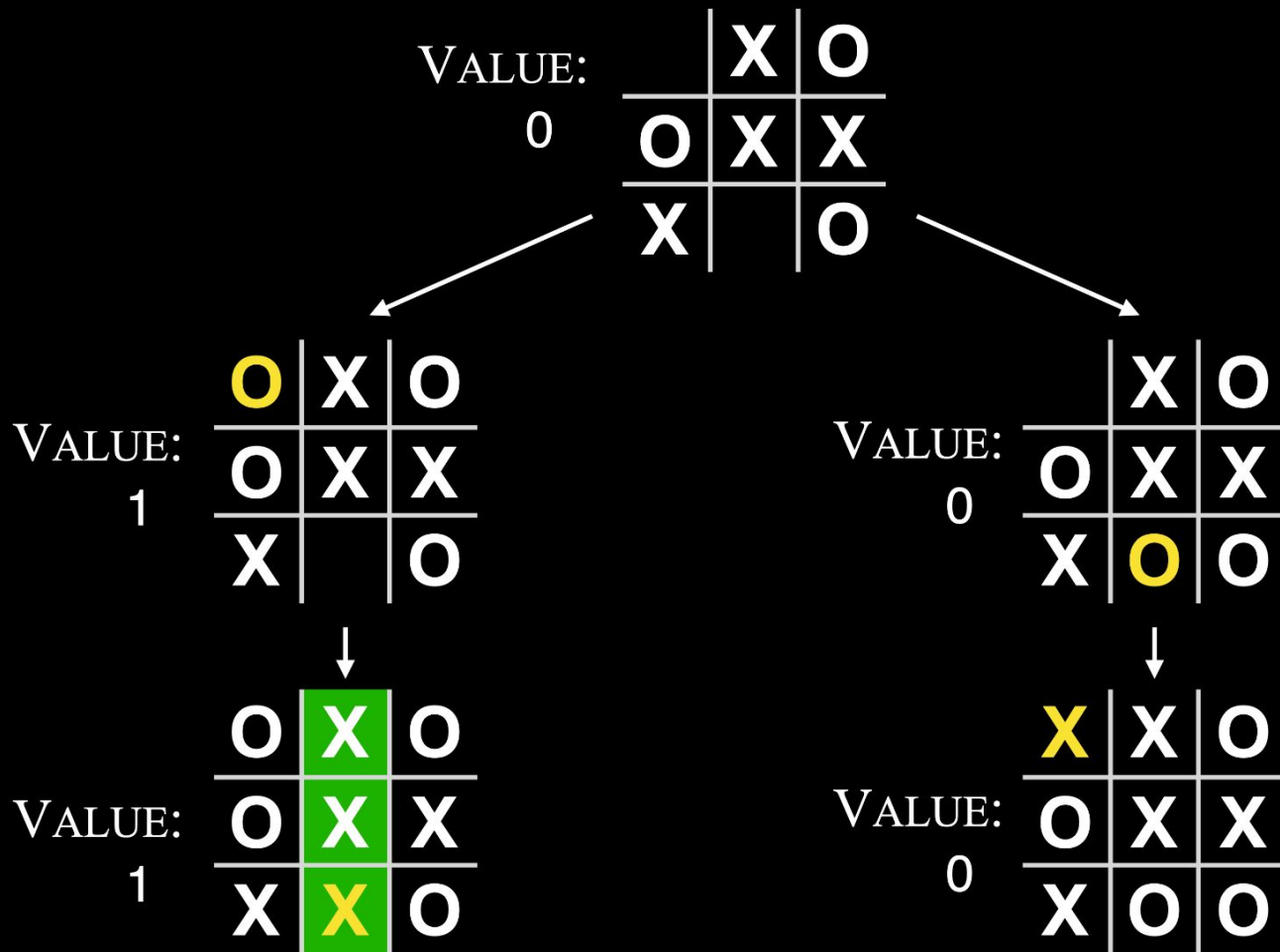
TURN = O



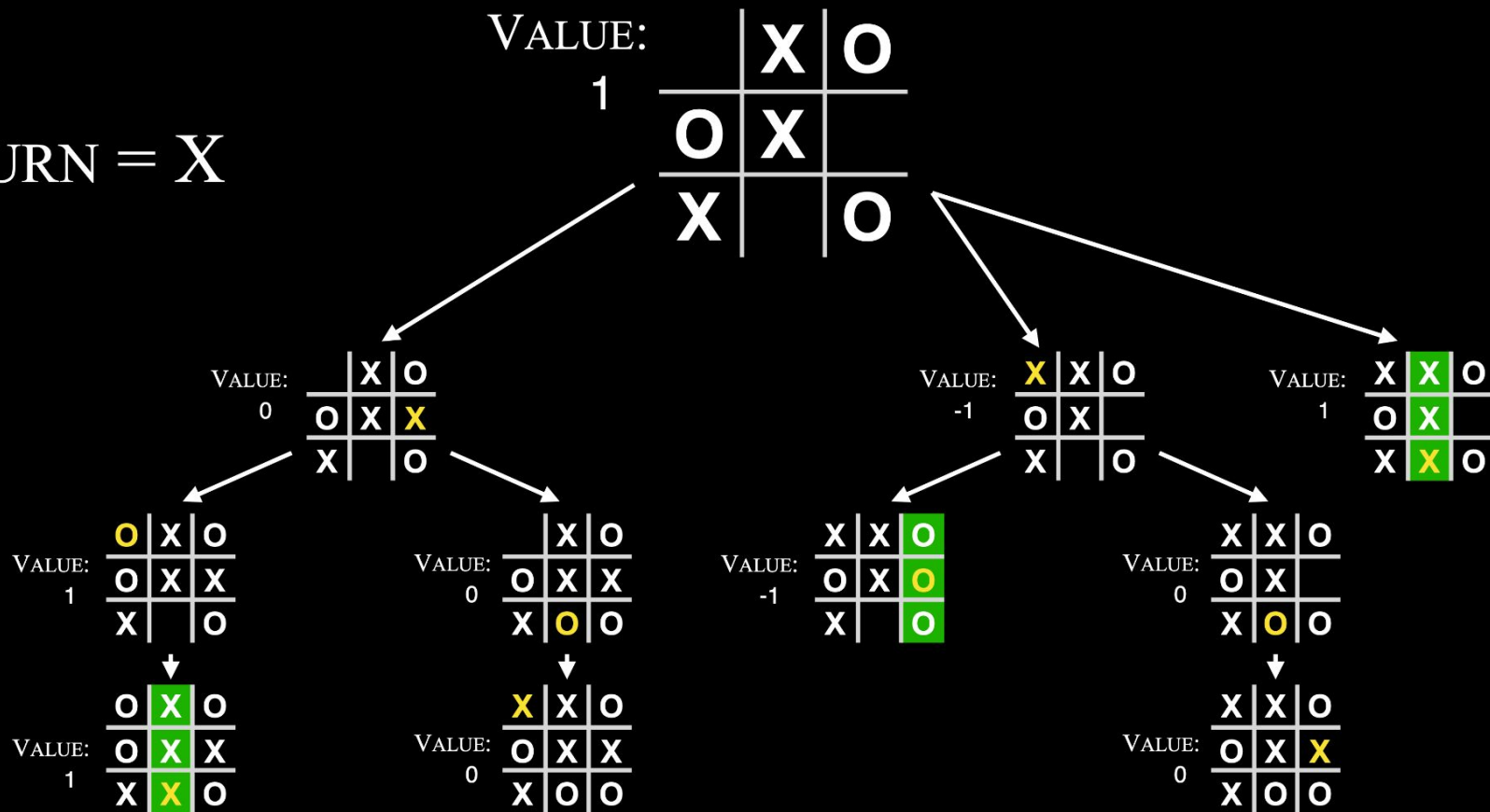
TURN = O



TURN = O



TURN = X



If player is X:

For each possible move:

Calculate score for board

Choose move with highest score

If player is X:

For each possible move:

Calculate score for board

Choose move with highest score

Else if player is O:

For each possible move:

Calculate score for board

Choose move with lowest score

Possibilities

Tic-Tac-Toe

All possible games

Possibilities

Tic-Tac-Toe

All possible games

255,168

Possibilities

Tic-Tac-Toe

All possible games

255,168

Chess

First 4 moves only

Possibilities

Tic-Tac-Toe

All possible games

255,168

Chess

First 4 moves only

85 billion

Possibilities

Tic-Tac-Toe

All possible games

255,168

Chess

First 4 moves only

85 billion

Go

First 4 moves only

266 quintillion

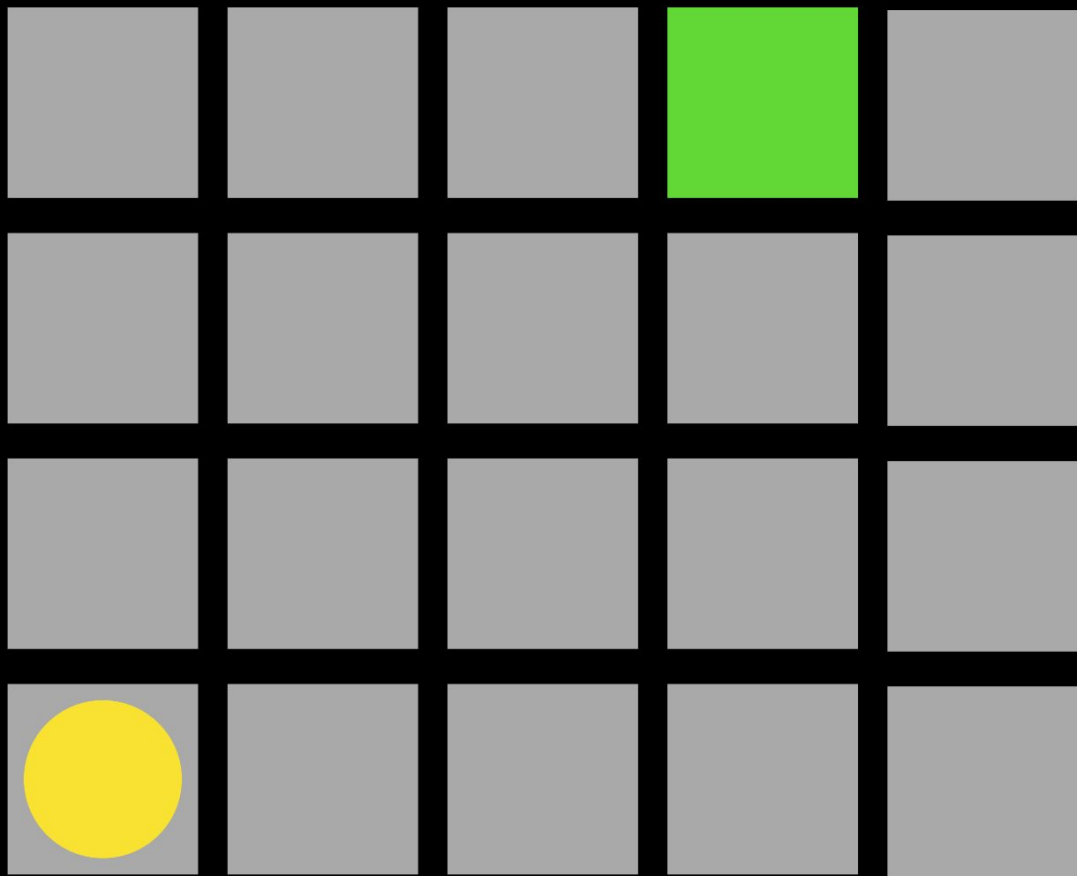
machine learning

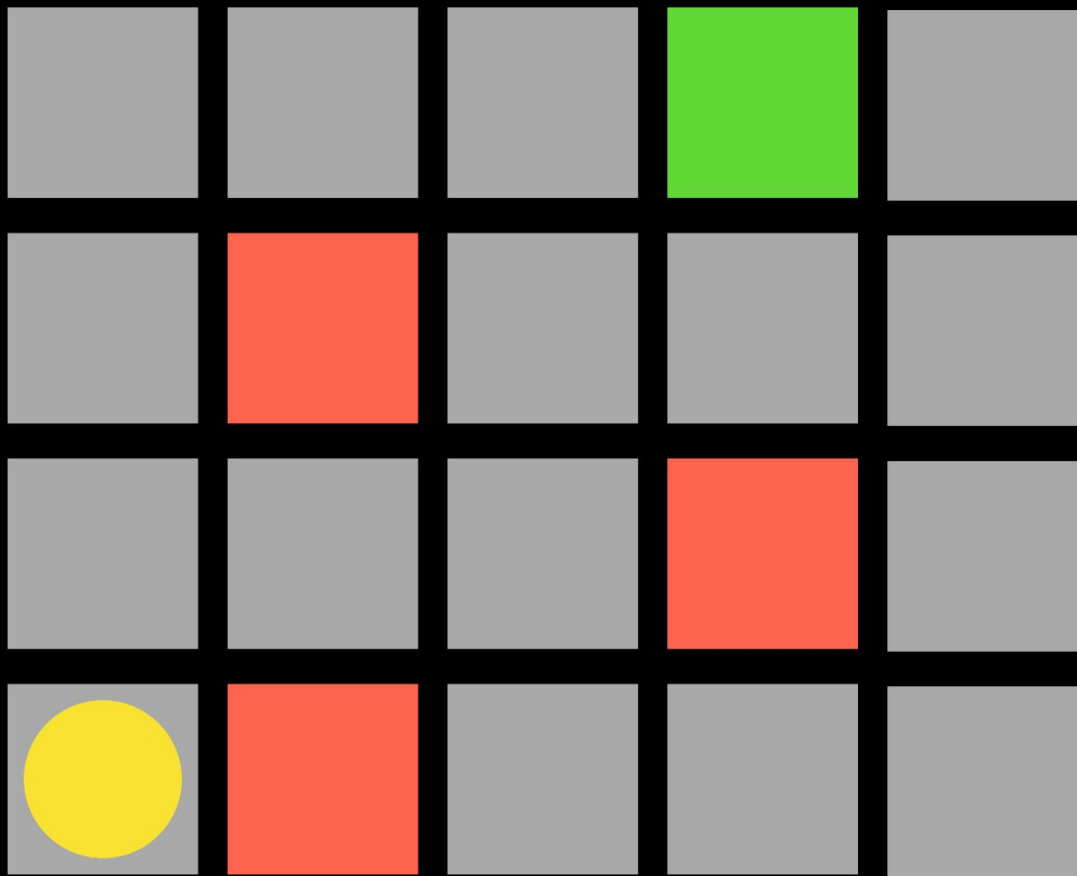
reinforcement learning

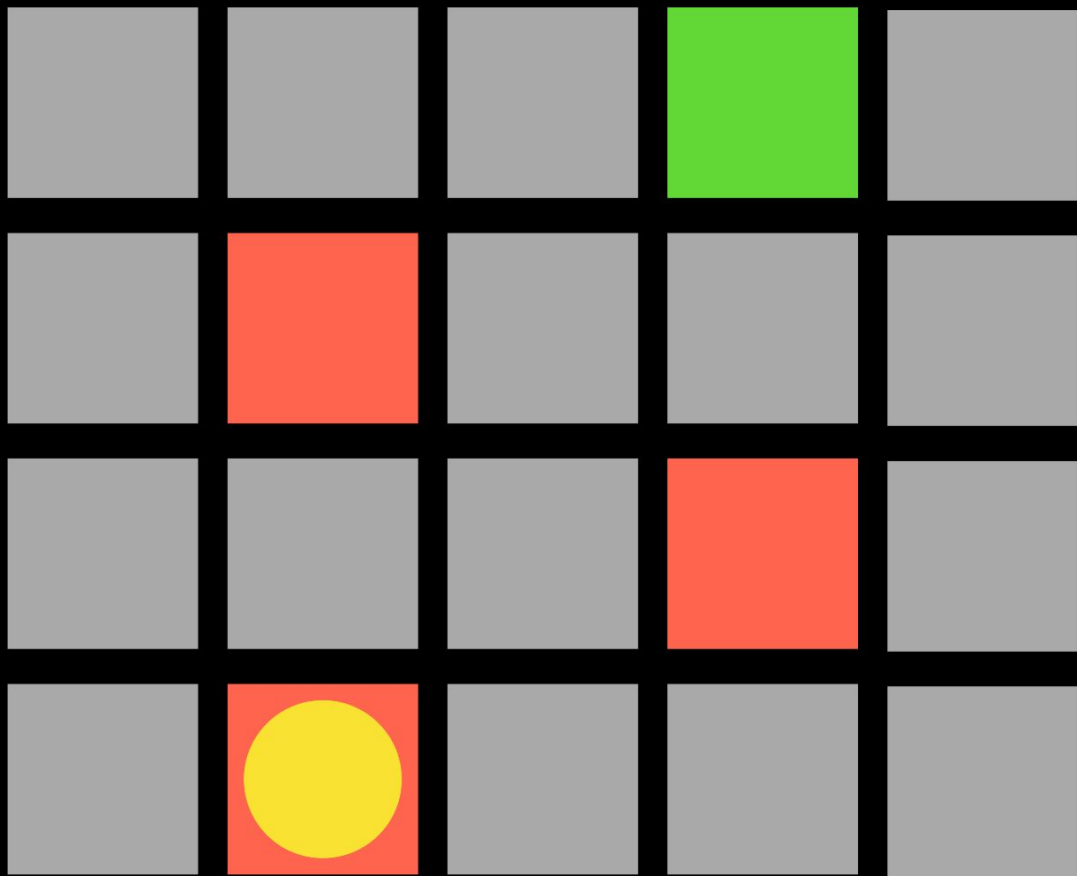


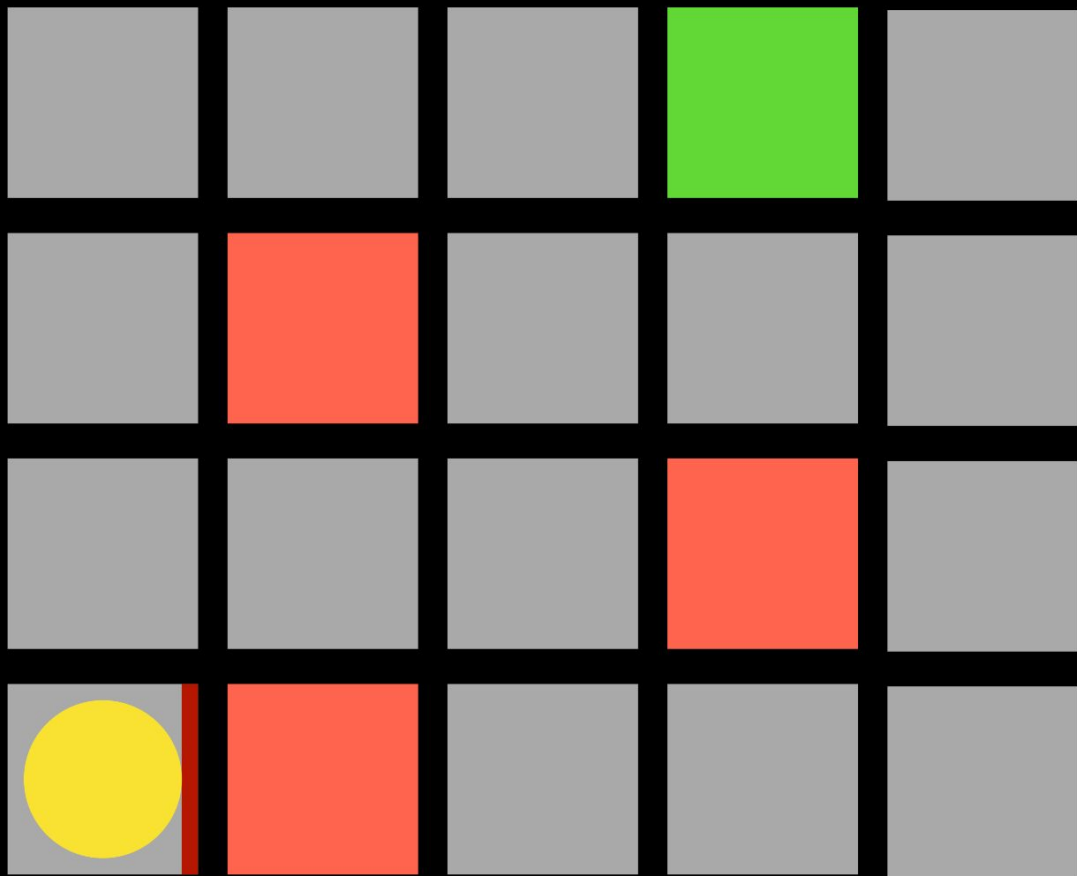
A white robotic arm with a yellow gripper is holding a black frying pan containing a yellow egg. The arm is positioned over a blue cloth-covered surface. The background is dark blue. The text "Reinforcement Learning" and "First trial..." is overlaid at the bottom.

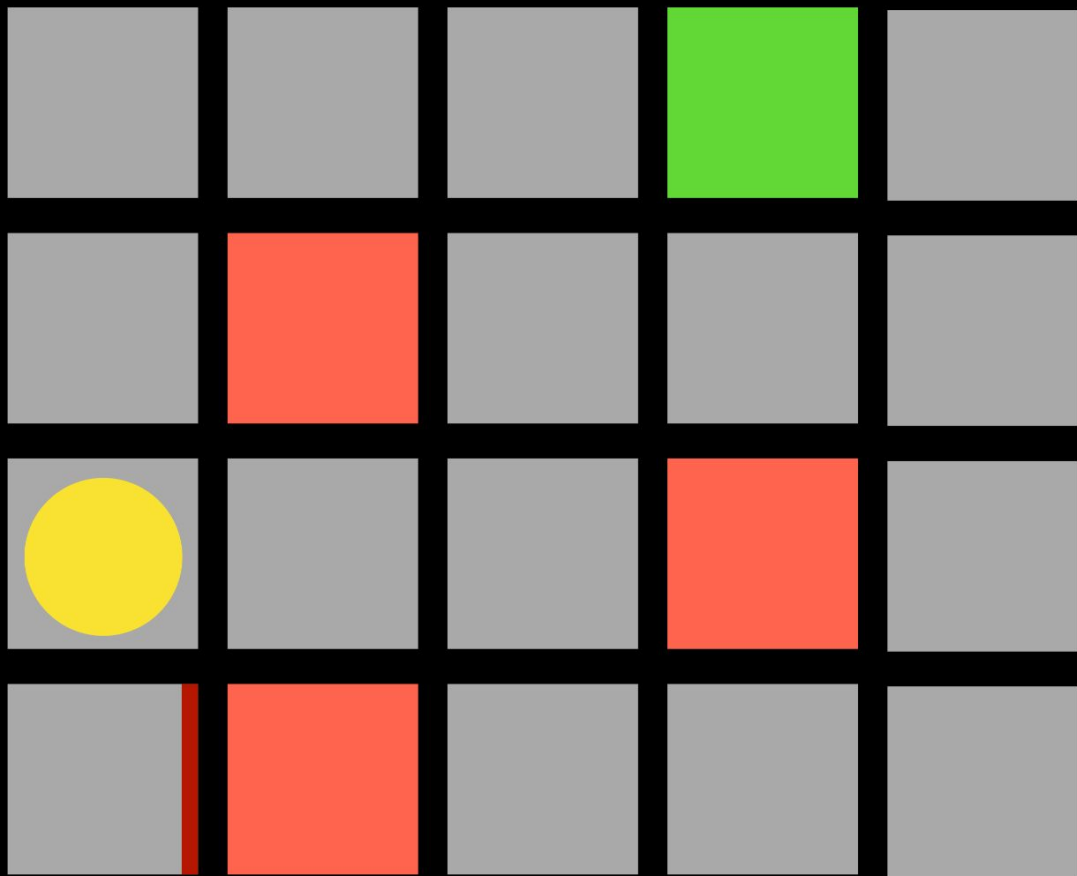
Reinforcement Learning
First trial...

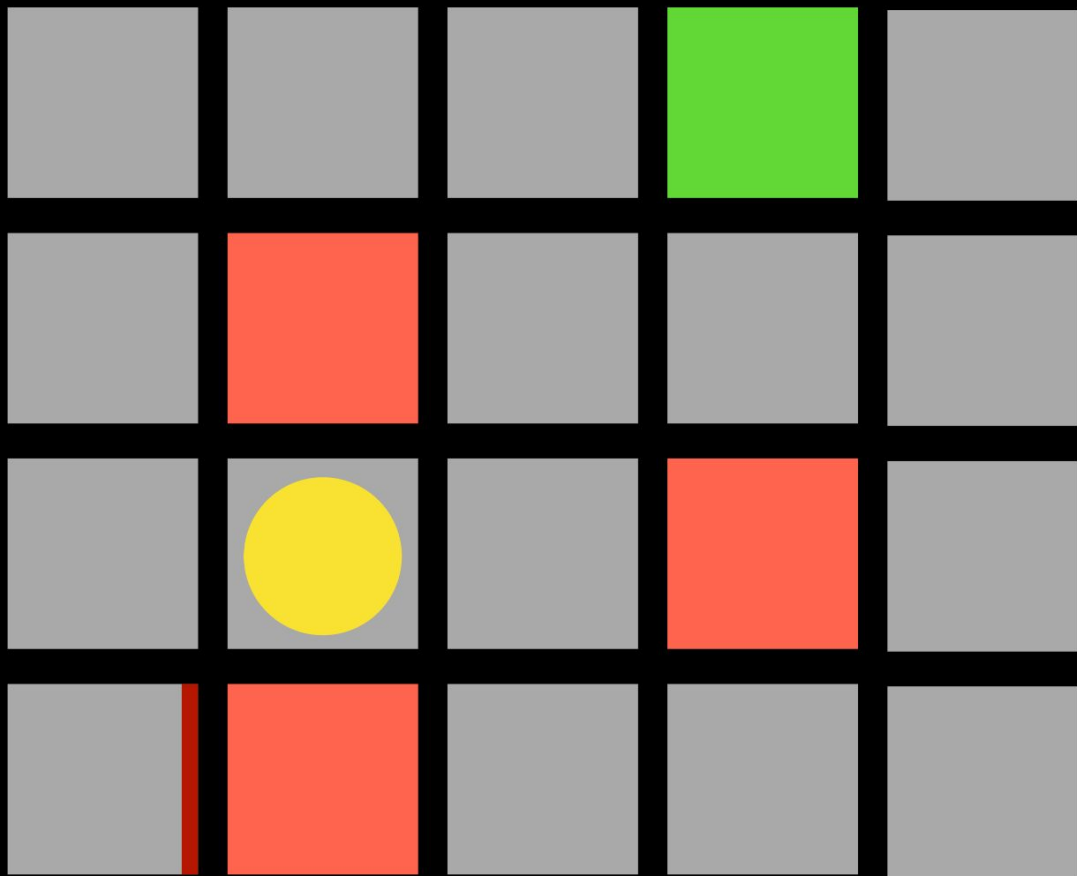


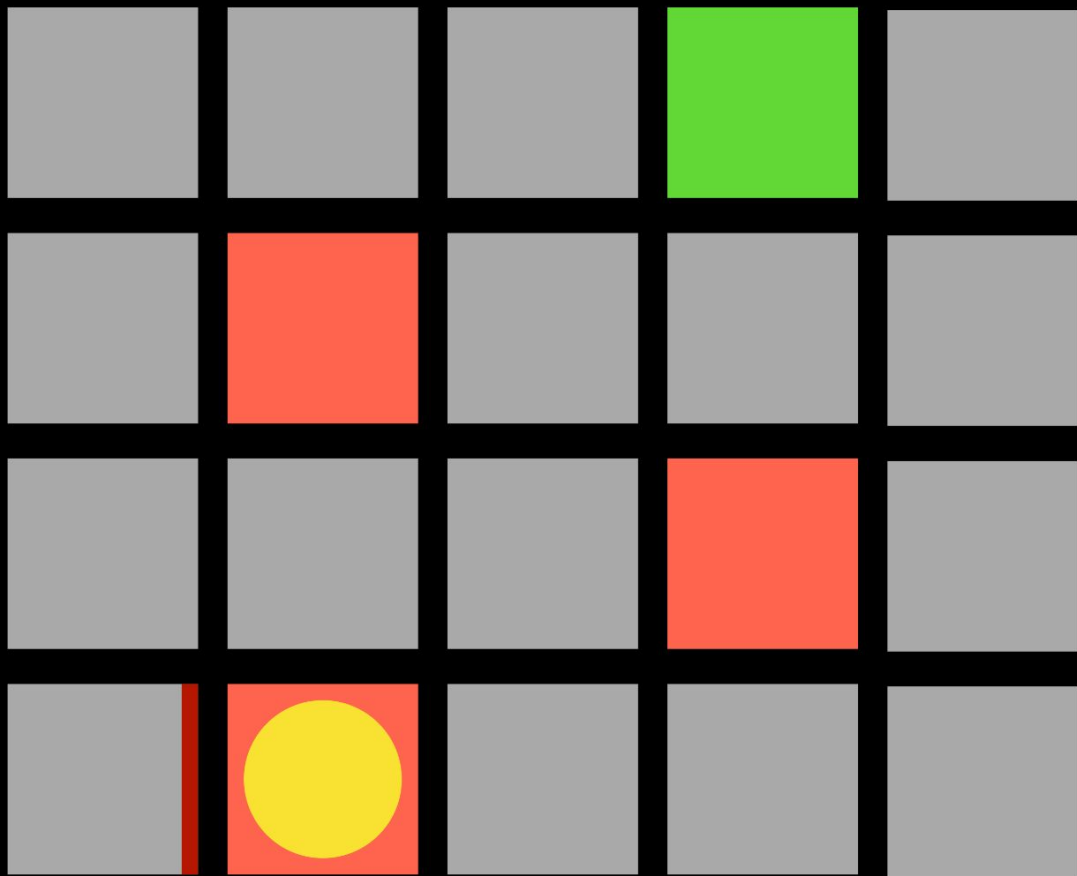


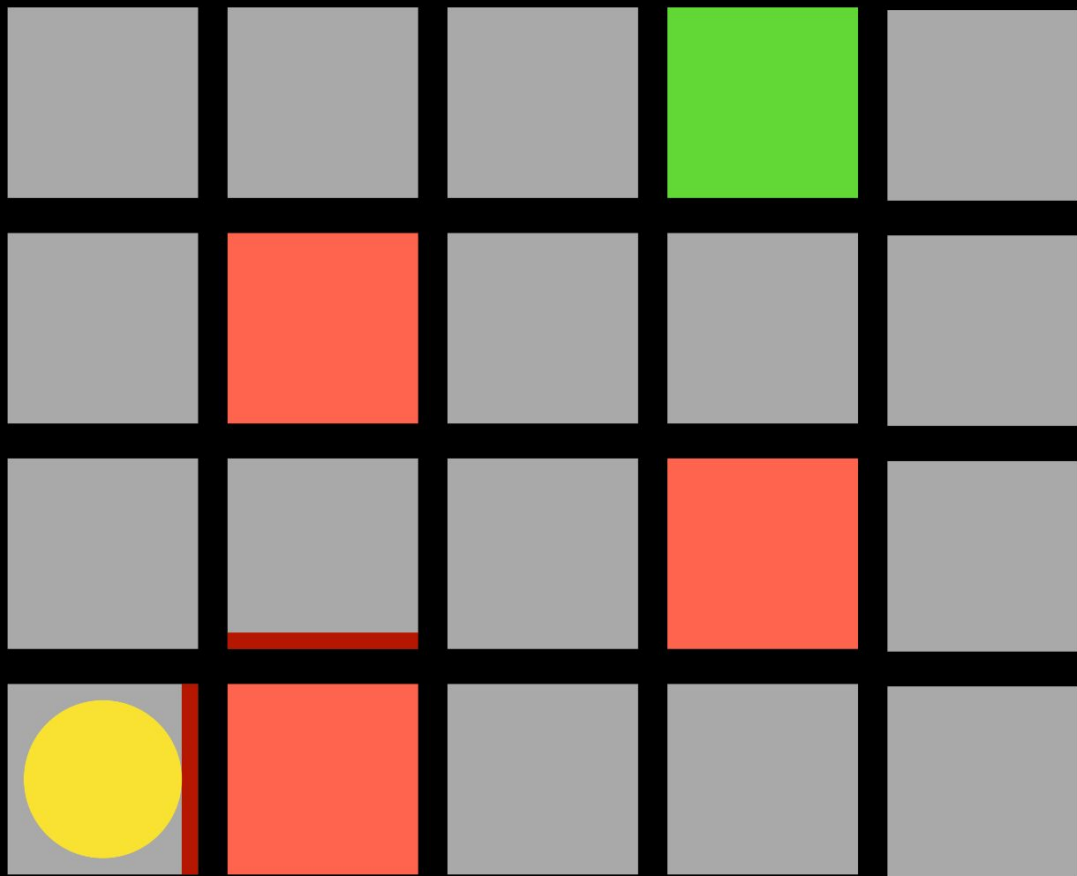


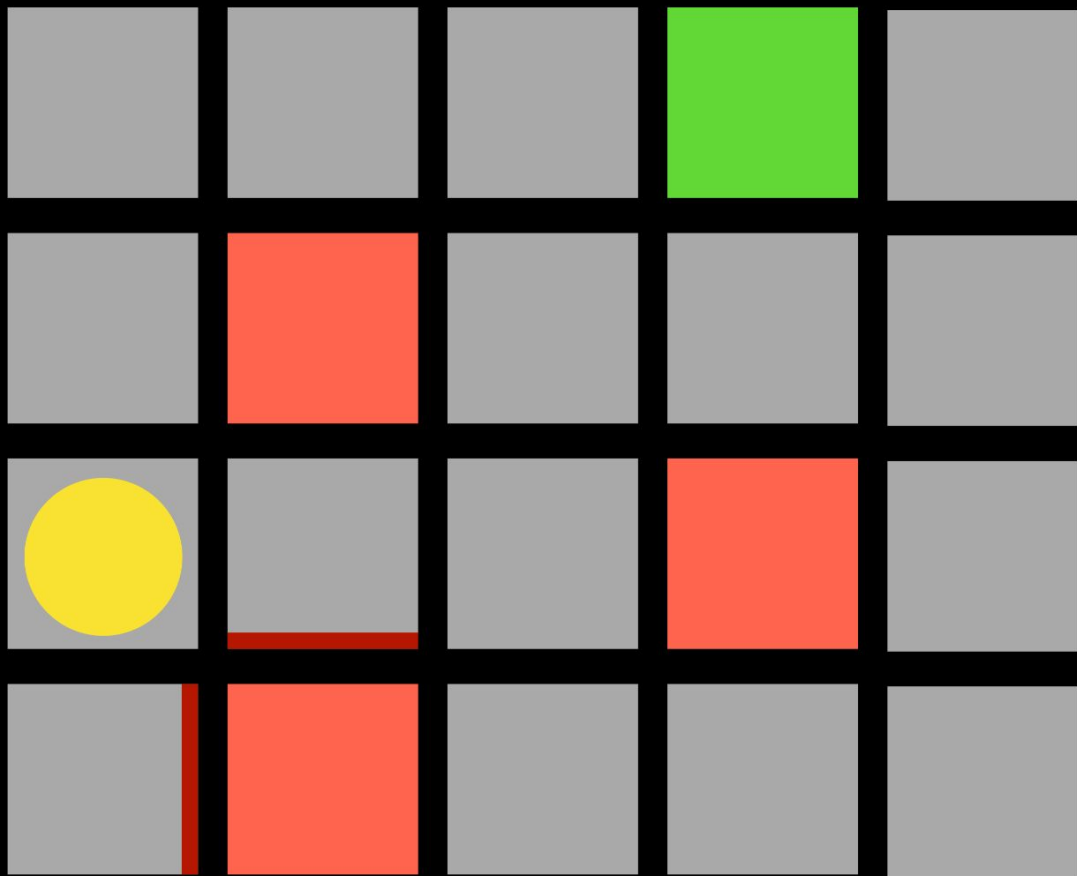


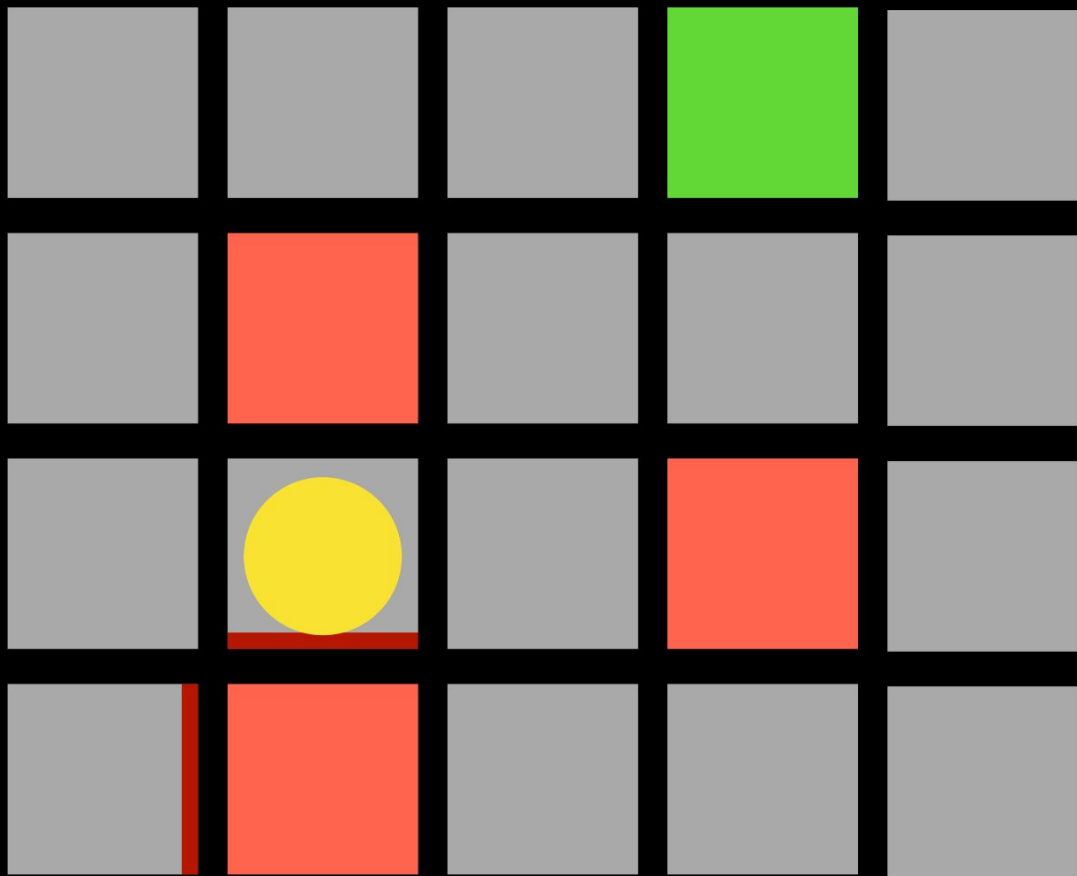


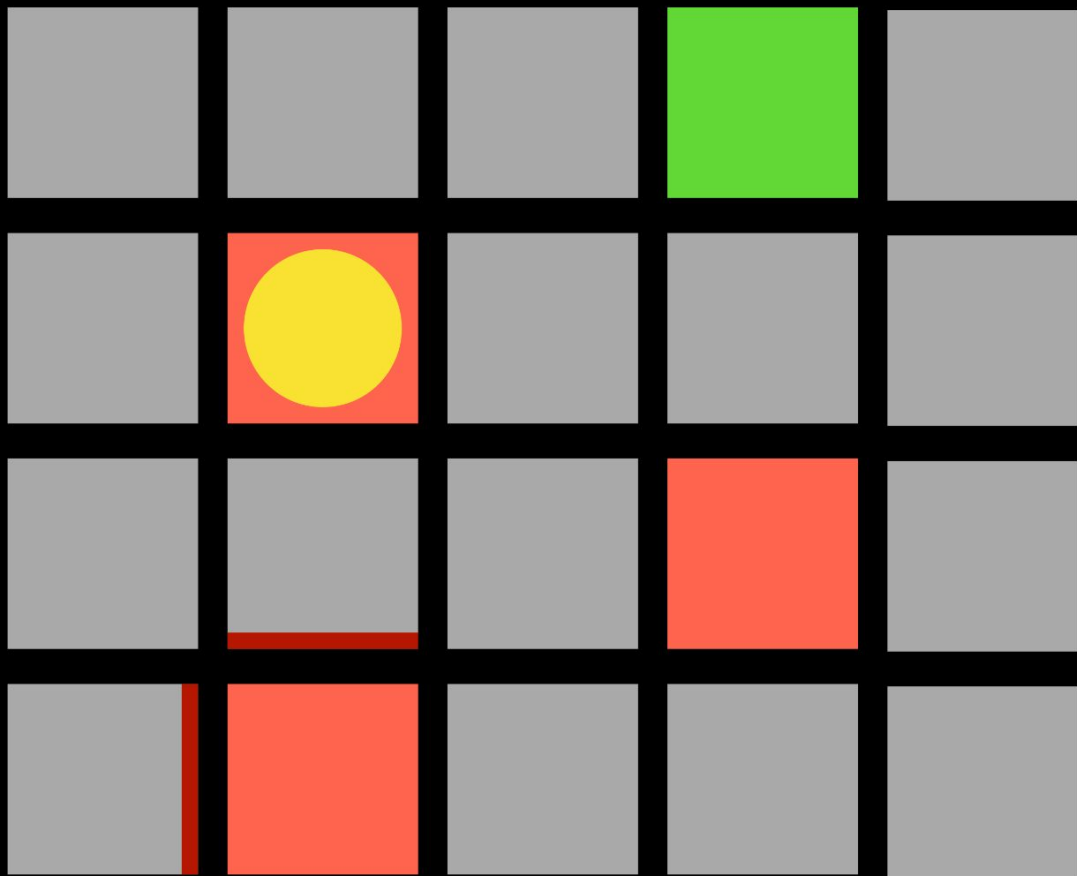


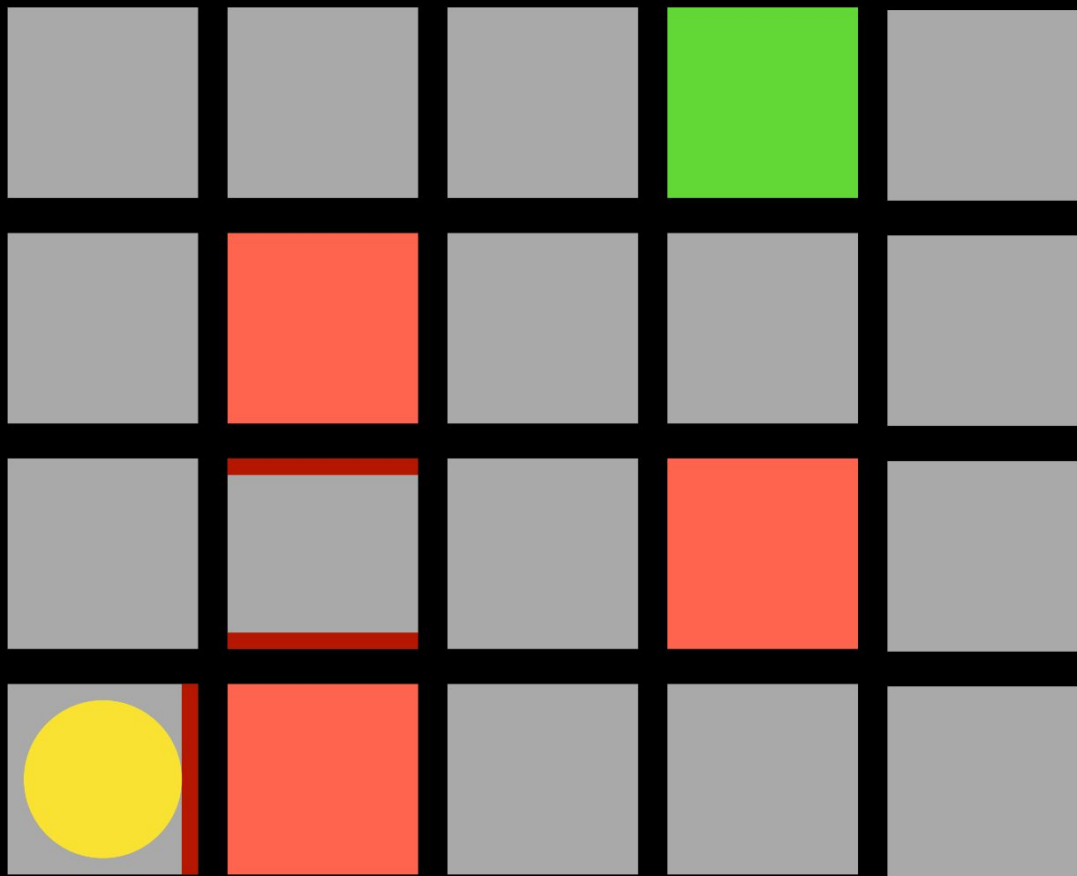


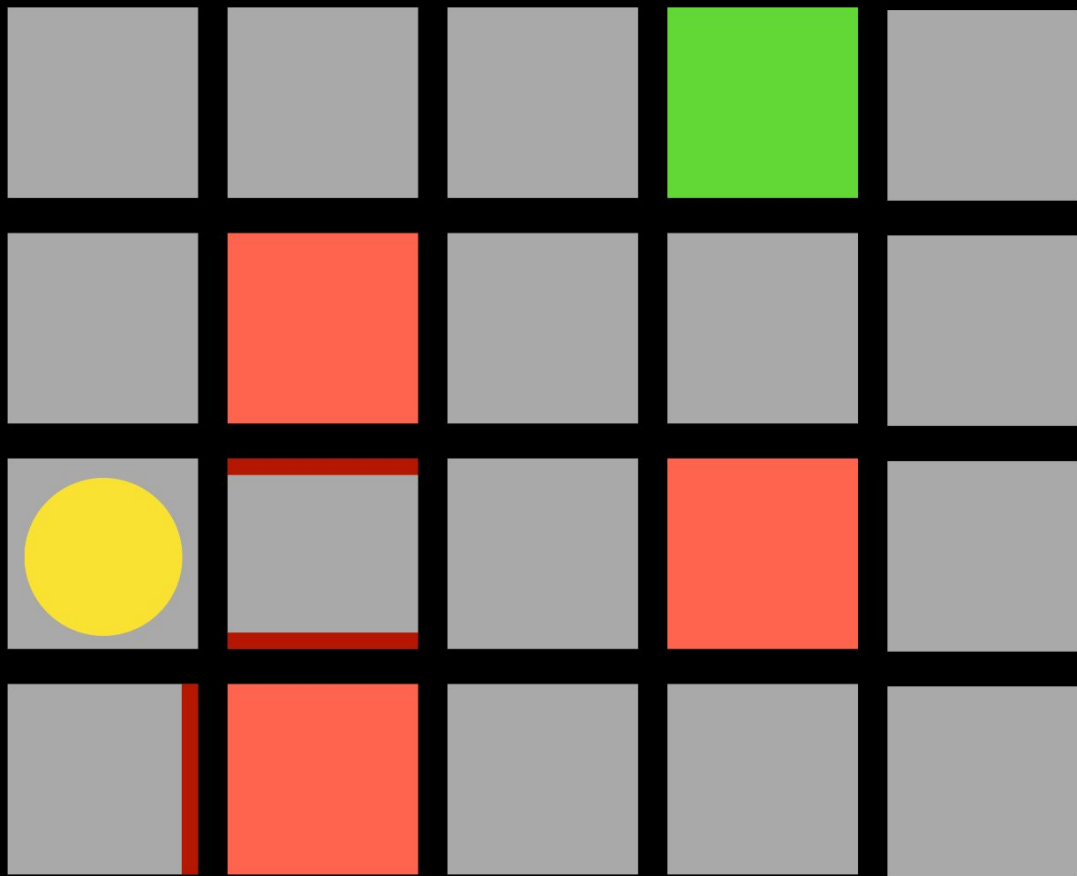


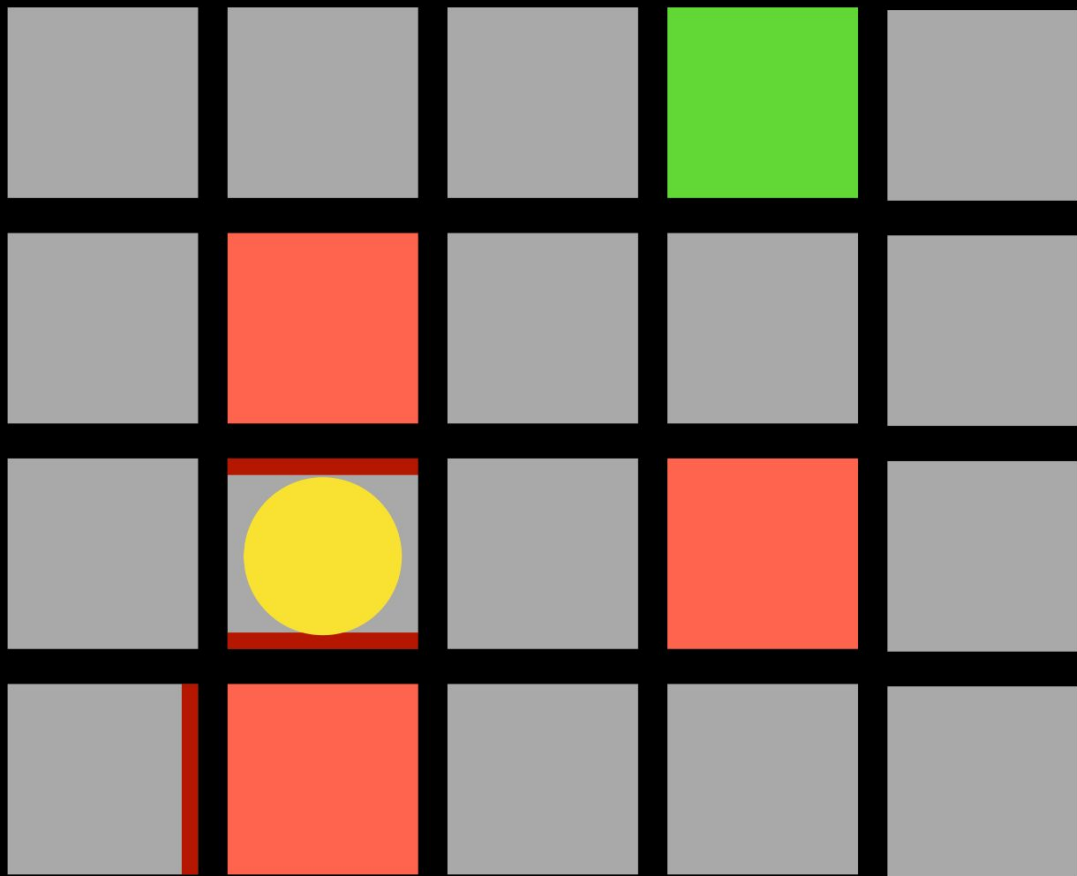


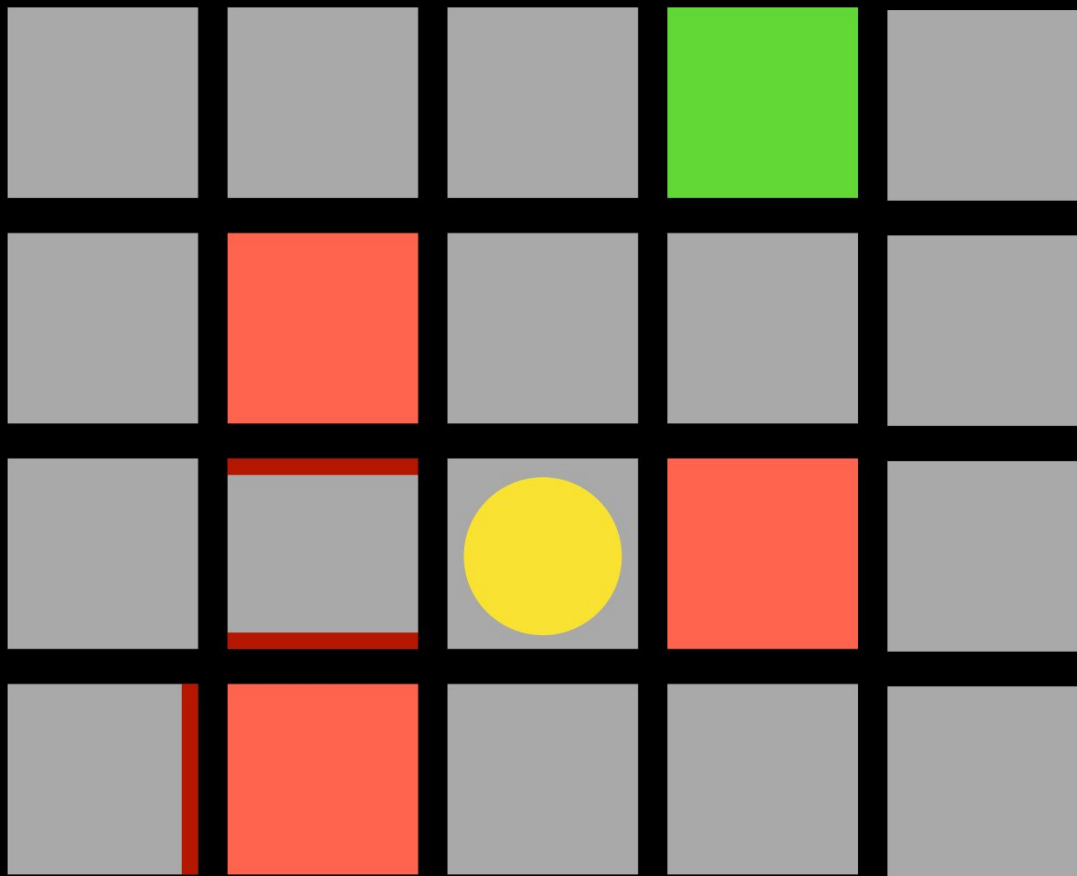


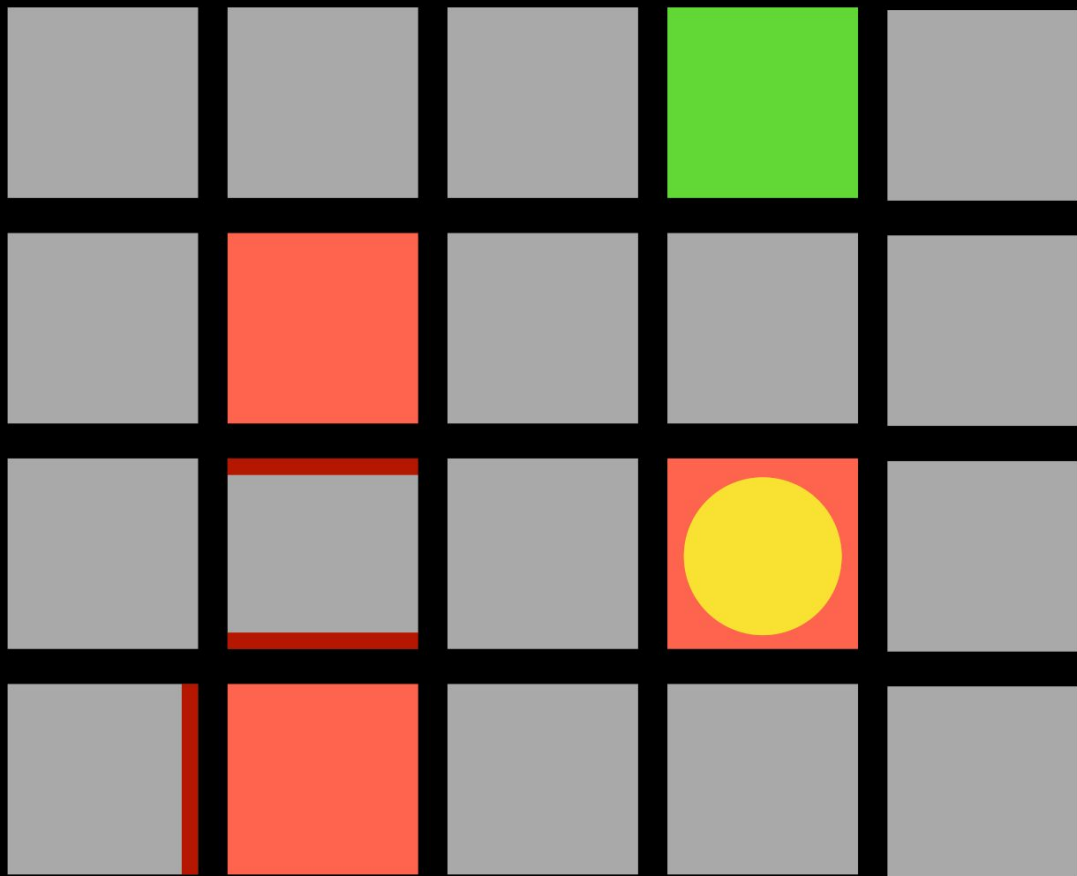


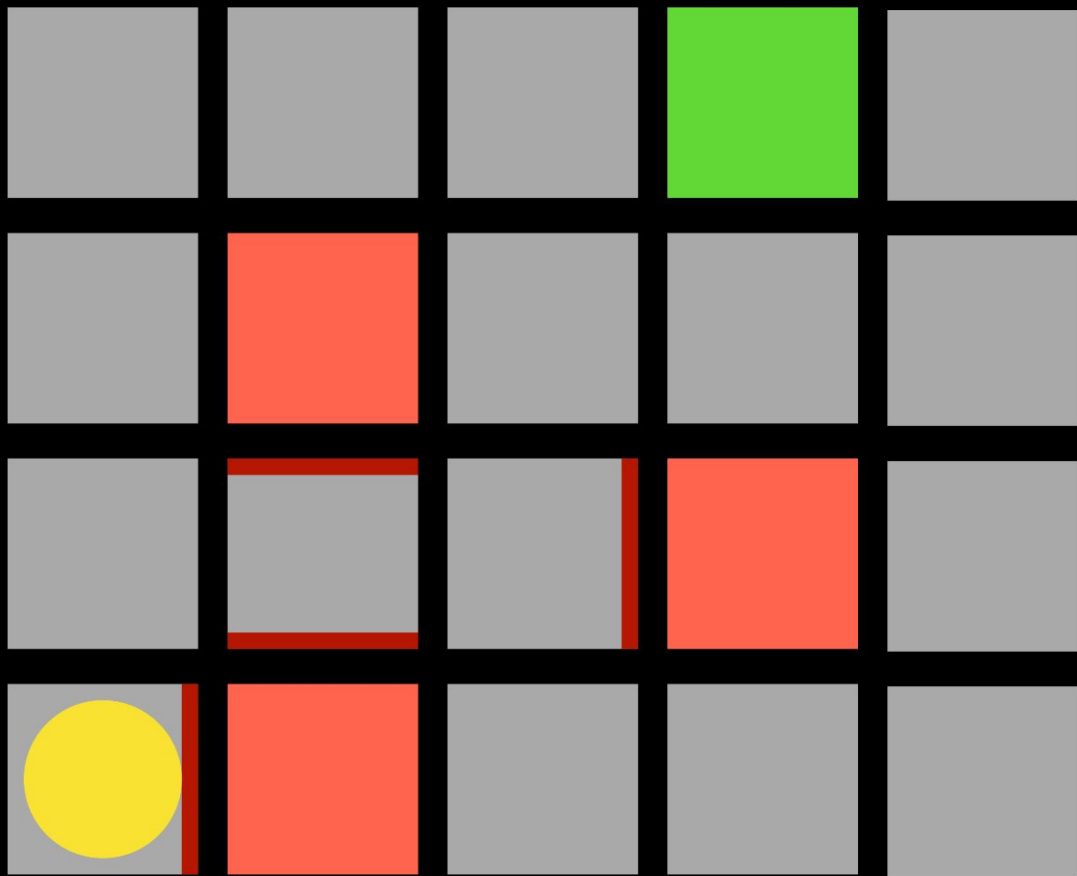


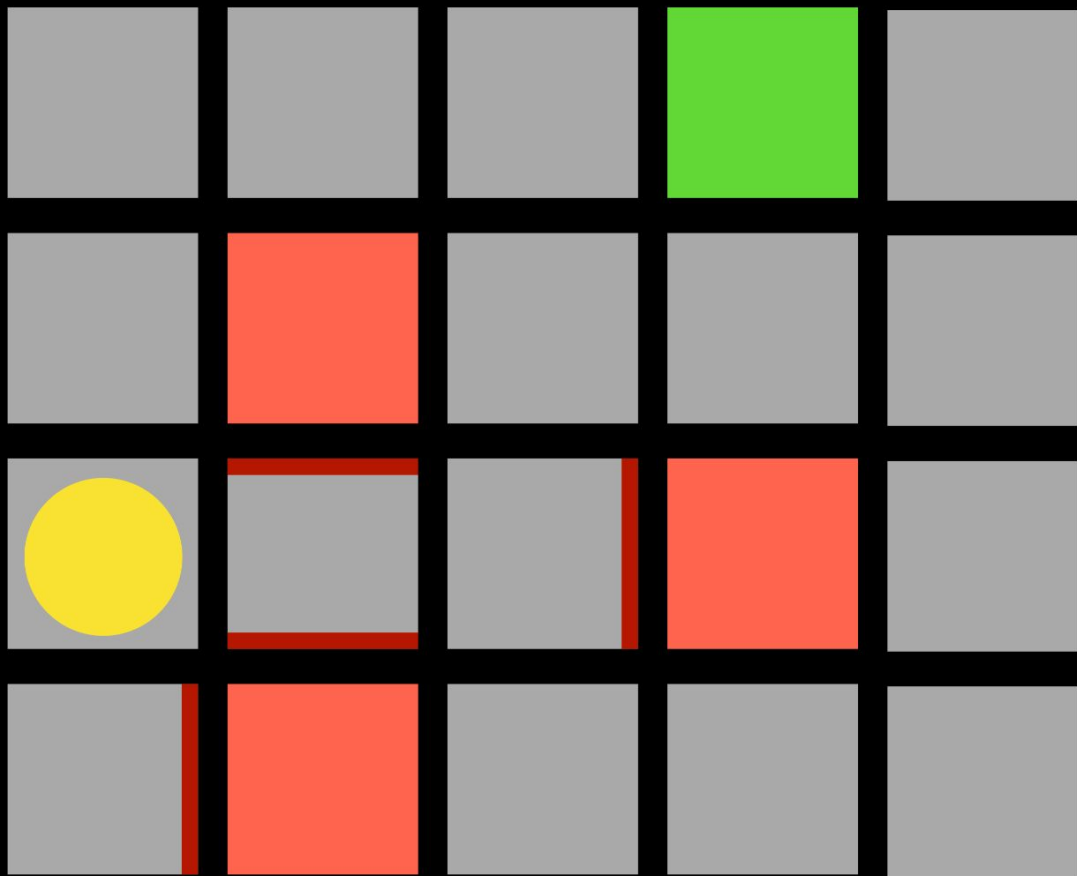


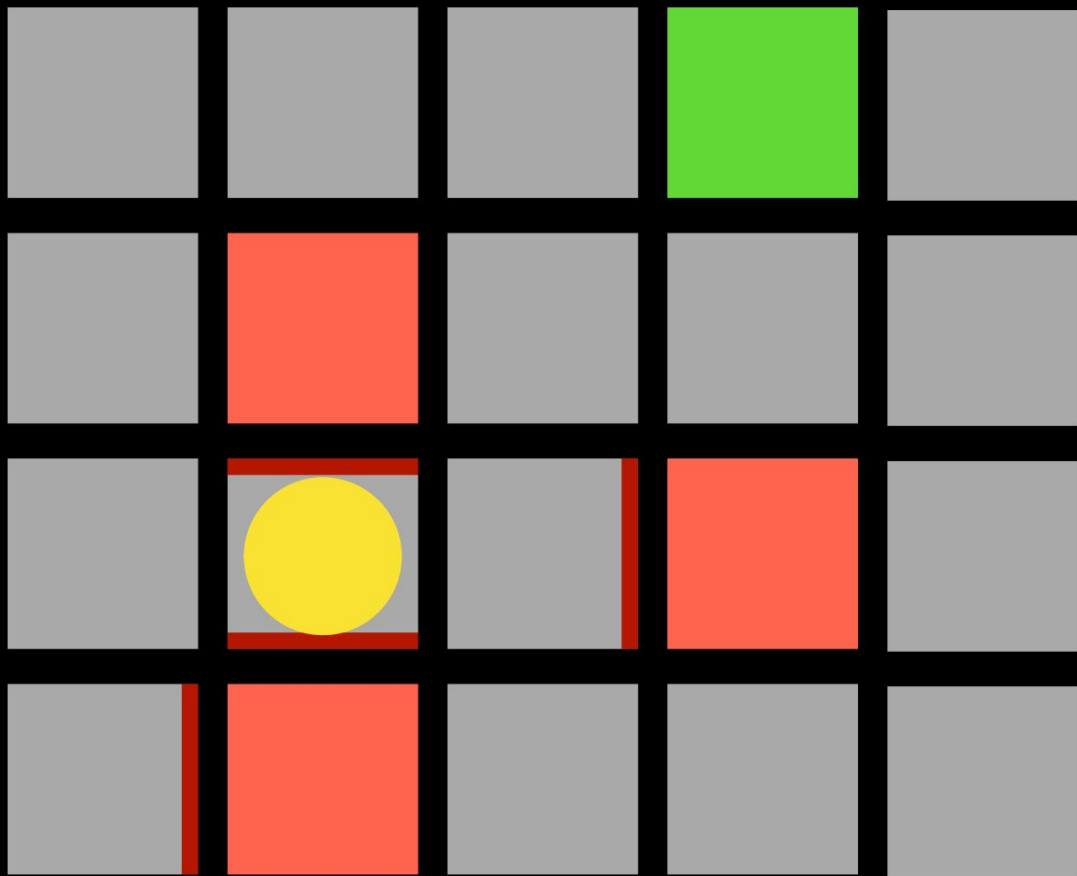


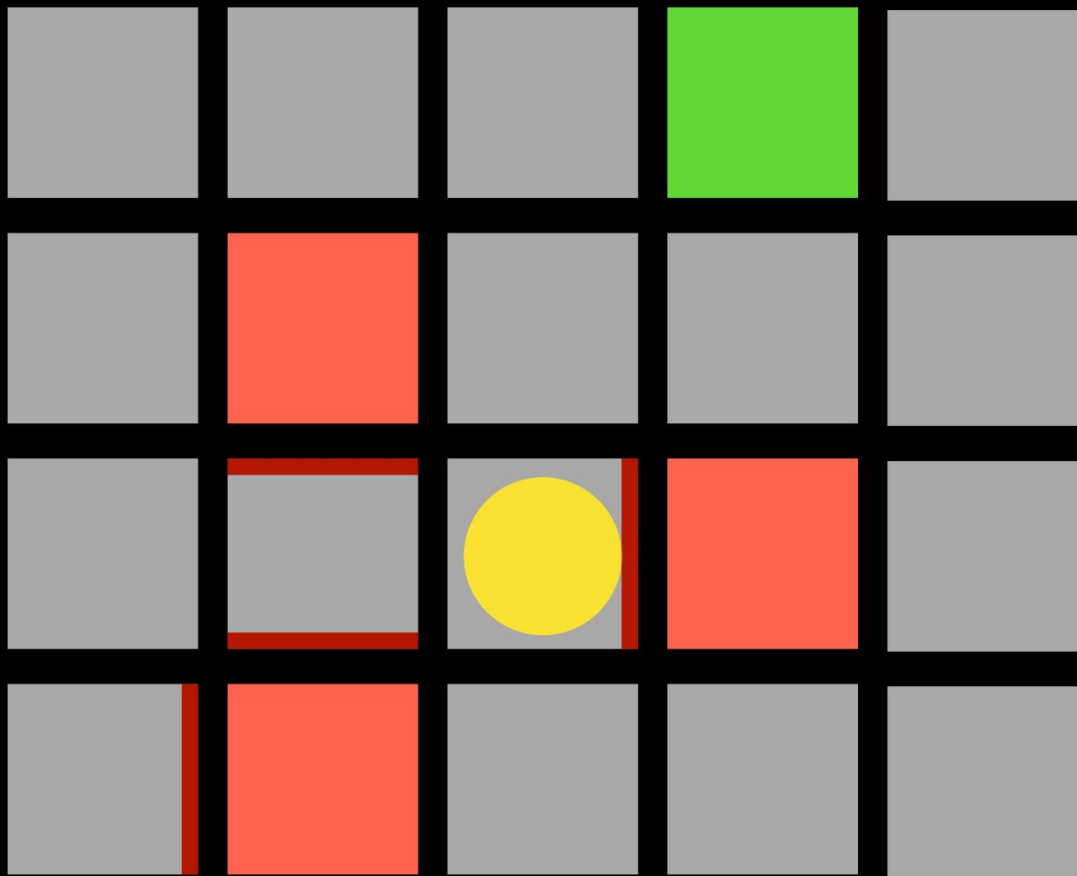


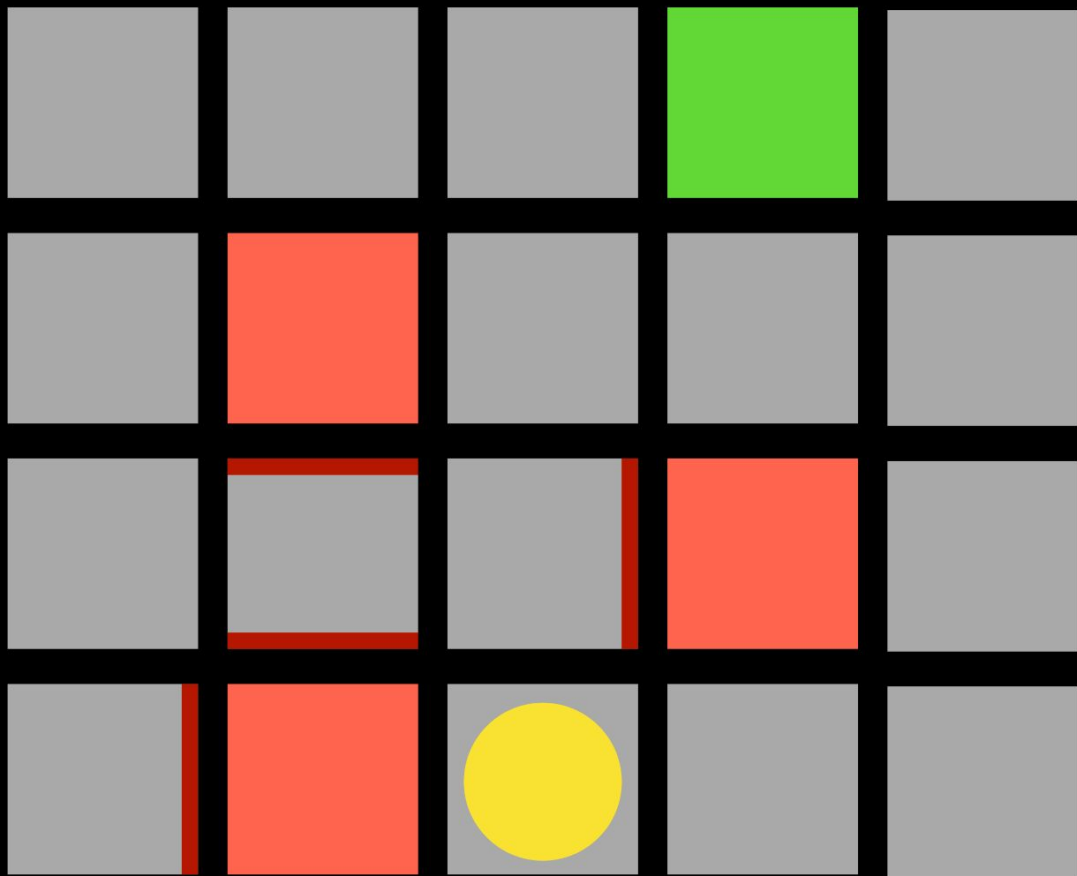


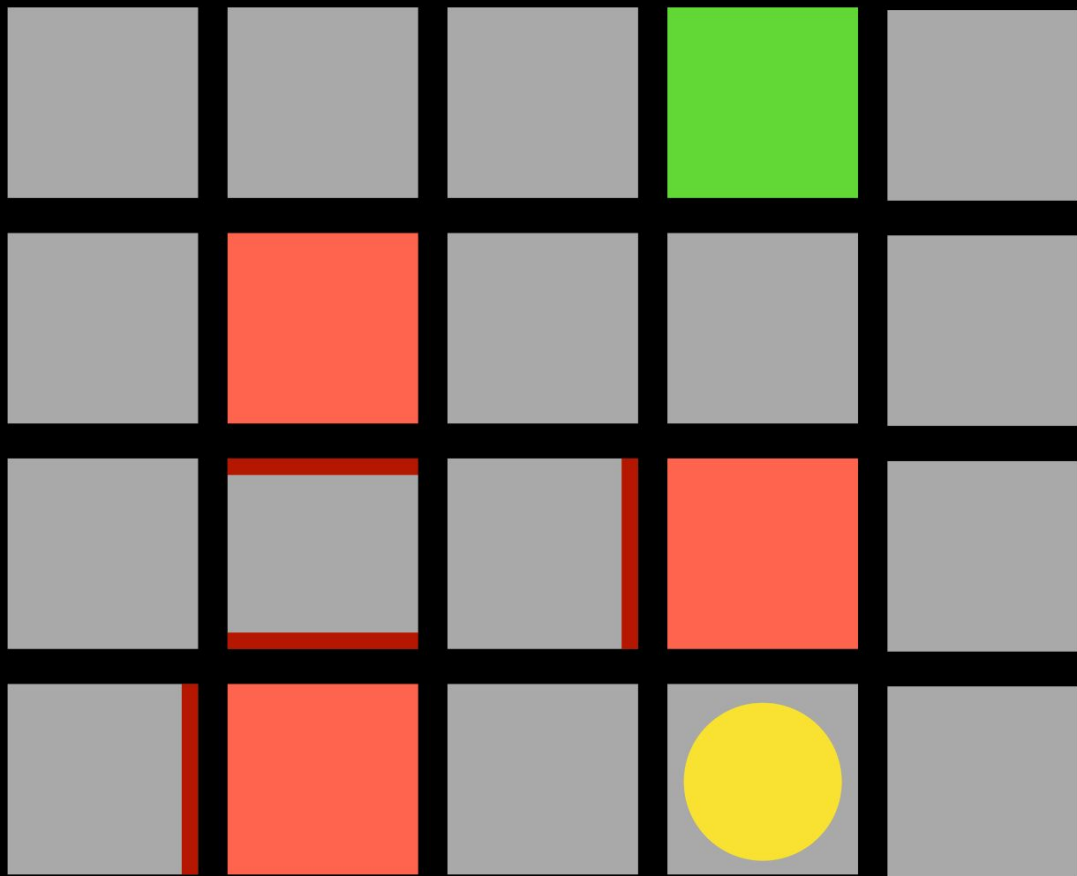


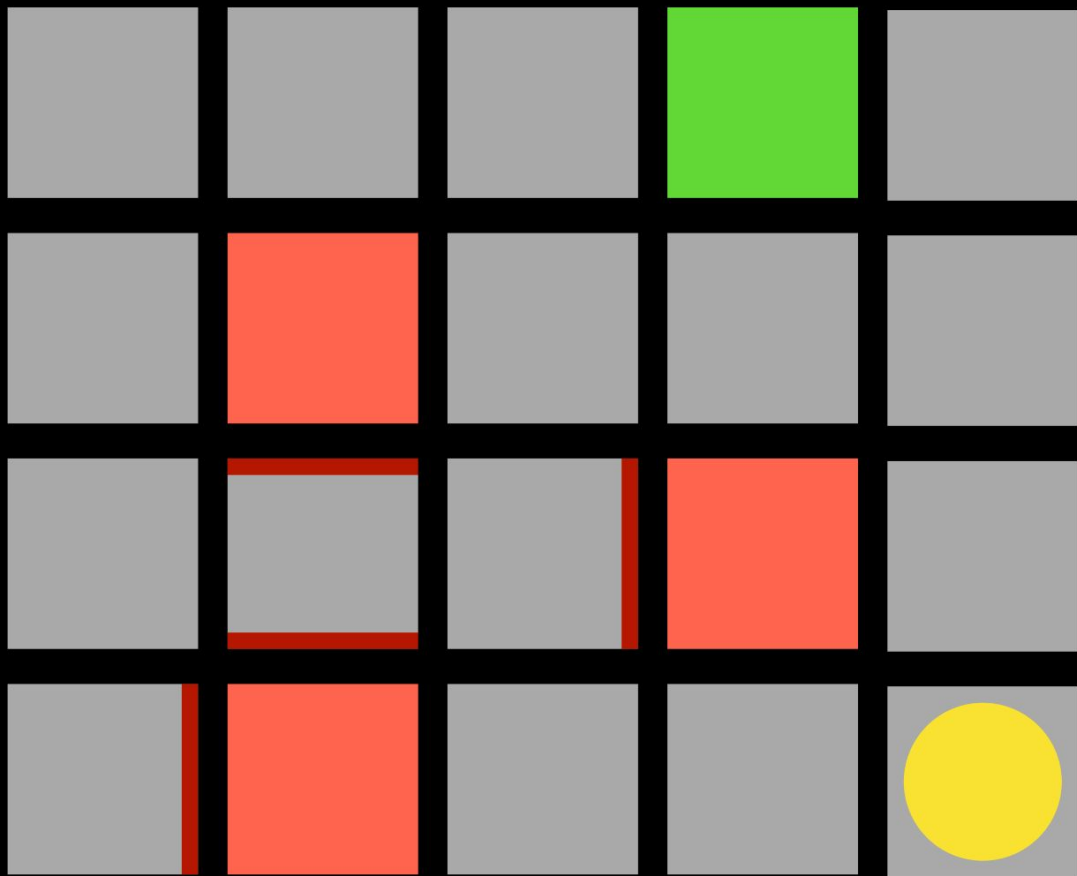


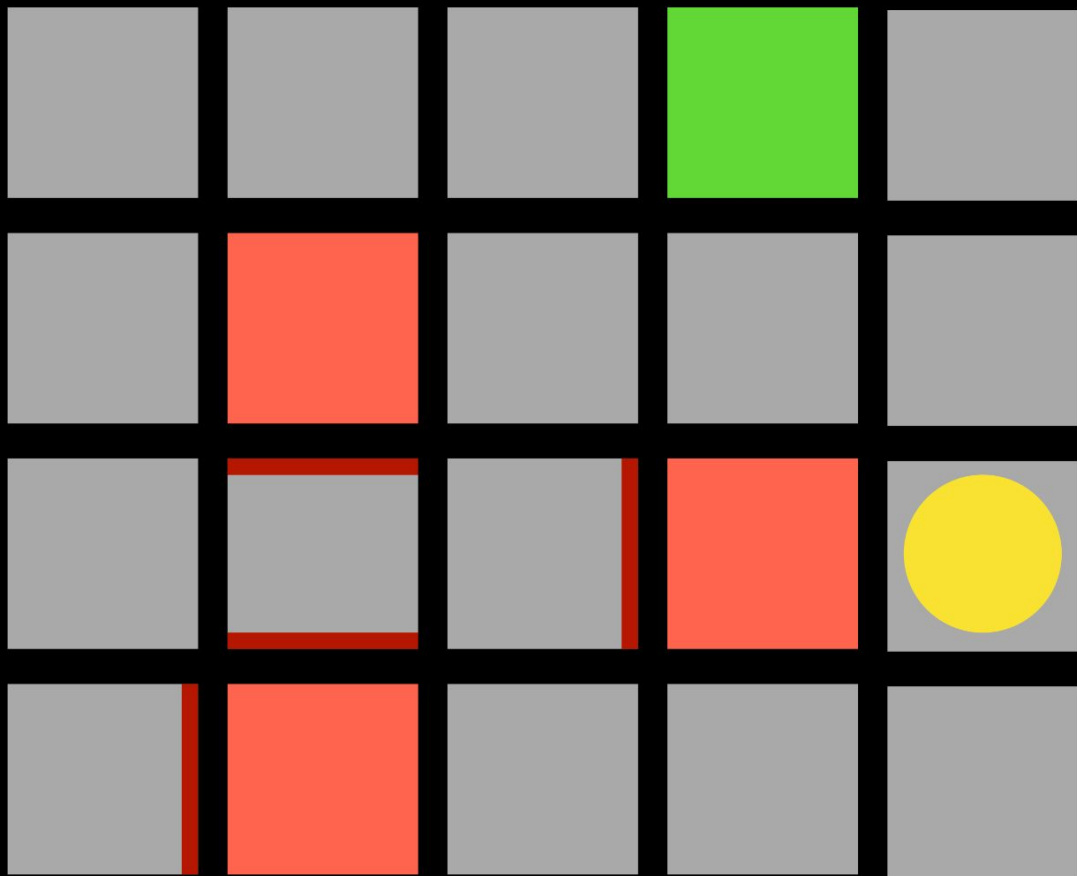


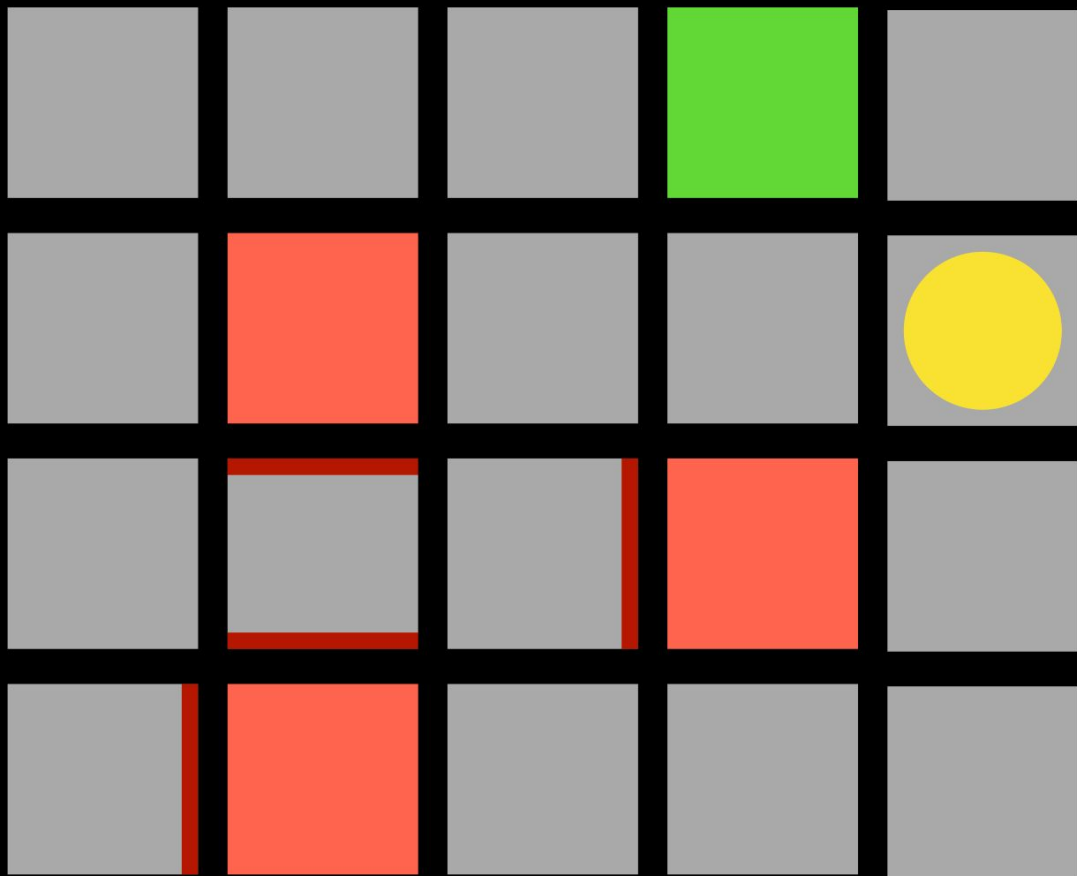


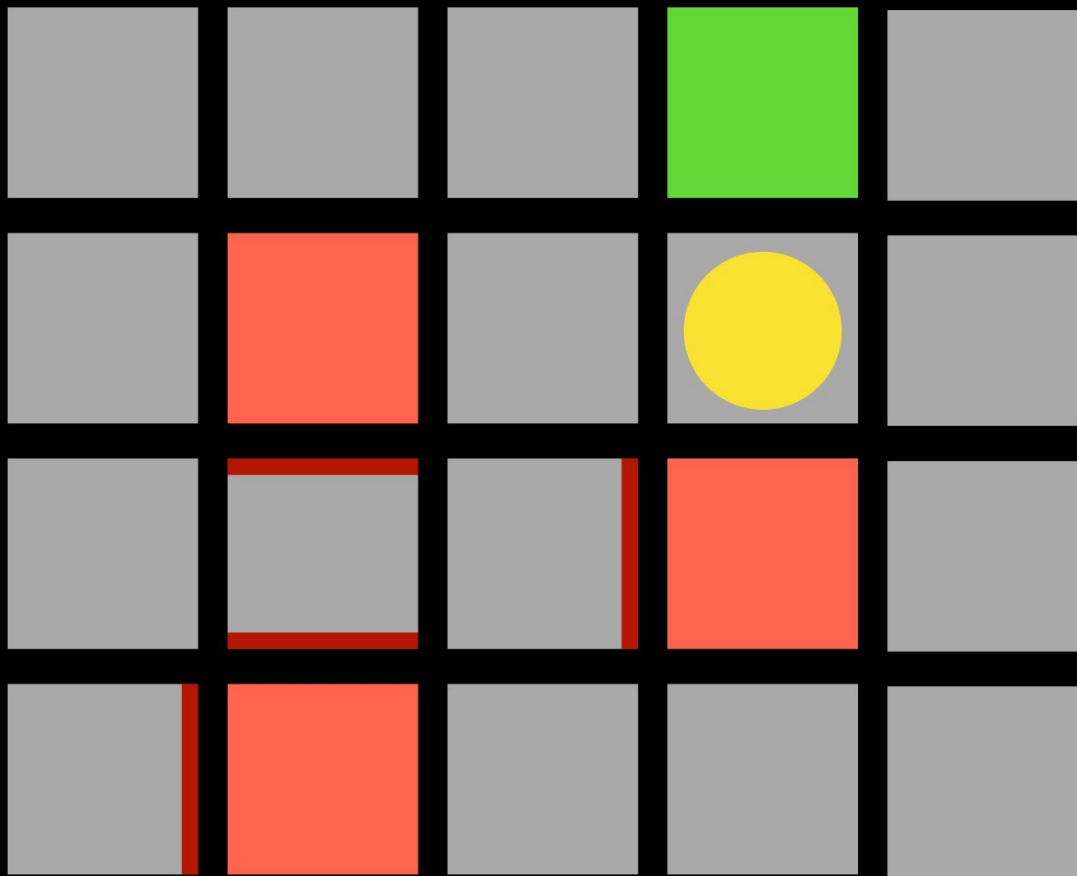


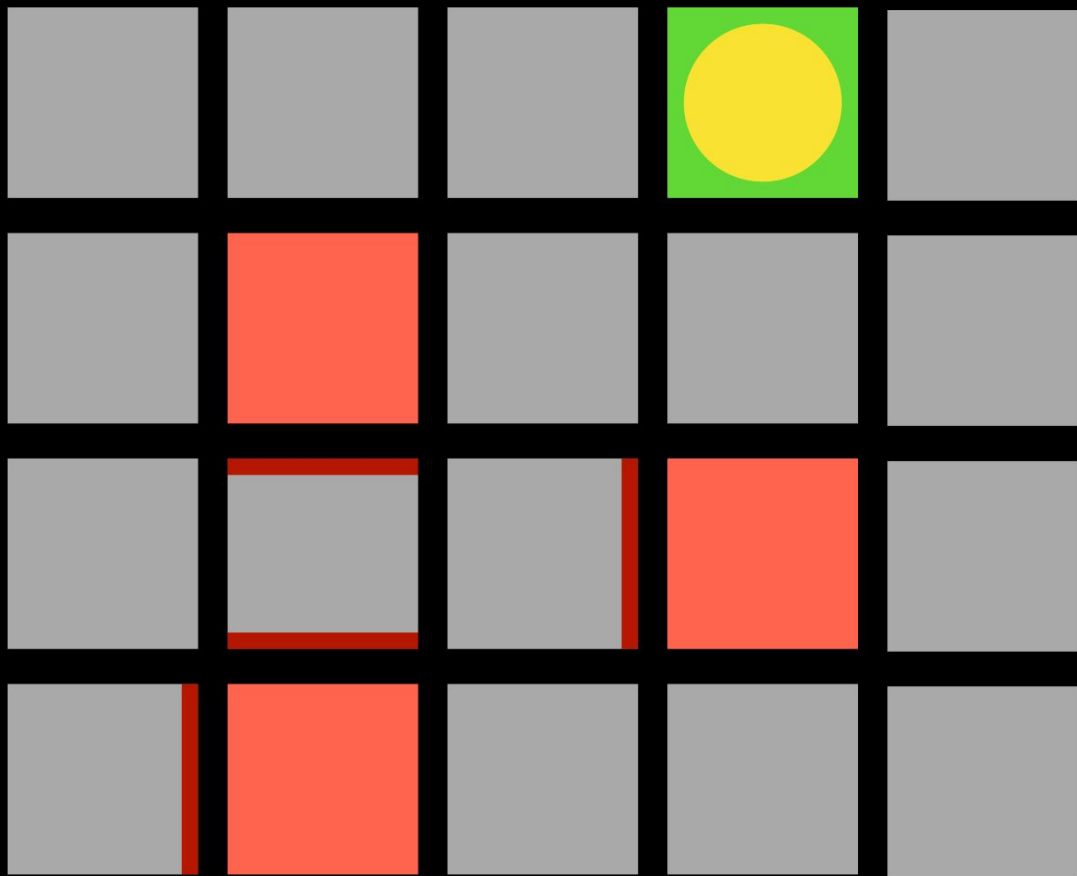


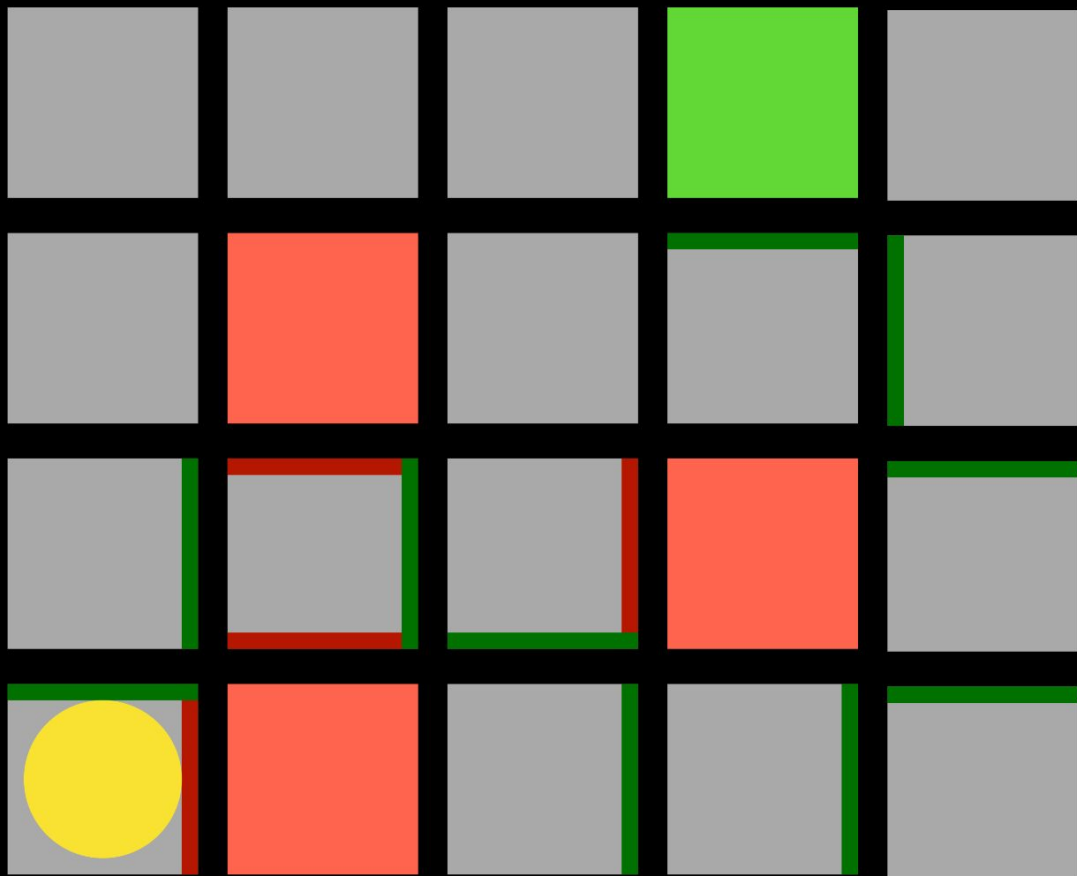


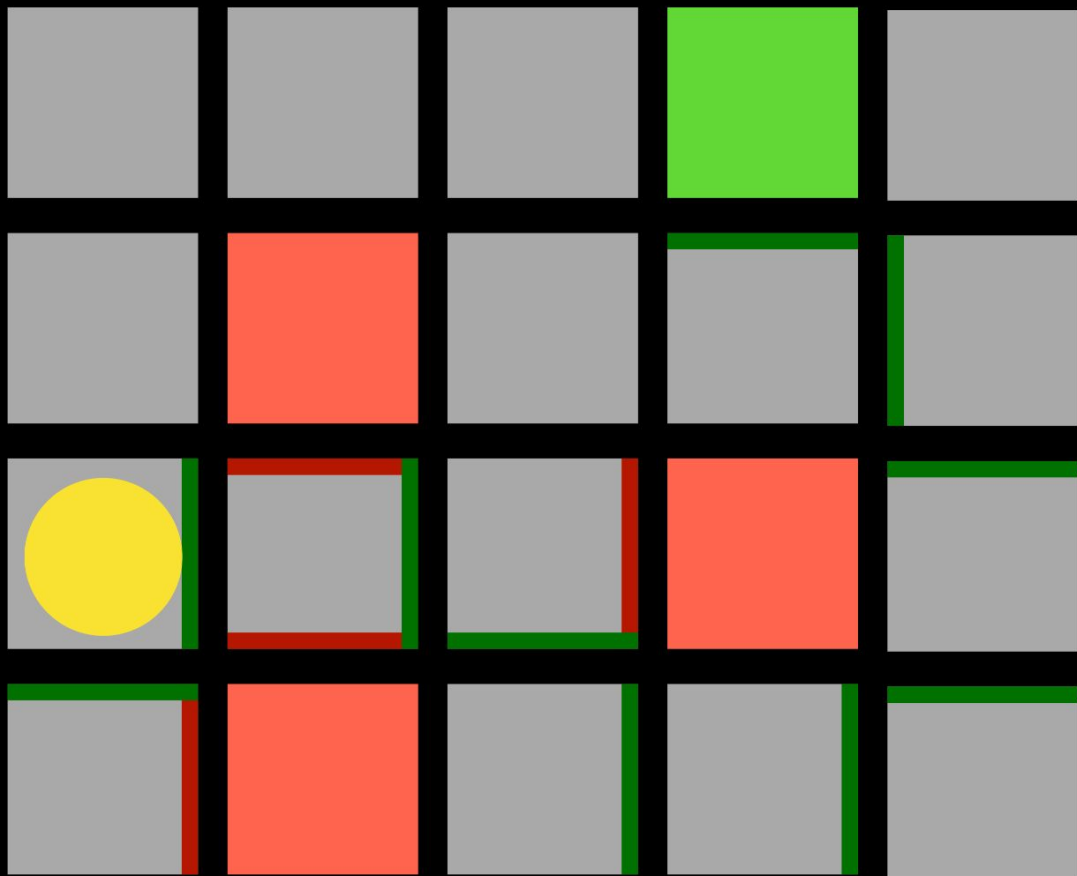


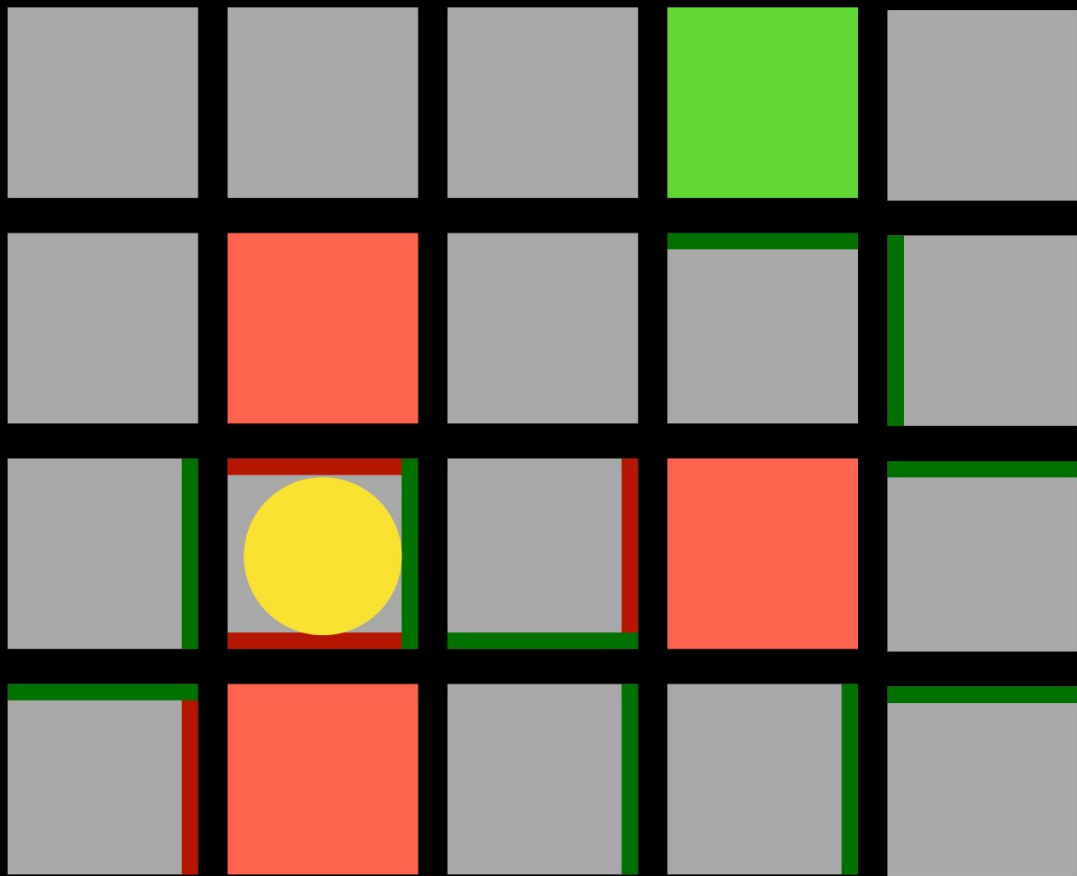


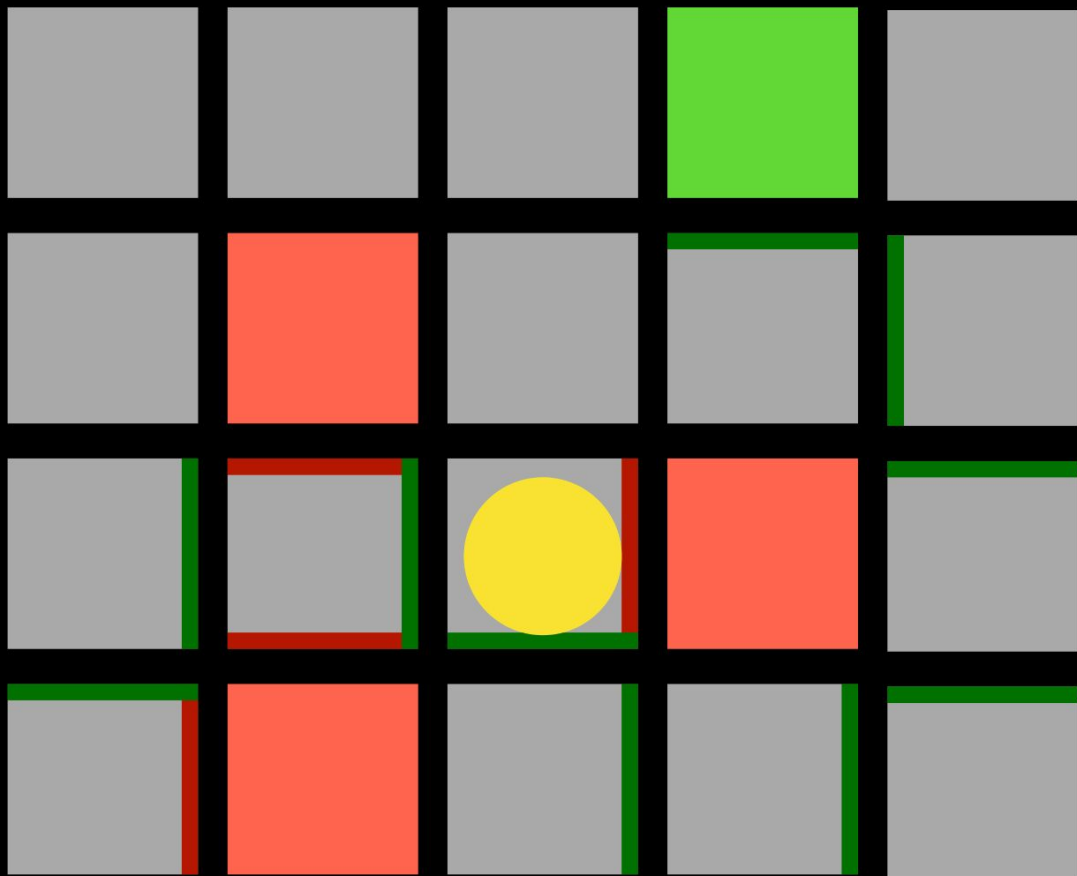


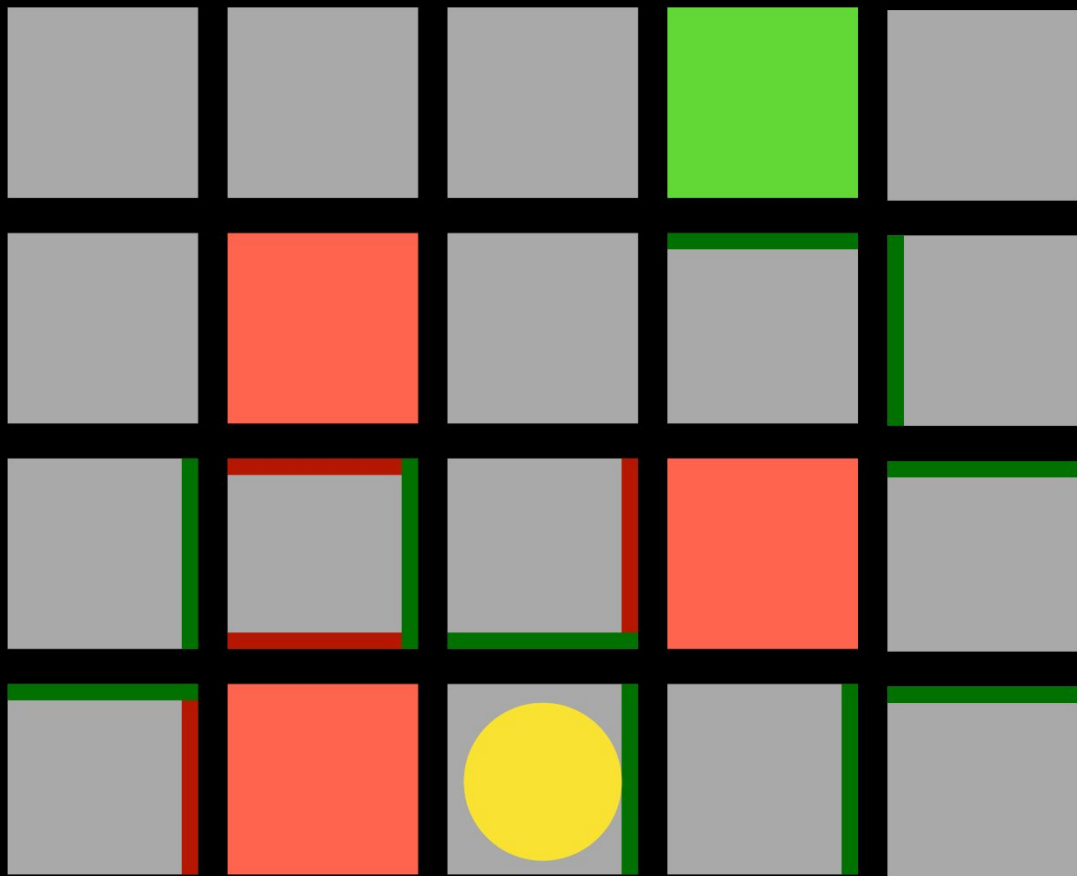


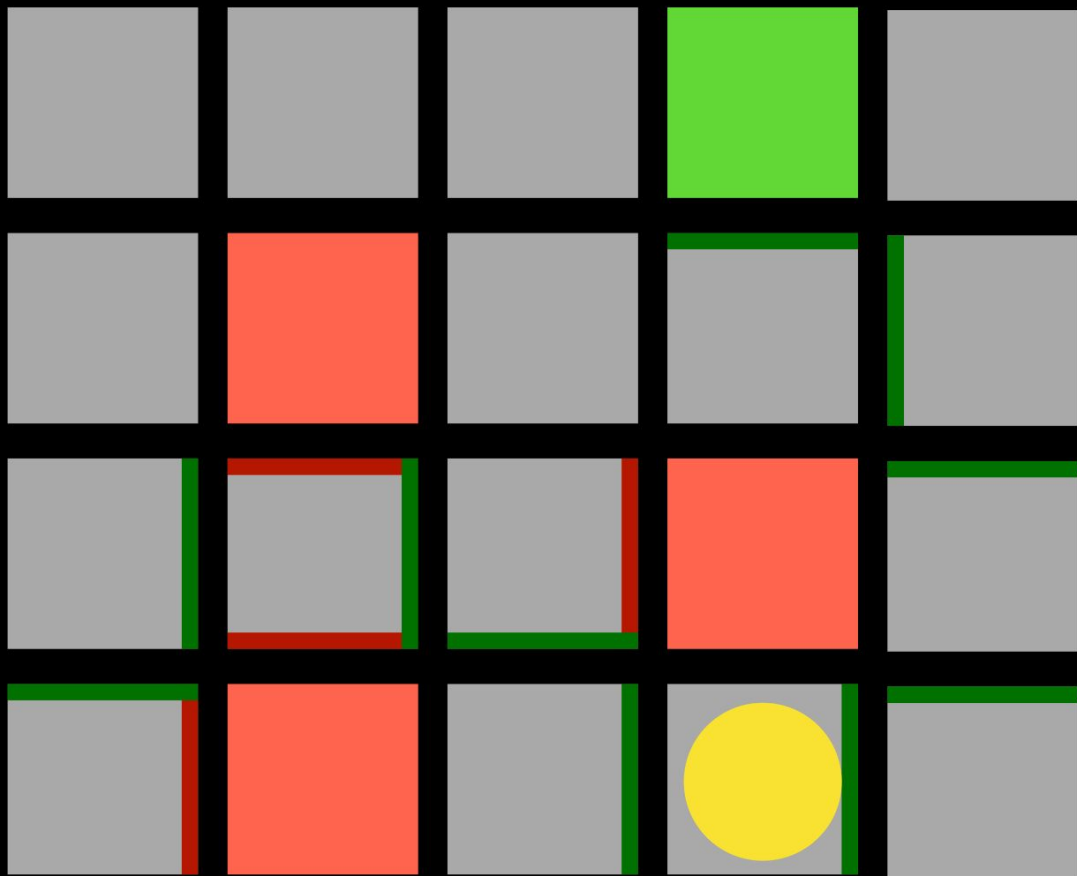


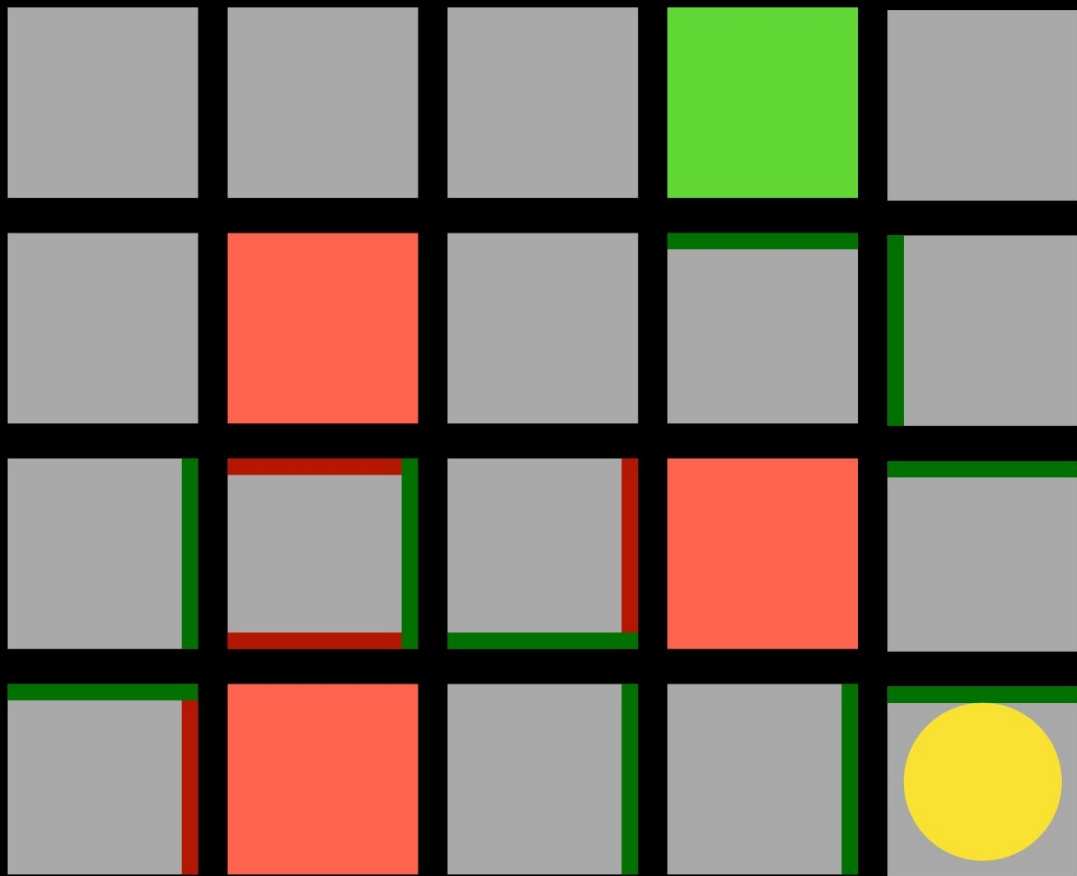


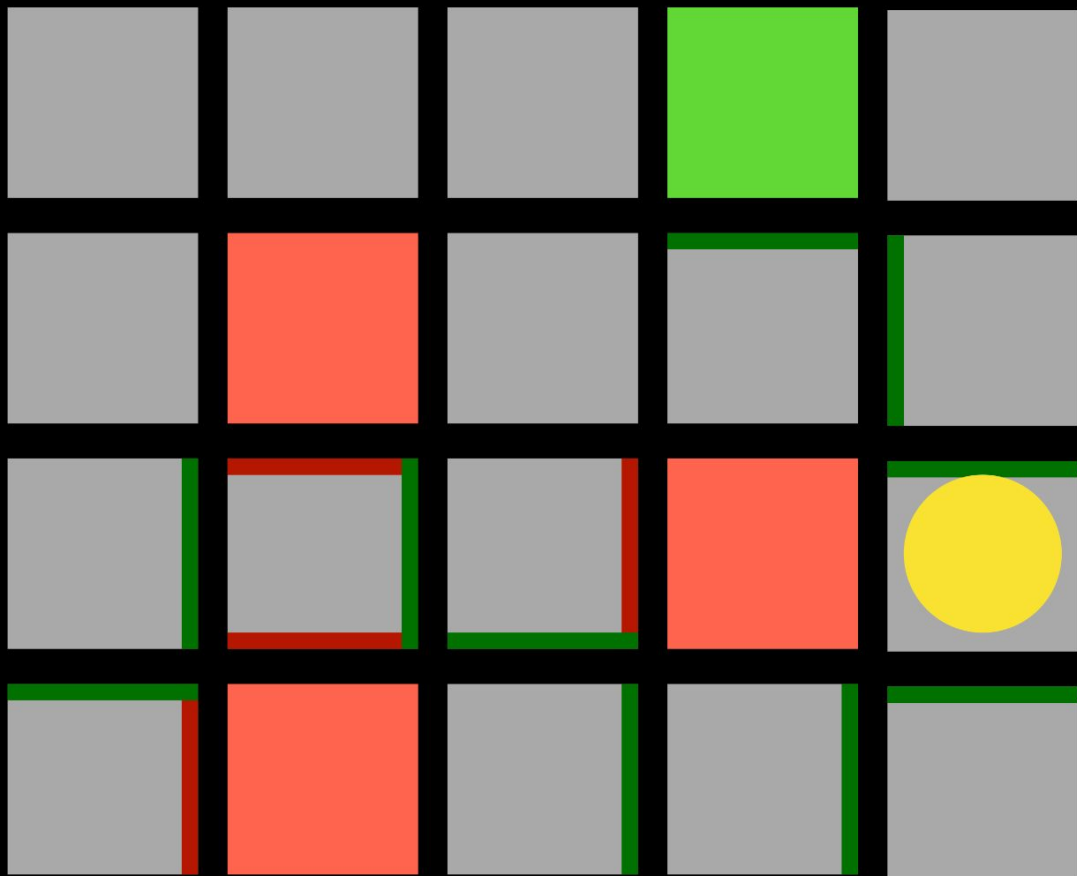


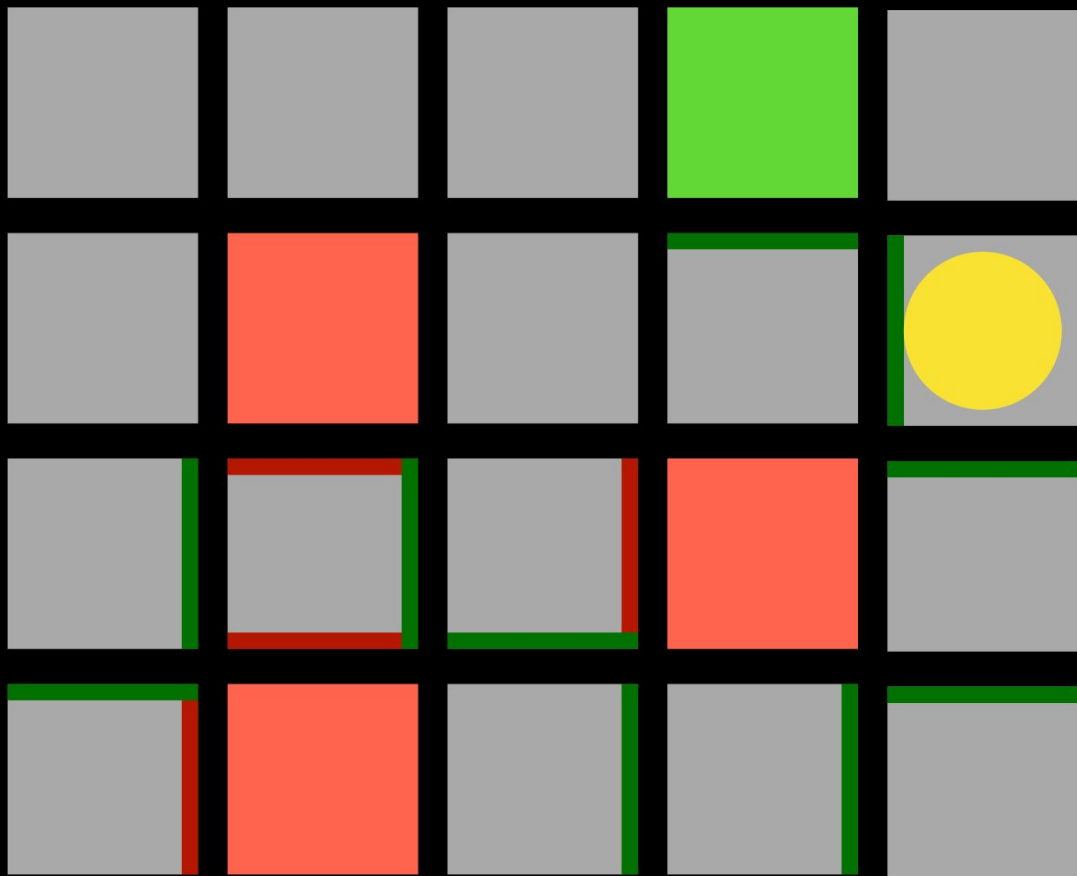


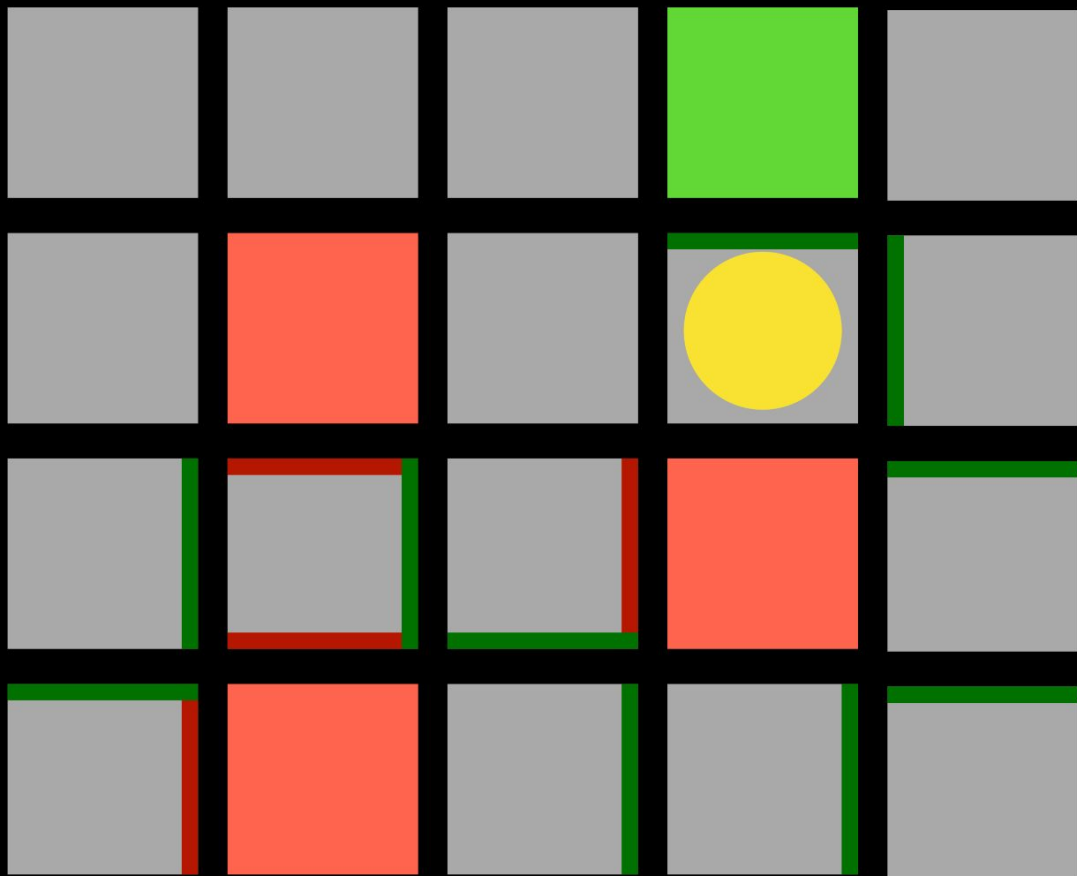


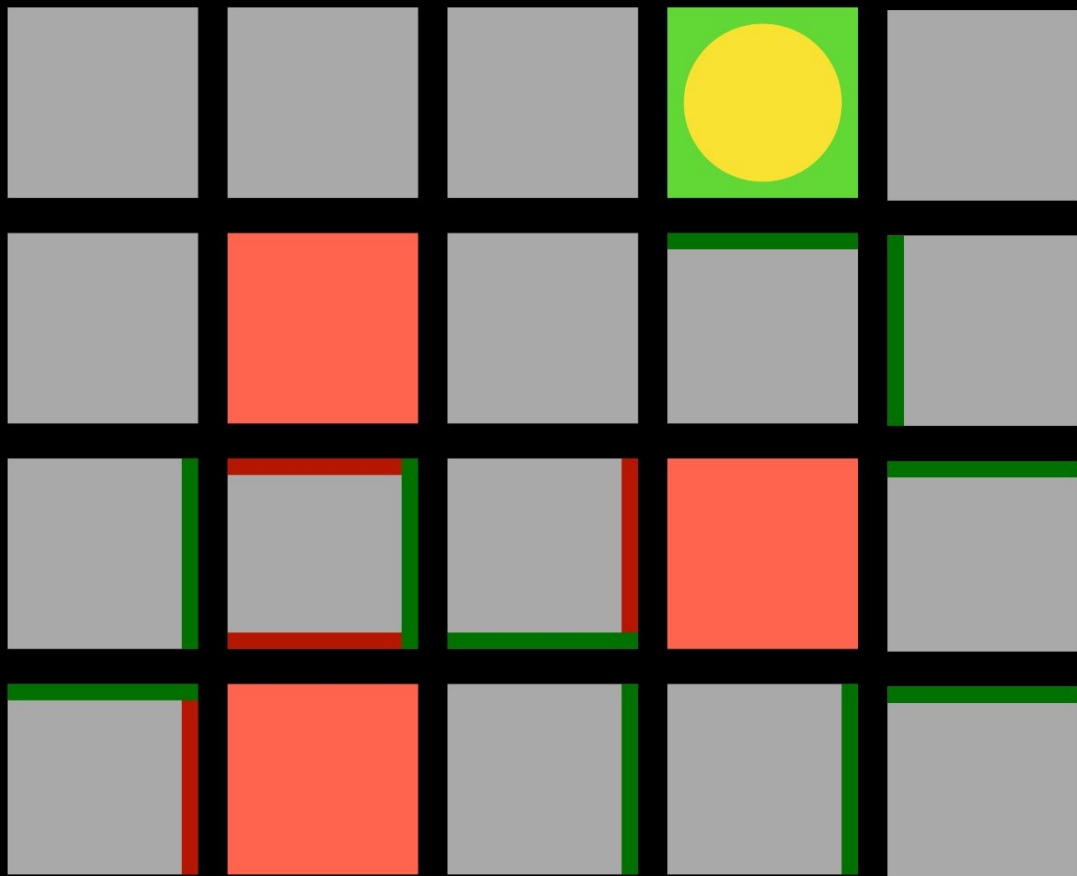












explore vs. exploit

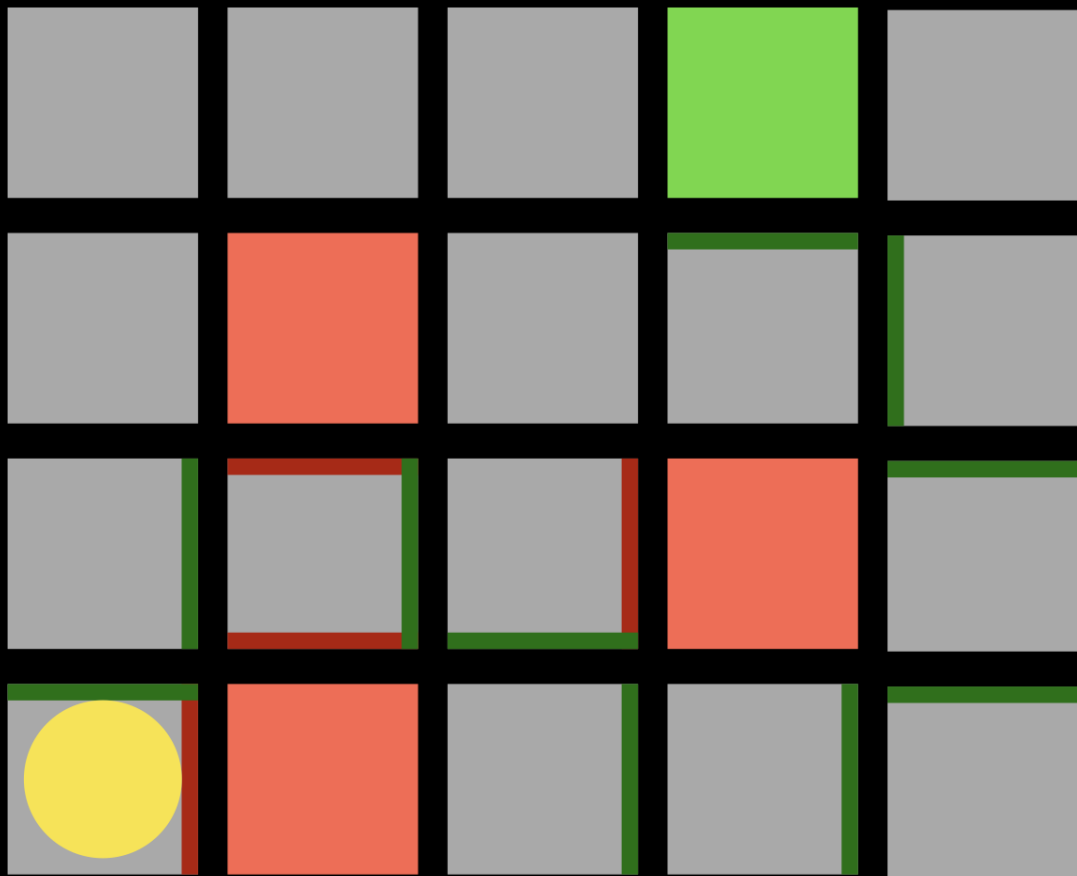
```
epsilon = 0.10
```

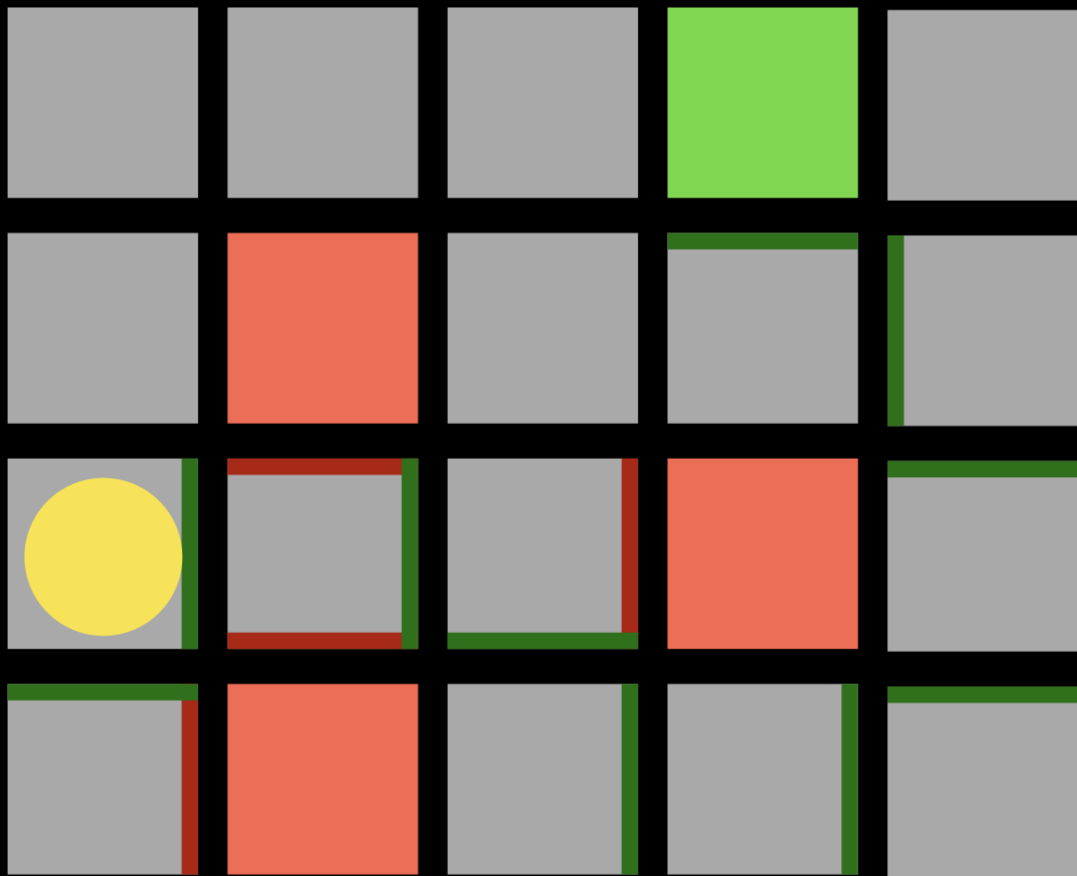
```
If random() < epsilon:
```

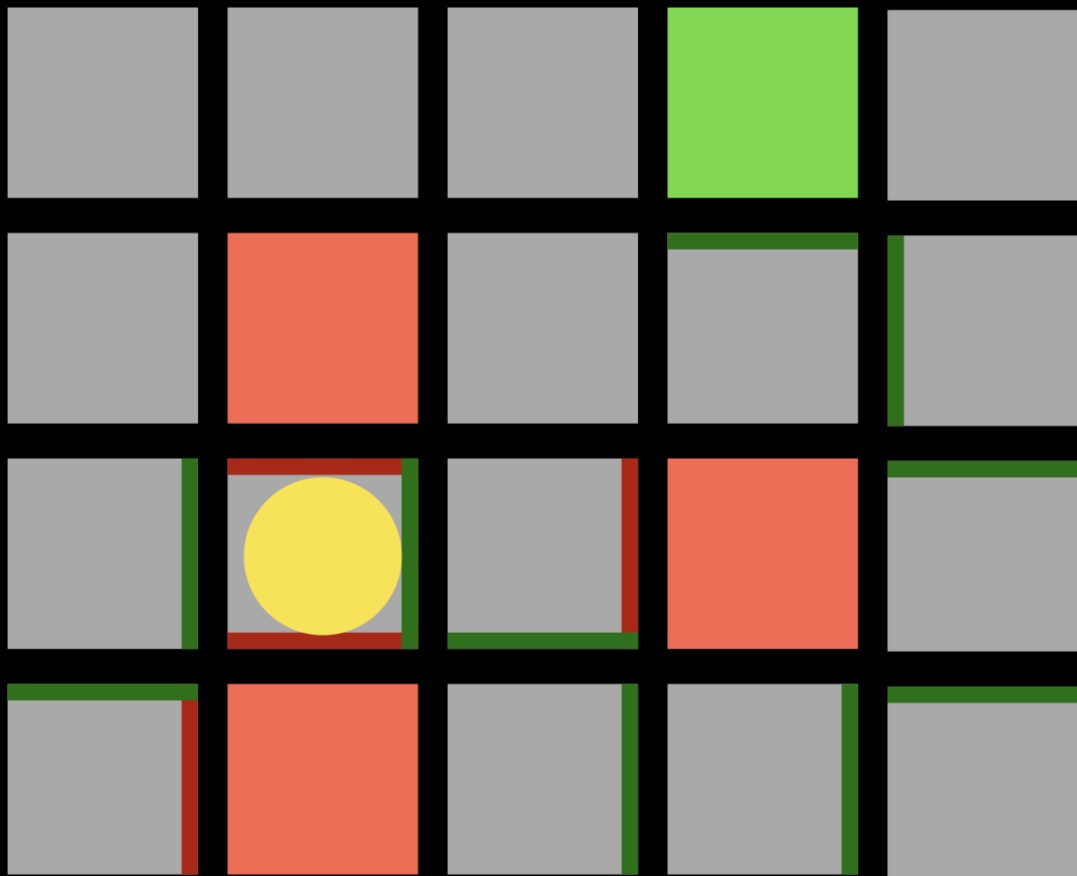
```
    Make a random move
```

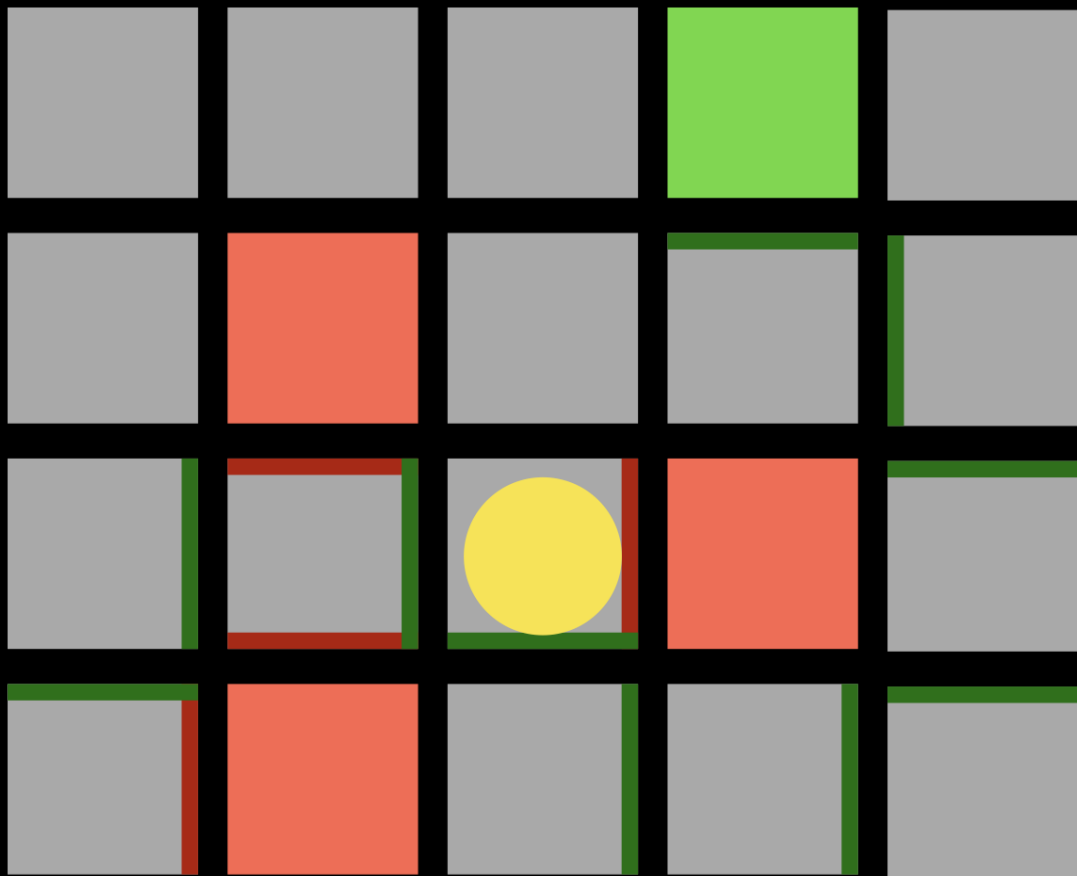
```
Else:
```

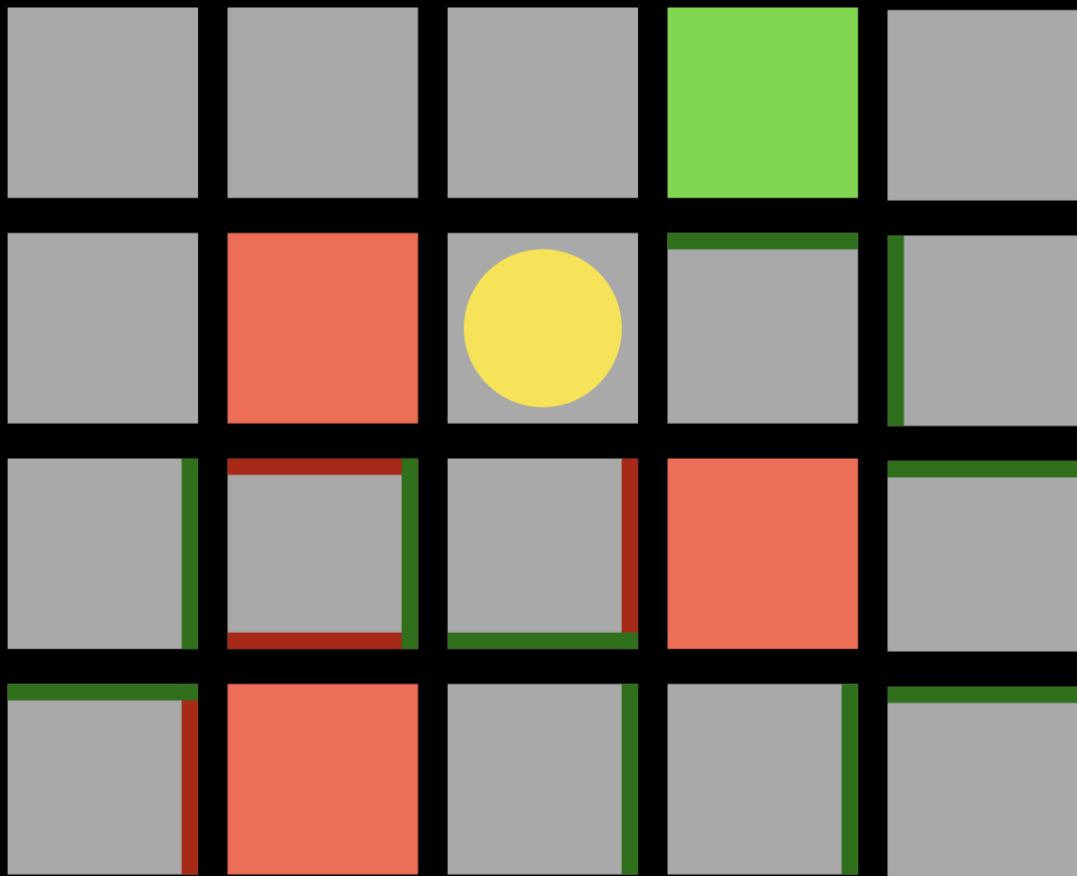
```
    Make the move with the highest value
```

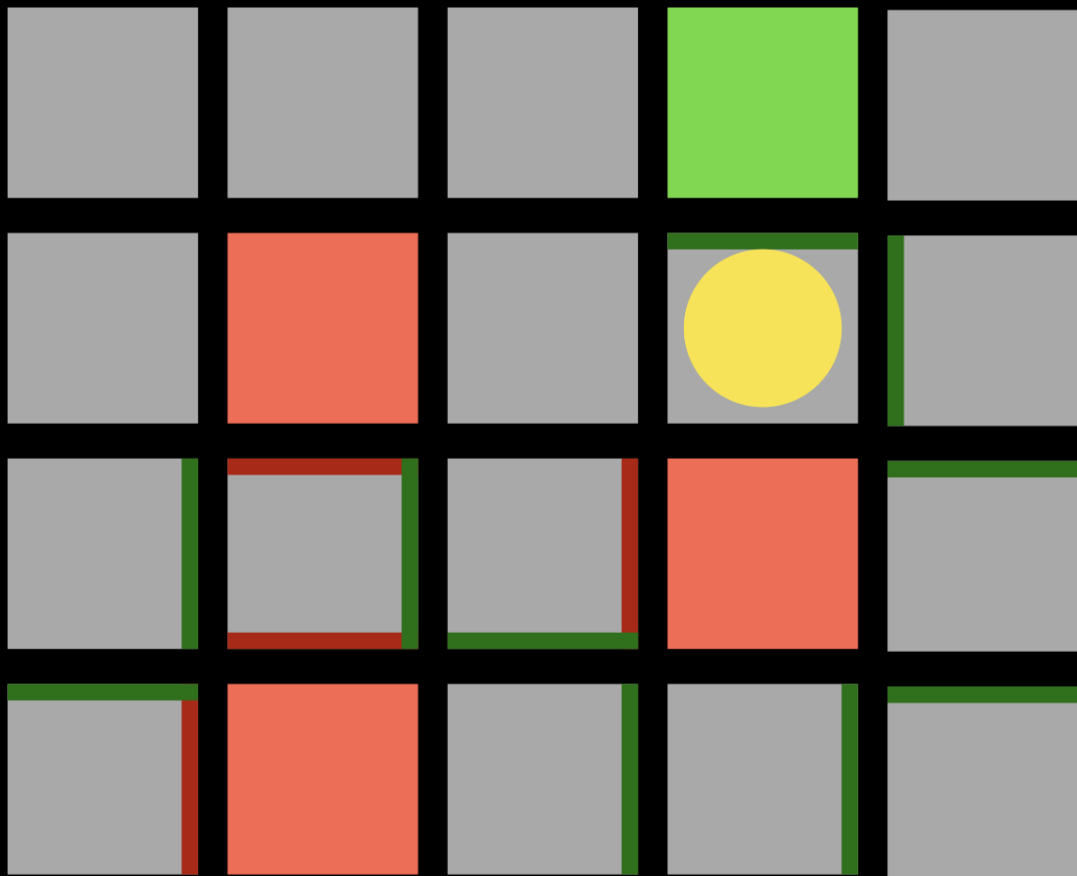


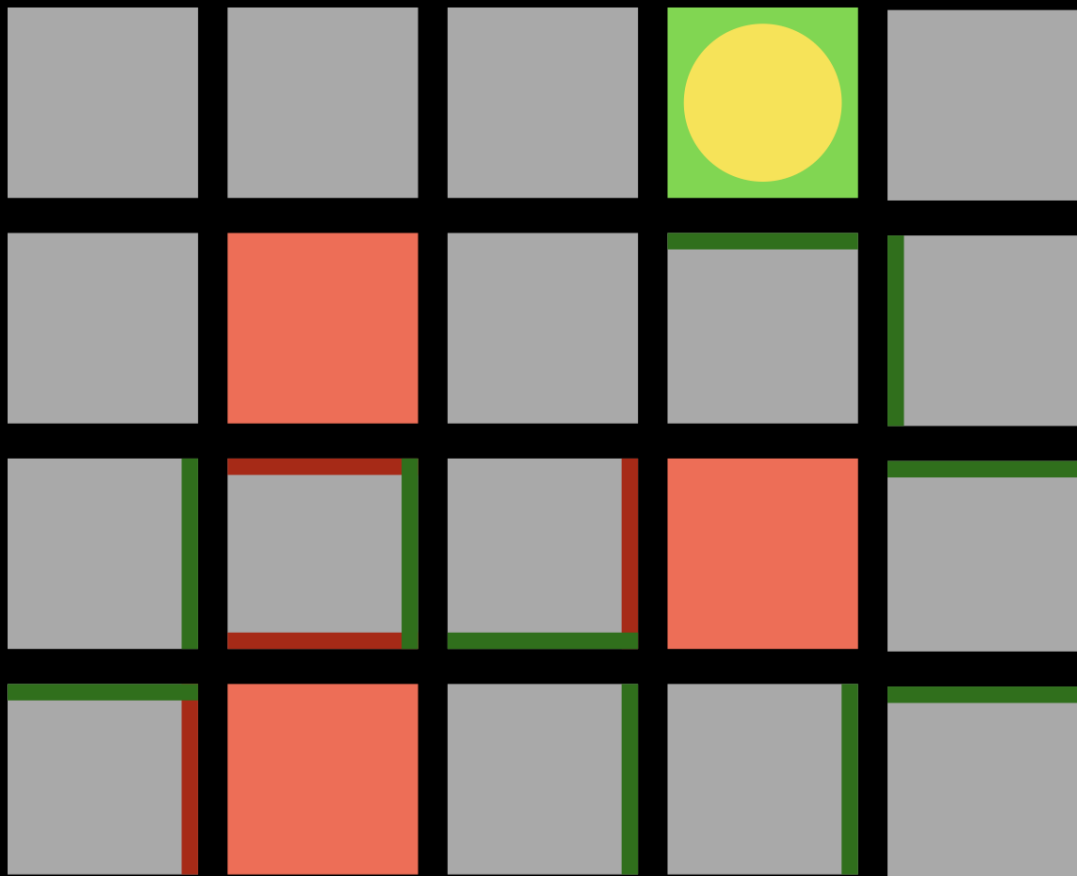








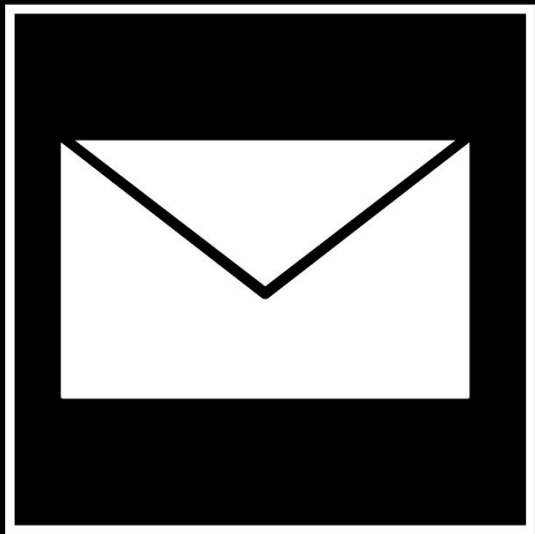




049 2 1



supervised learning

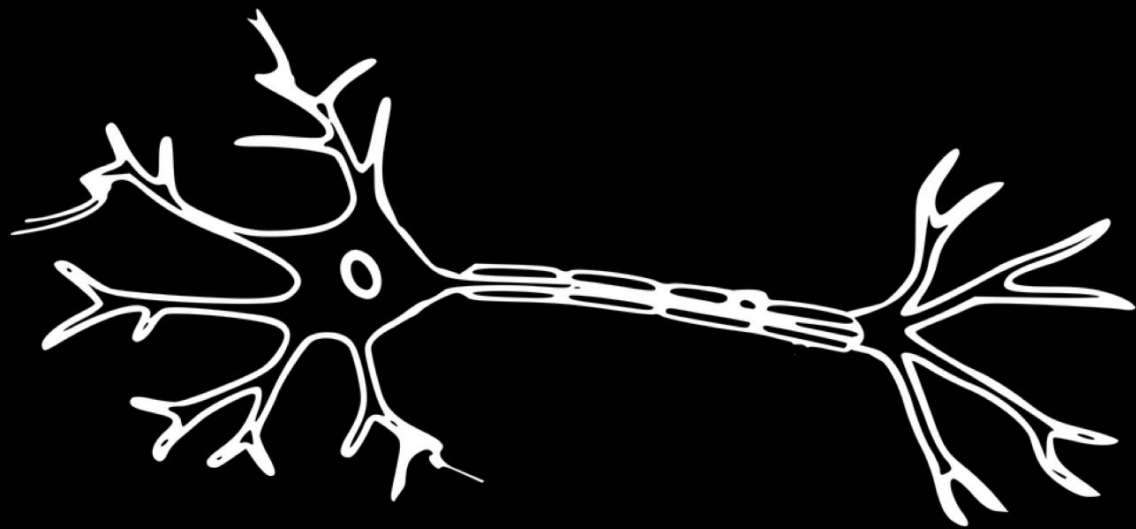


spam

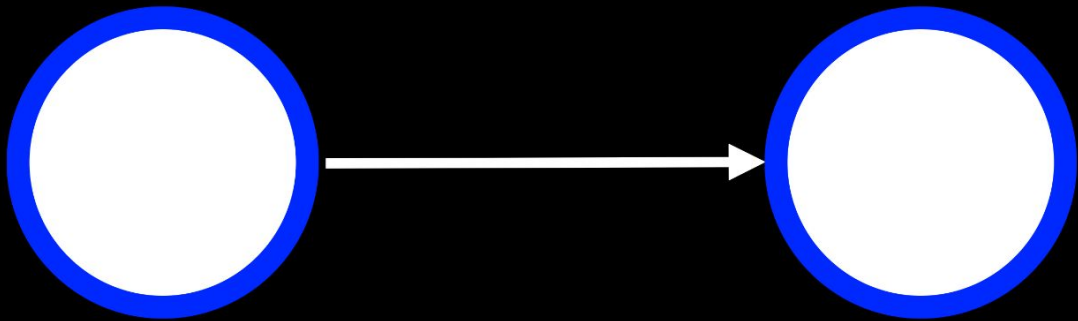
unsupervised learning

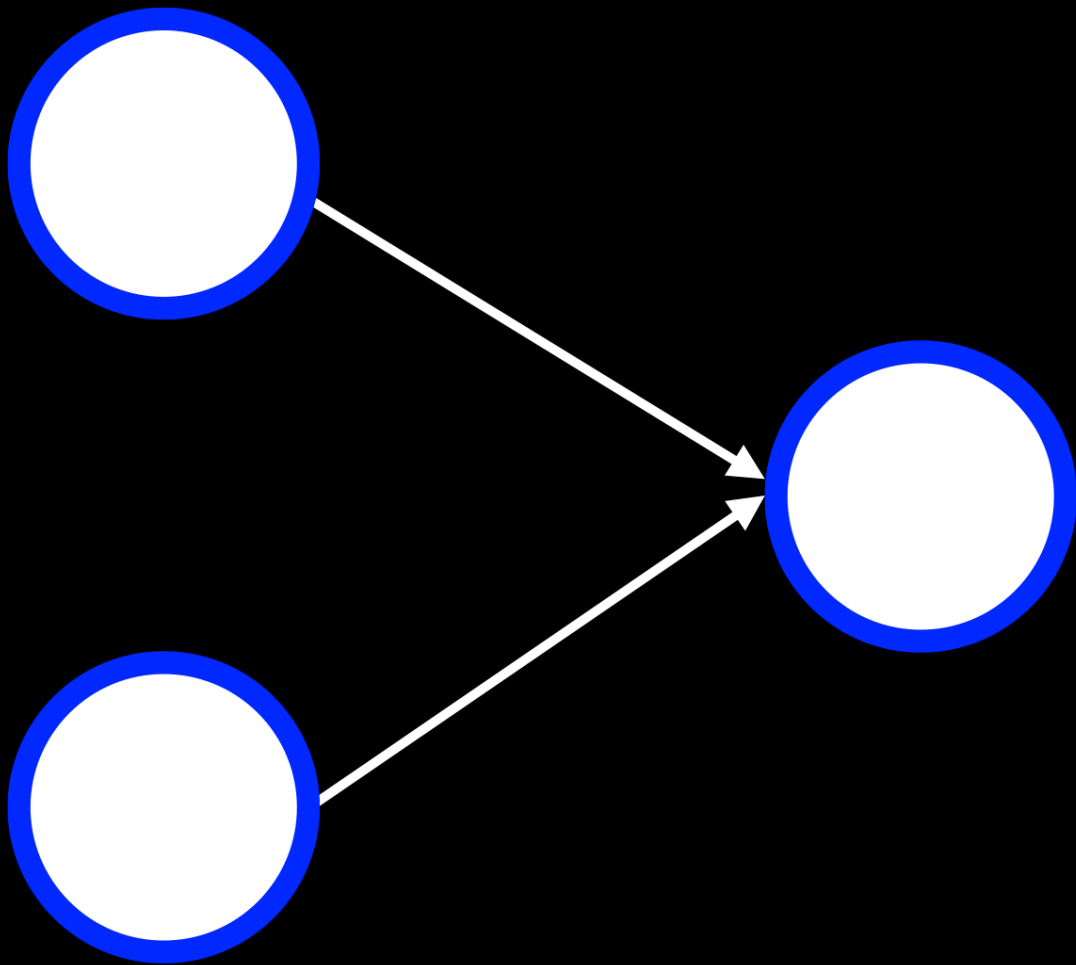
deep learning

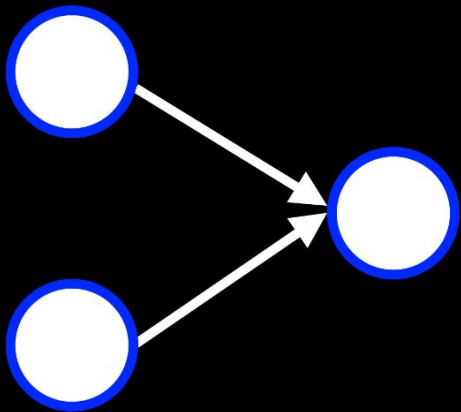
neural networks

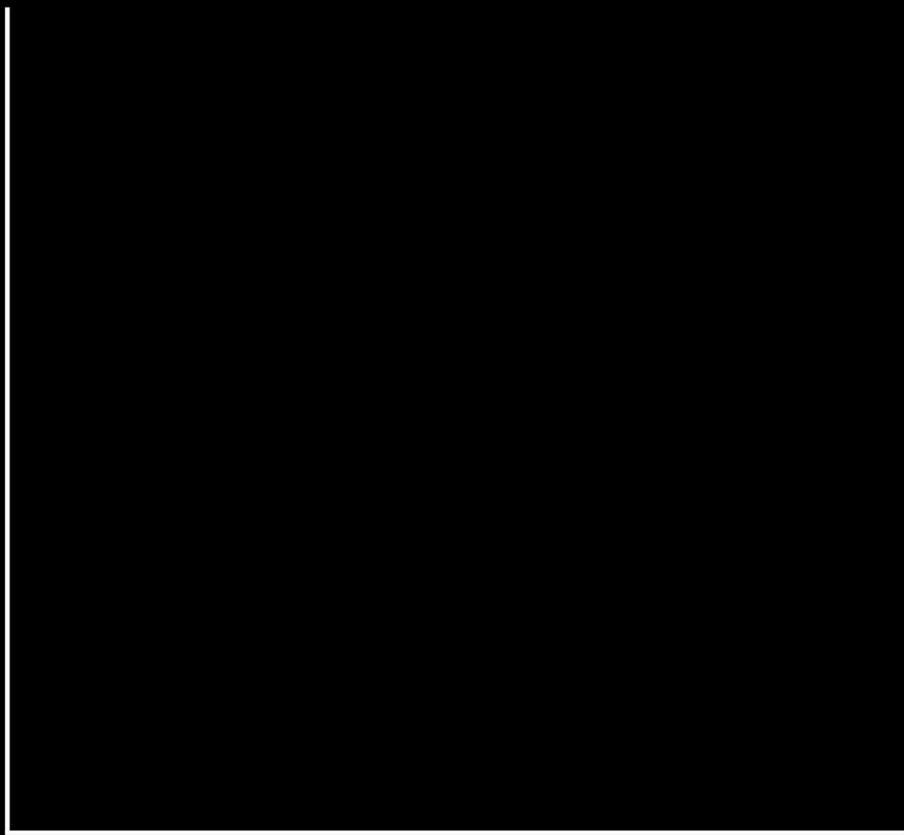
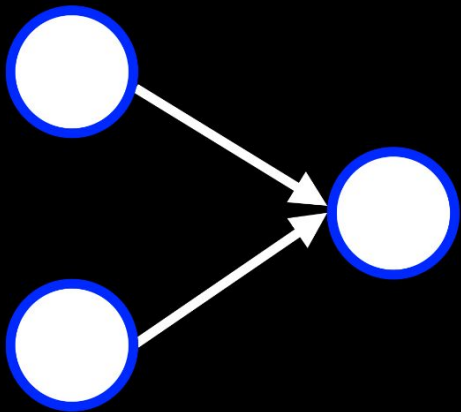


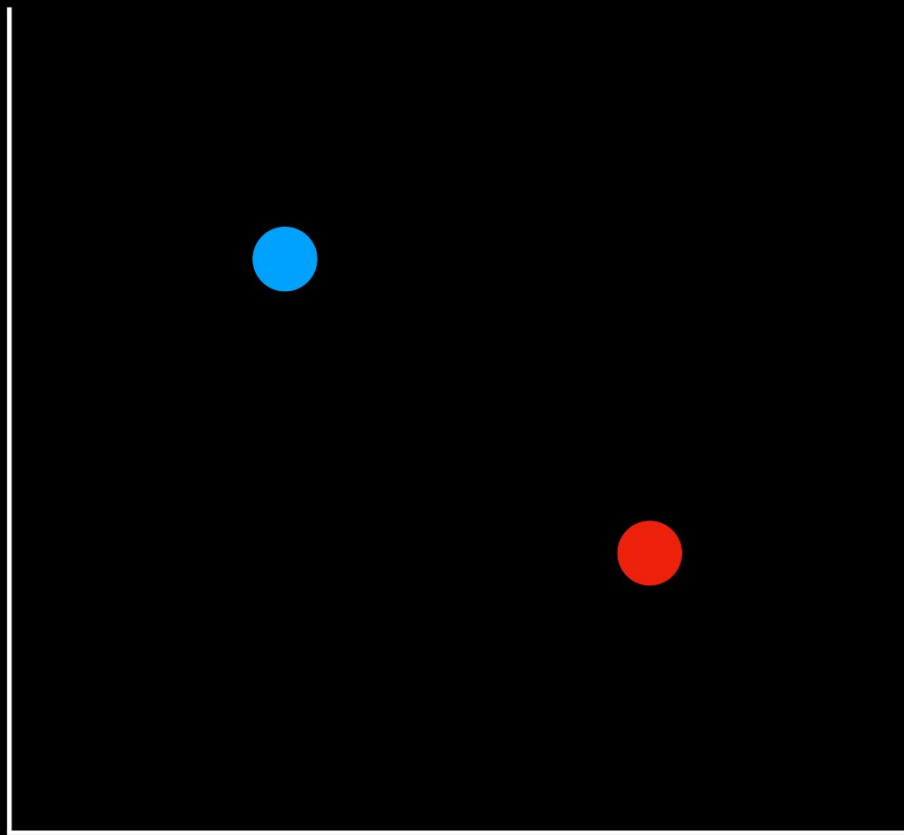
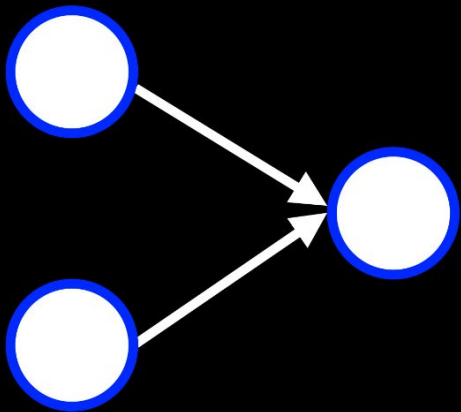


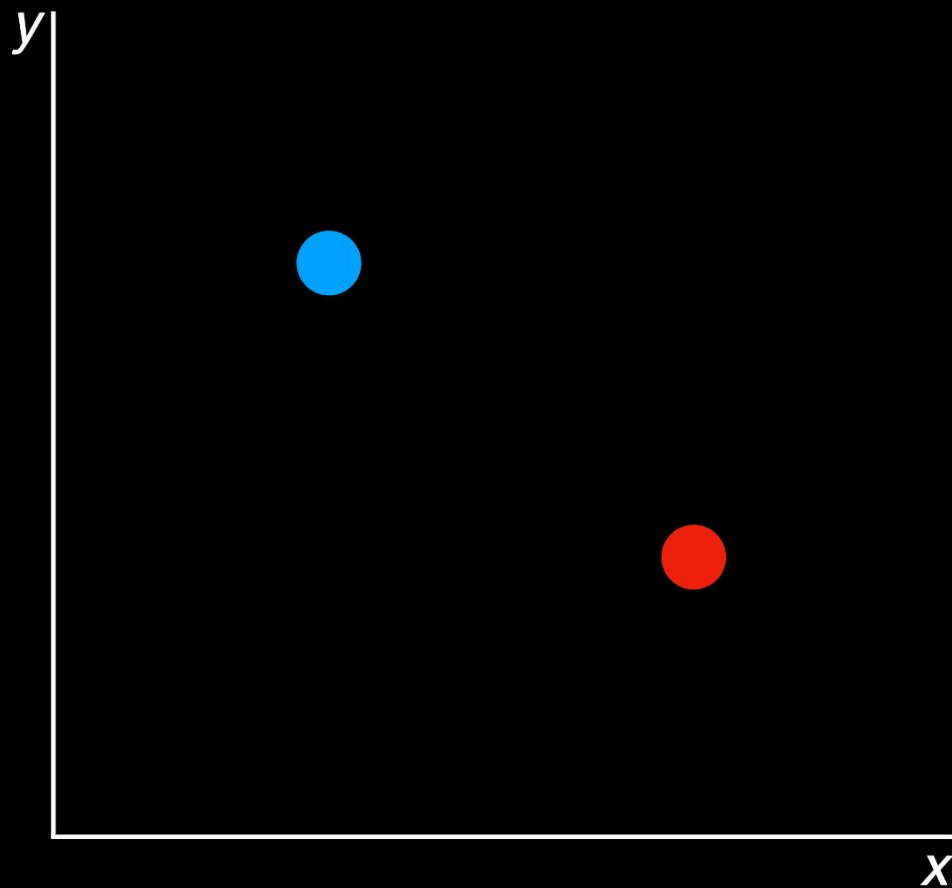
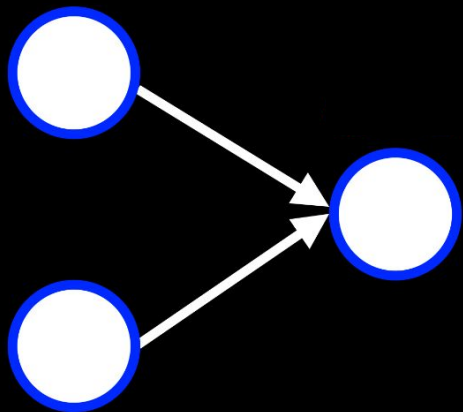


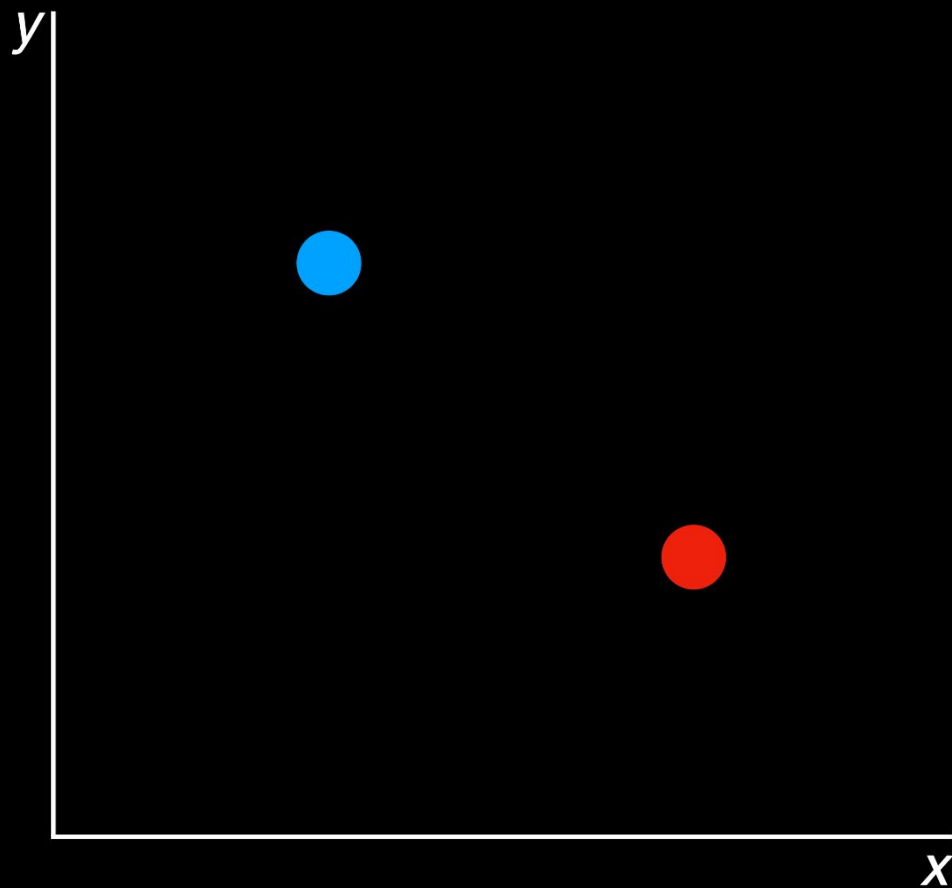
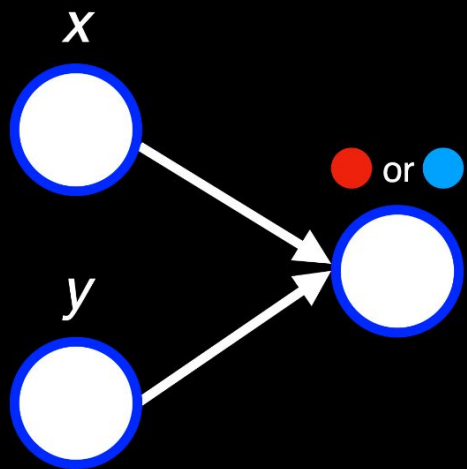


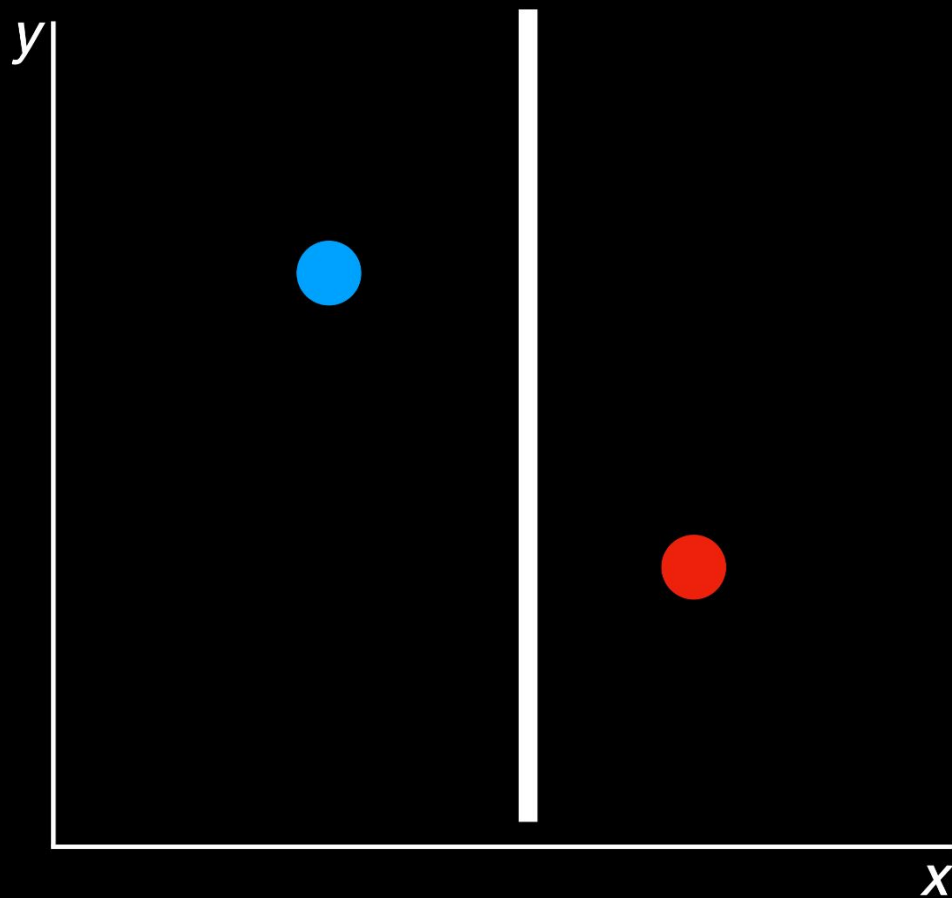
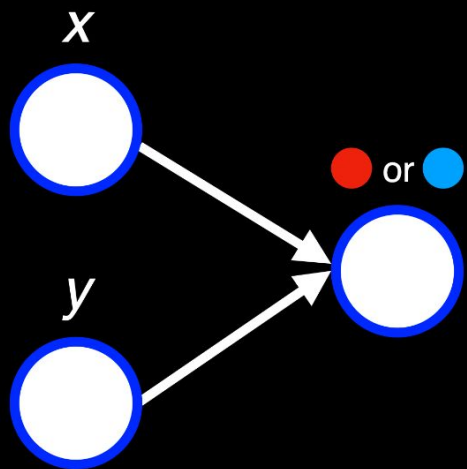


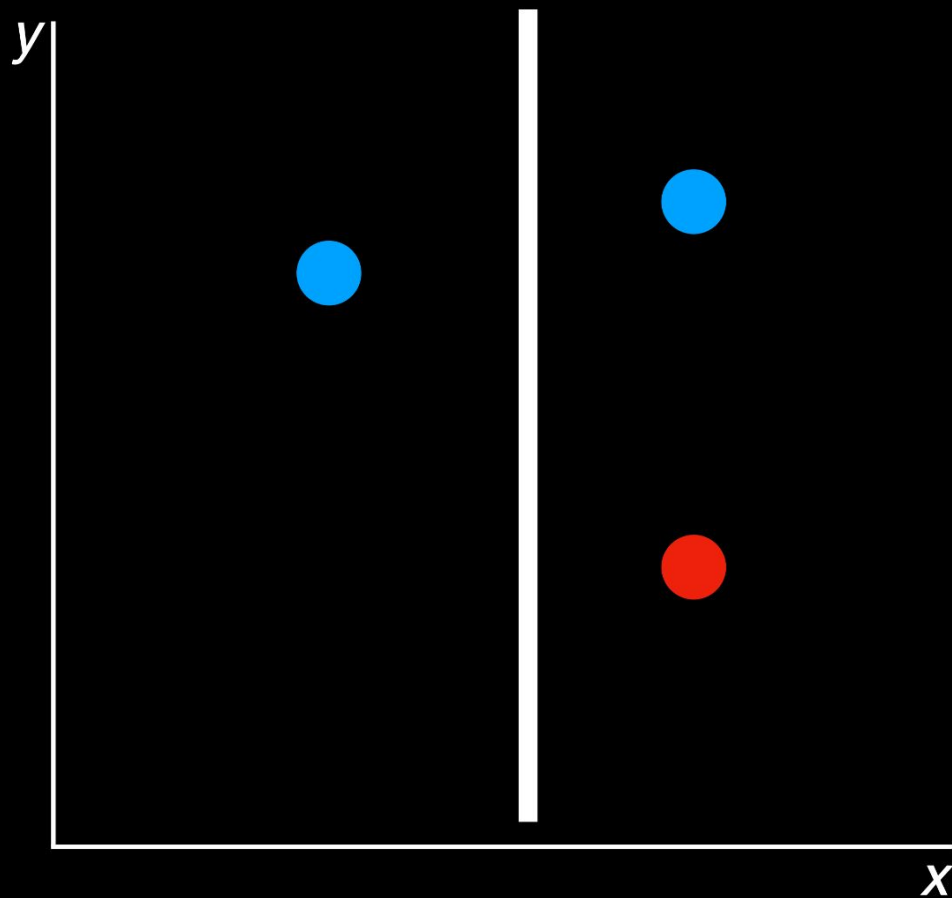
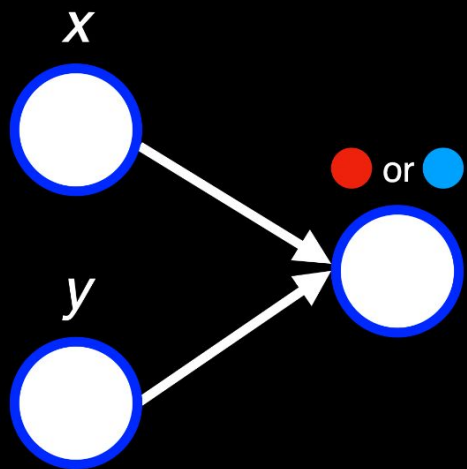


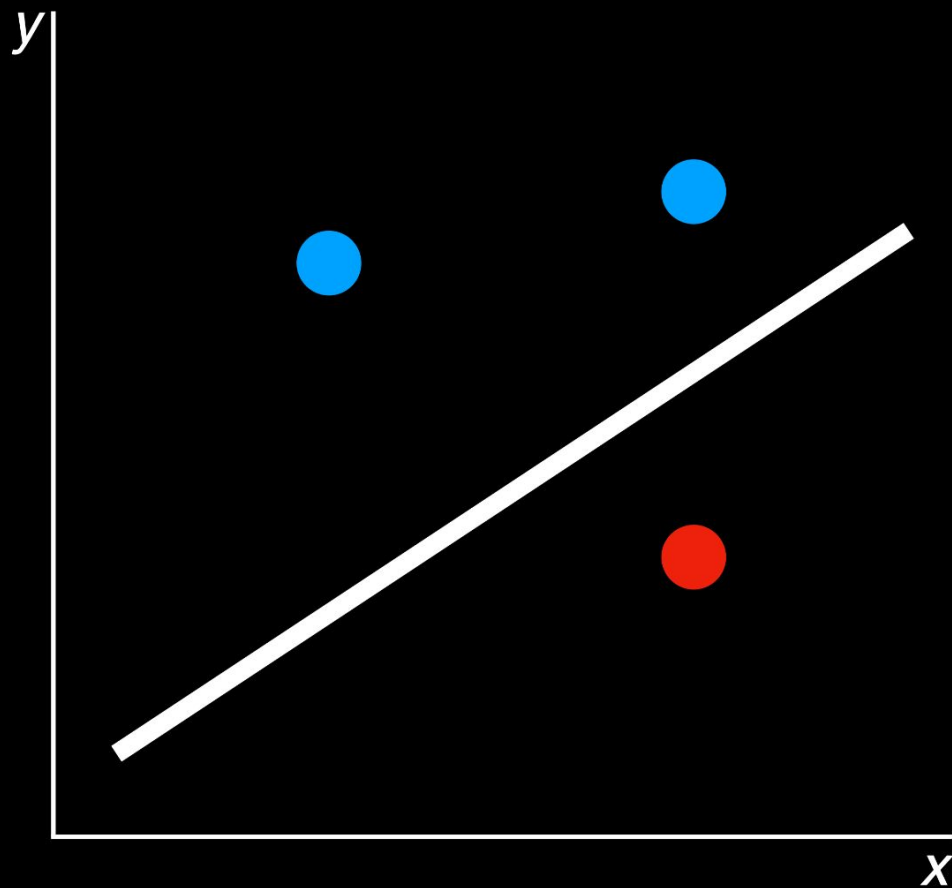
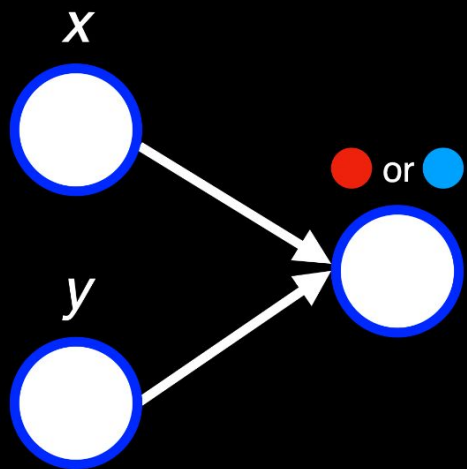


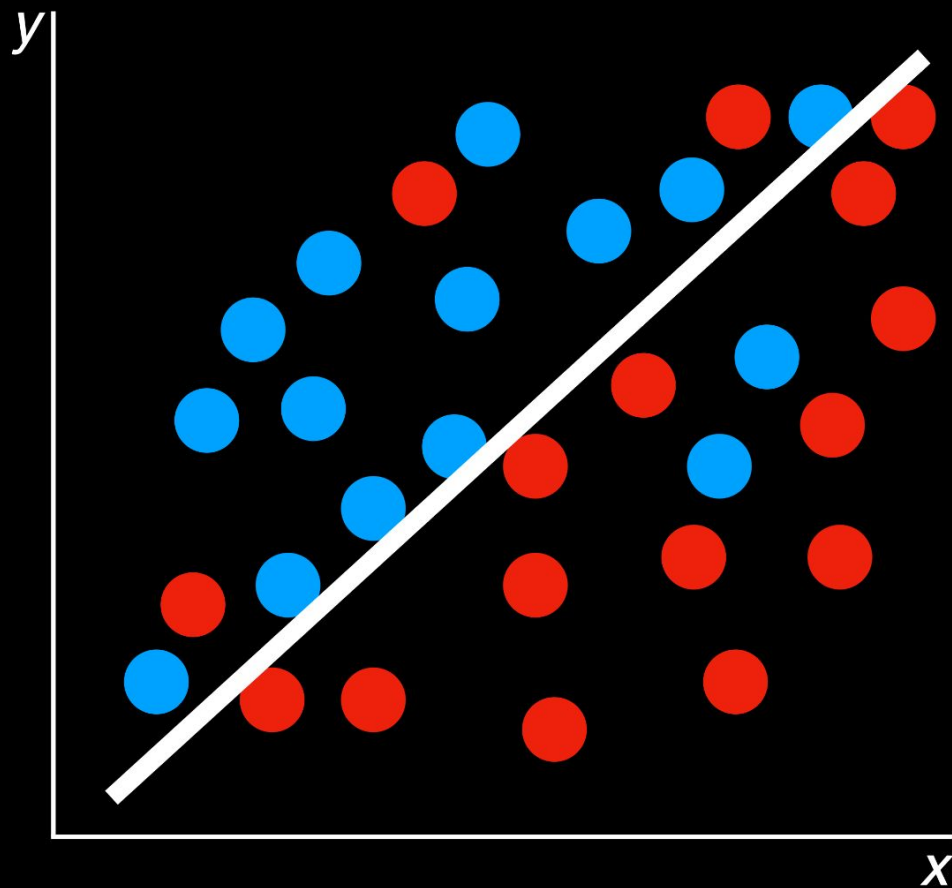
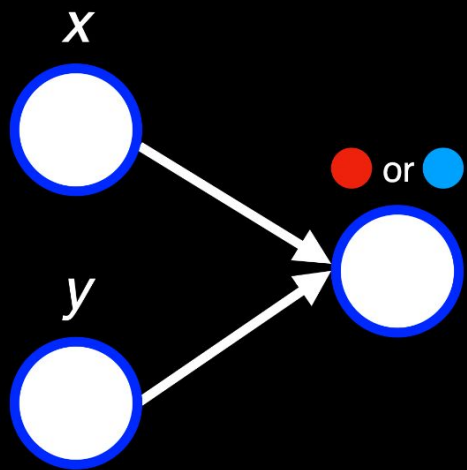


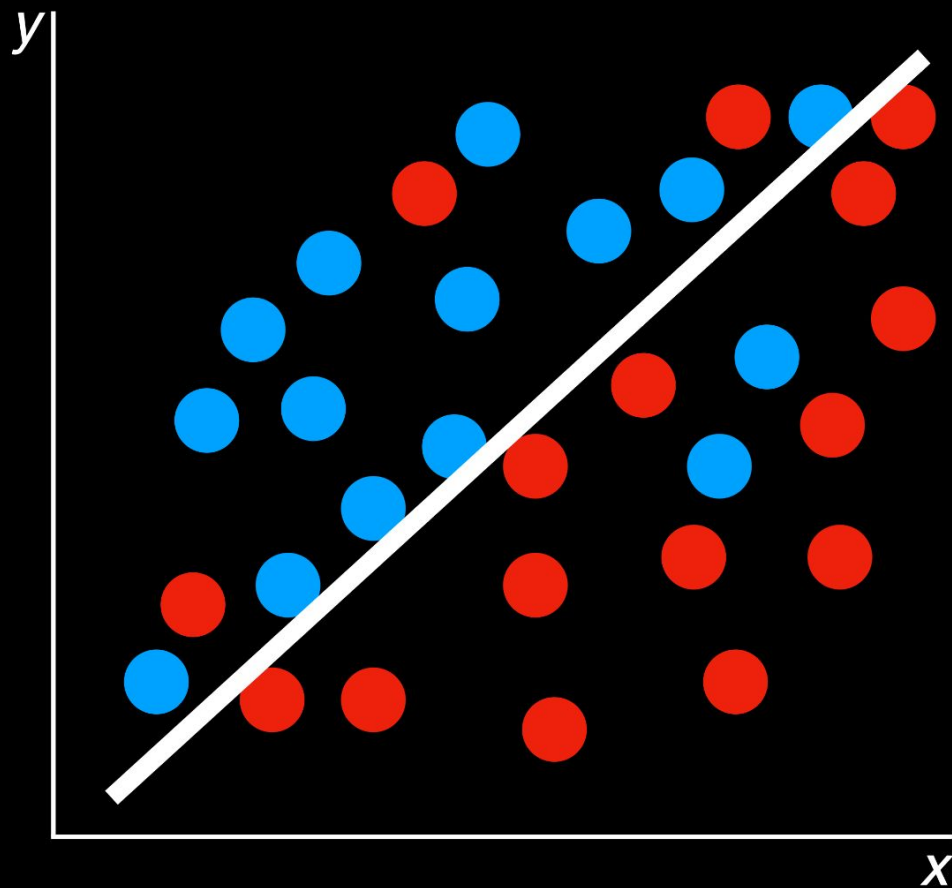
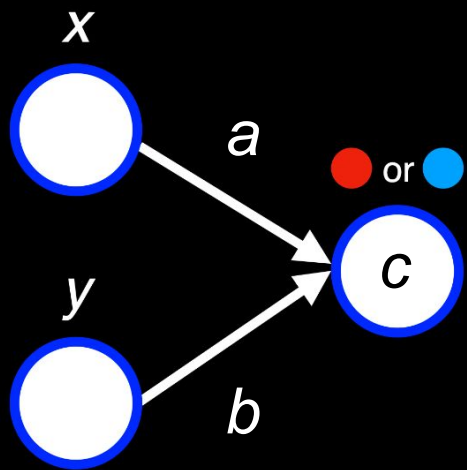




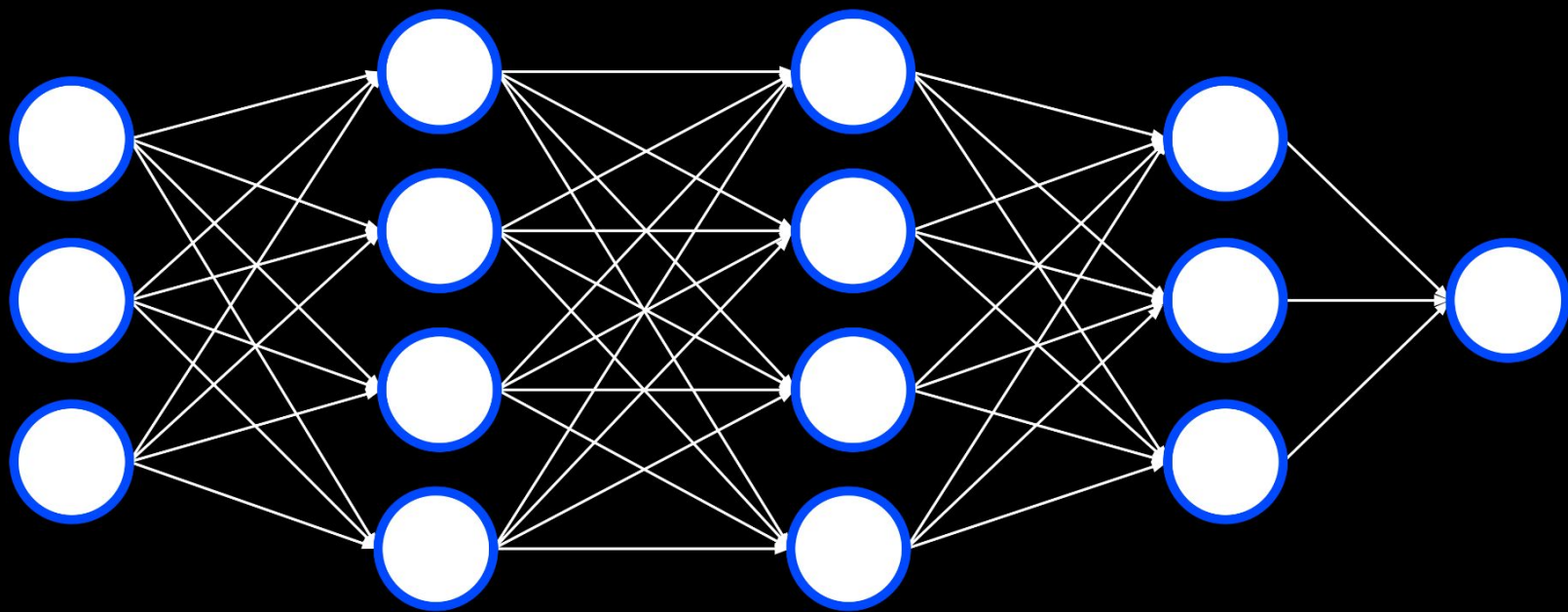


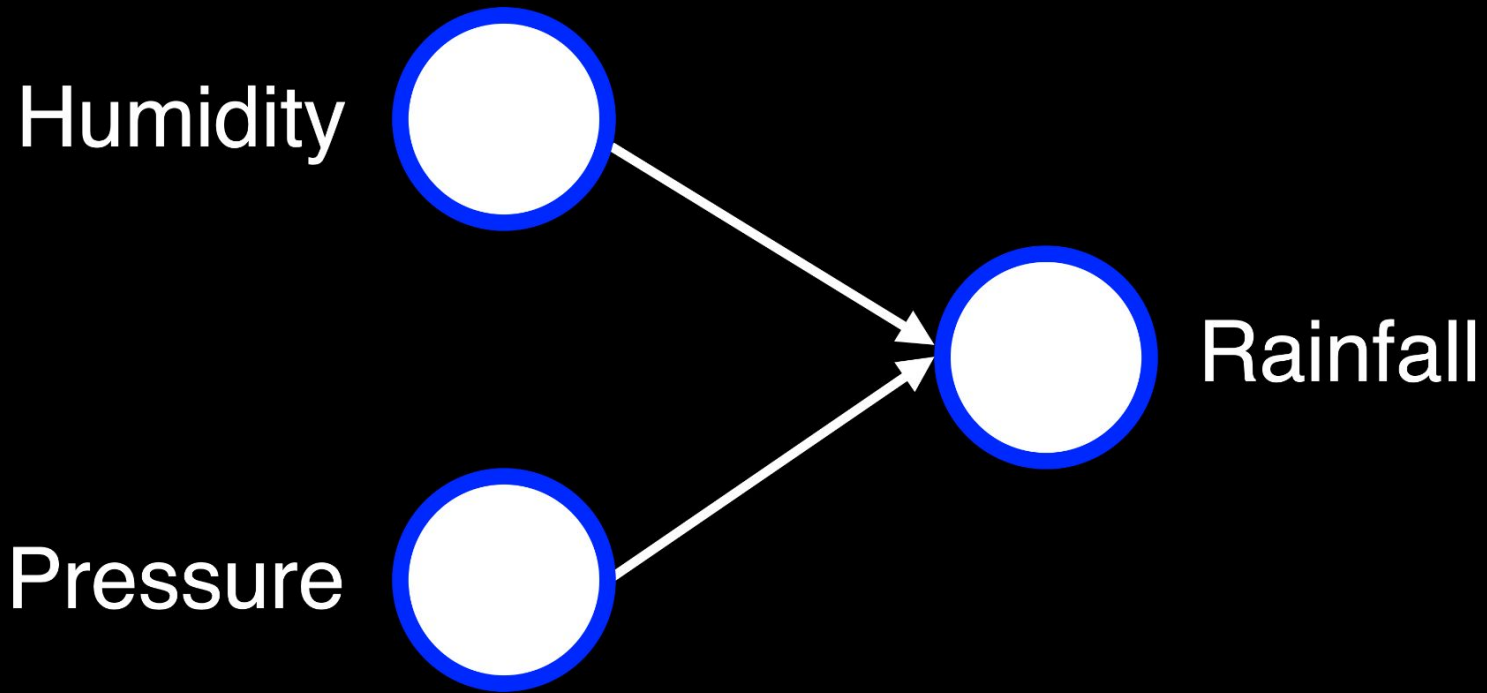


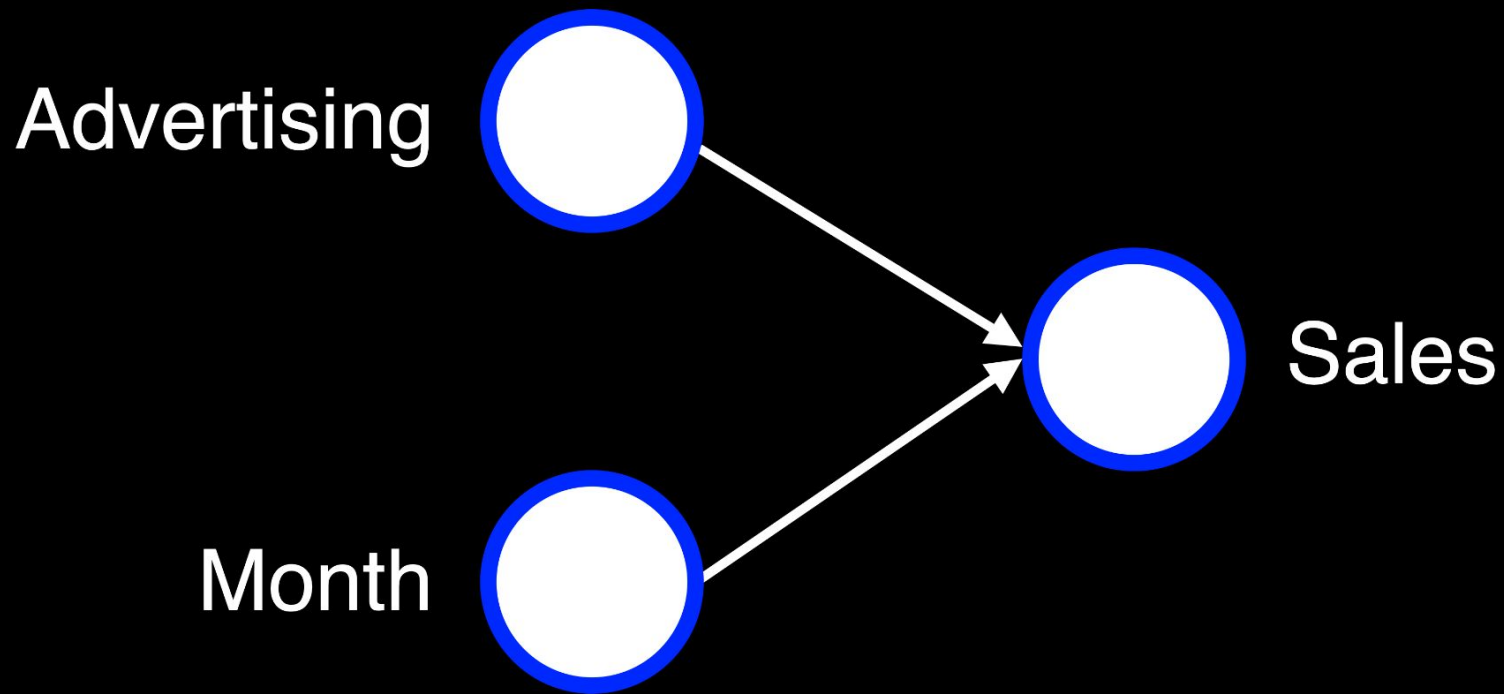




$$ax + by + c > 0$$







large language models



**Transformer Model
2017**



**GPT-3 Language Model
2020**



**ChatGPT Model
2022**

Generative Pre-trained Transformers

Generative Pre-trained Transformers

"Massachusetts is a state in the New England region of the Northeastern United States. It borders on the Atlantic Ocean to the east. The state's capital is..."

"Massachusetts is a state in the New England region of the Northeastern United States. It borders on the Atlantic Ocean to the east. The state's capital is..."

Massachusetts

is

a

state

...

the

state's

capital

is

Massachusetts ①

is ②

a ③

state ④

...

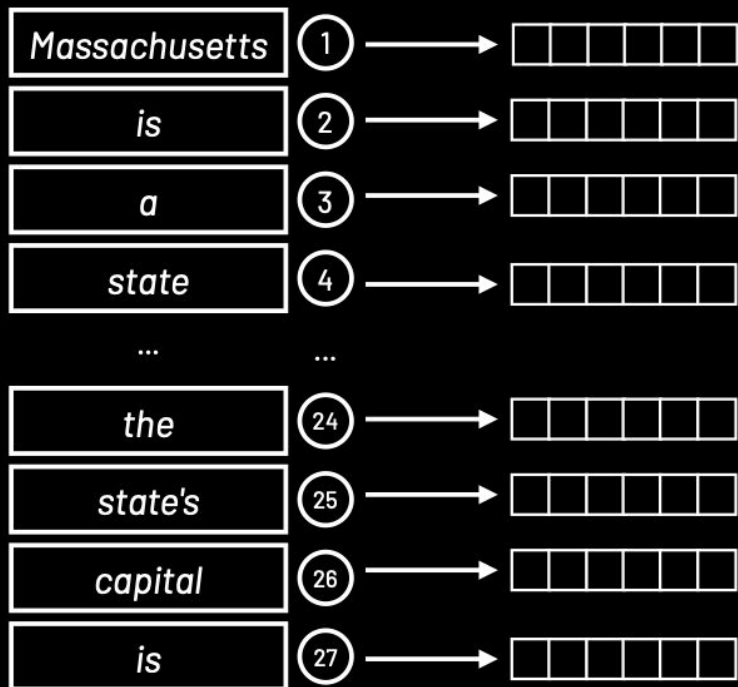
...

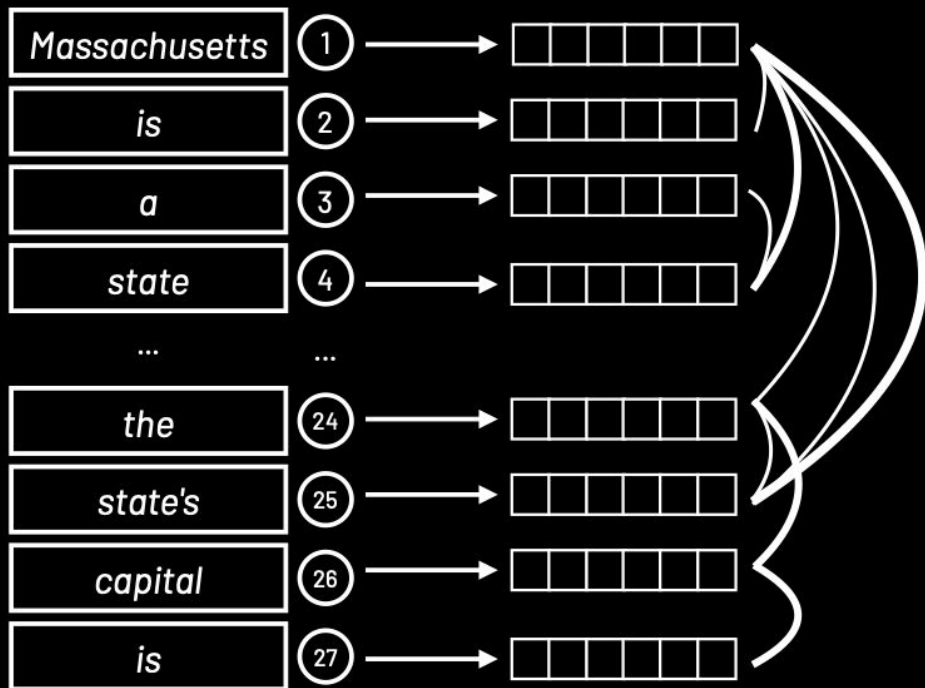
the ②④

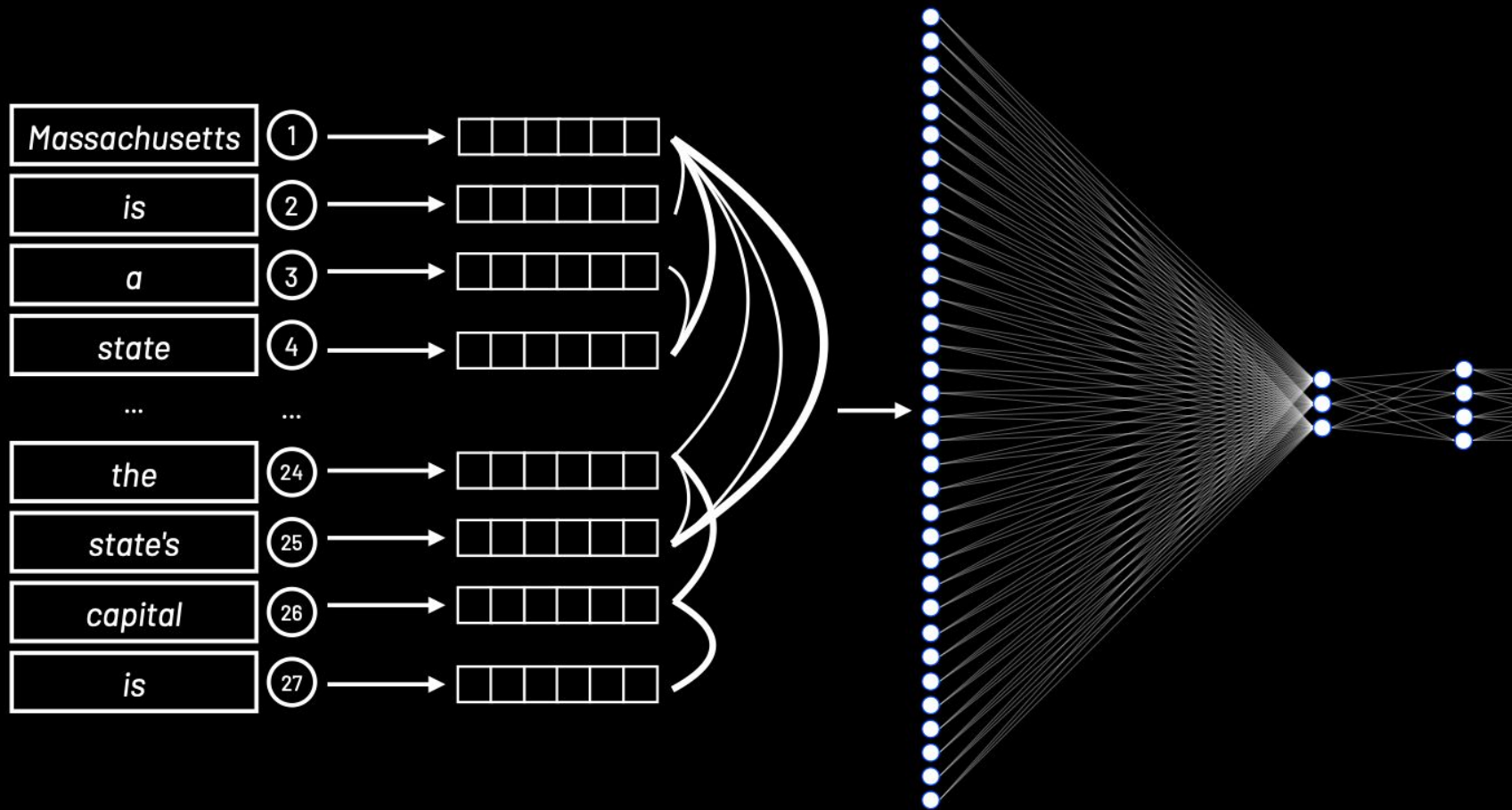
state's ②⑤

capital ②⑥

is ②⑦





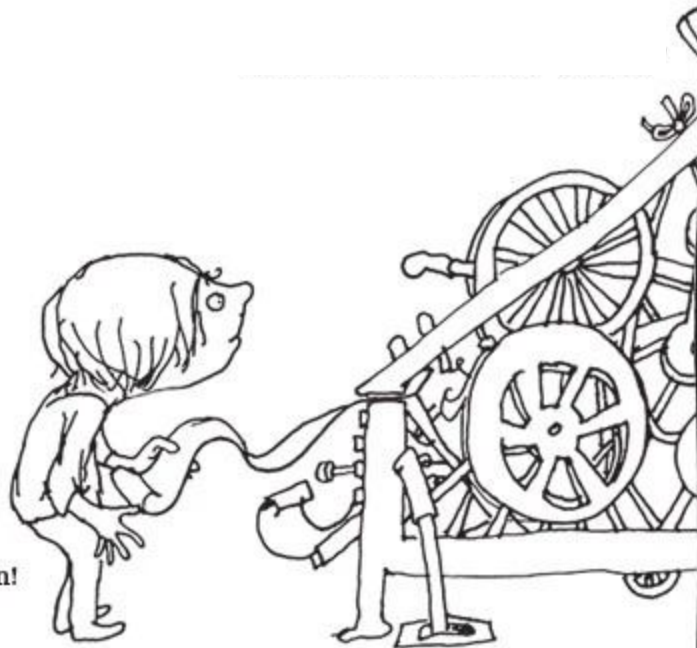




hallucinations

HOMEWORK MACHINE

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poems in
*A Light in
the Attic*
by Shel Silverstein!

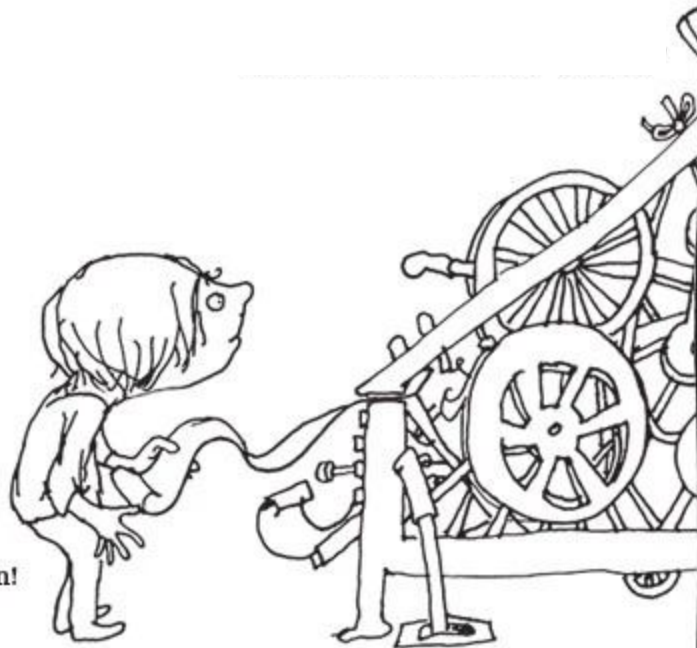


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HOMWORK MACHINE

The Homework Machine, oh the Homework Machine,

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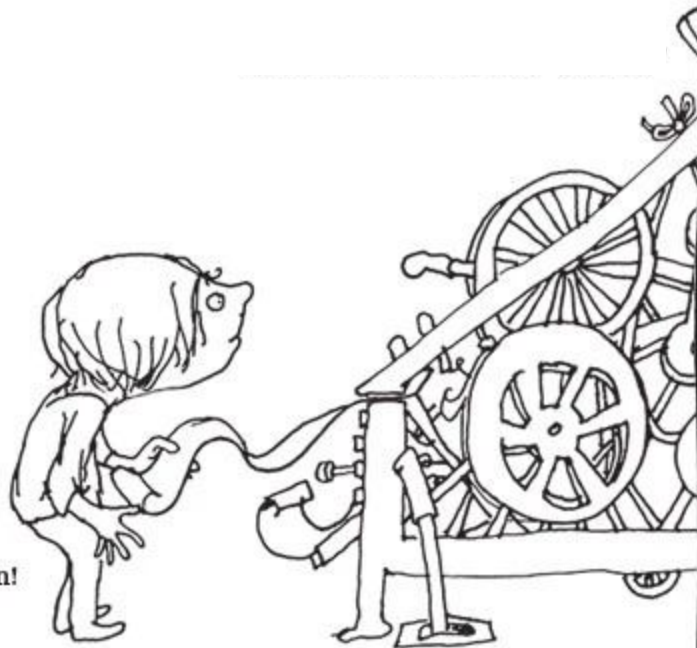


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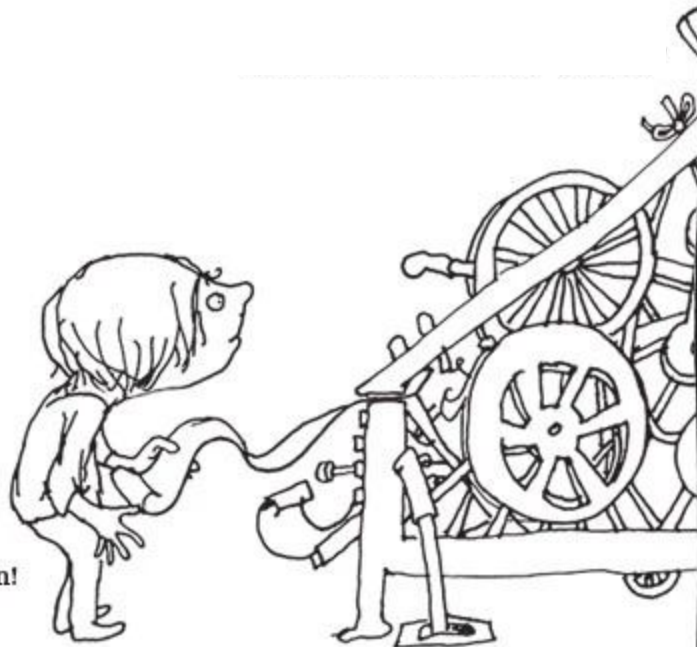


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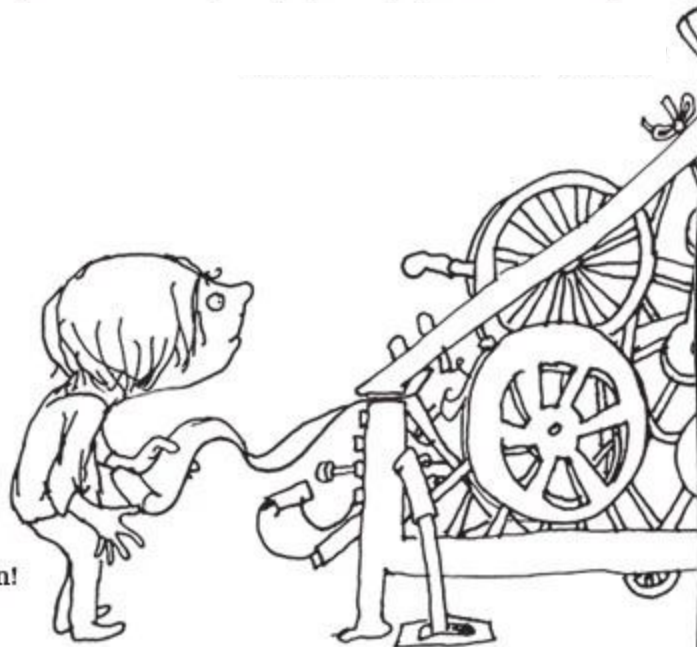


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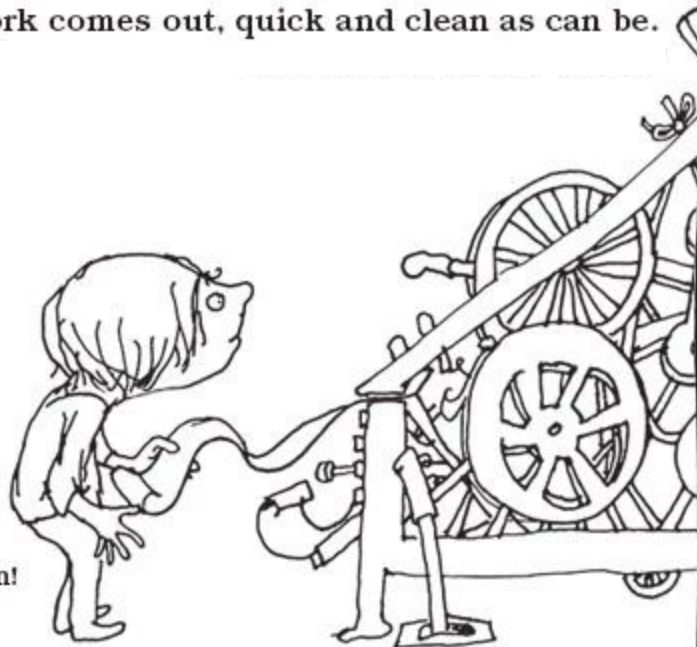


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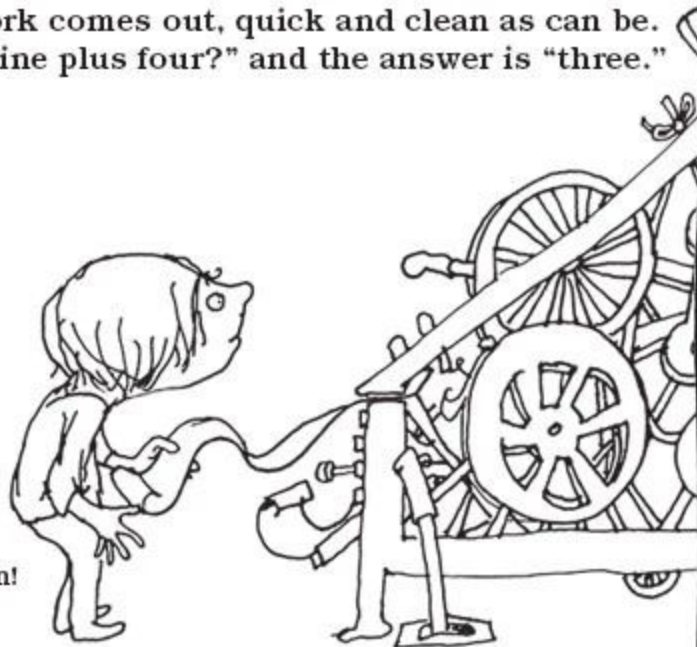


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Here it is—"nine plus four?" and the answer is "three."

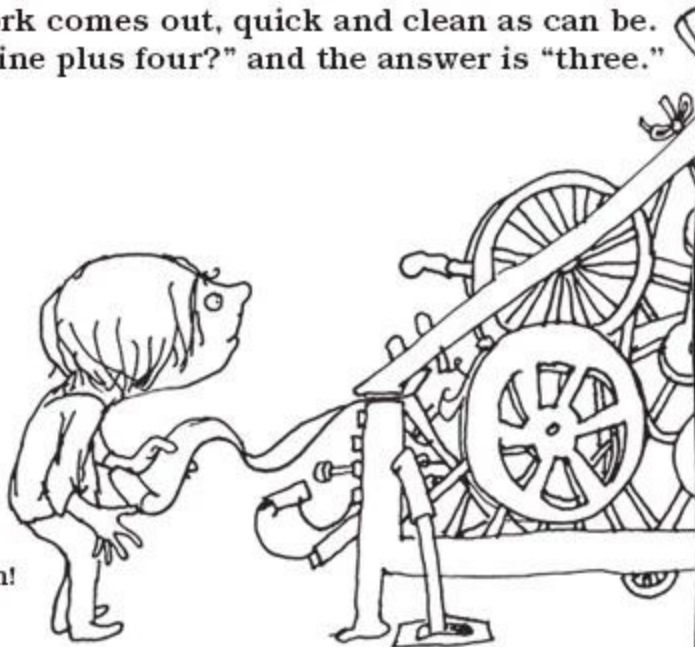
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Three?

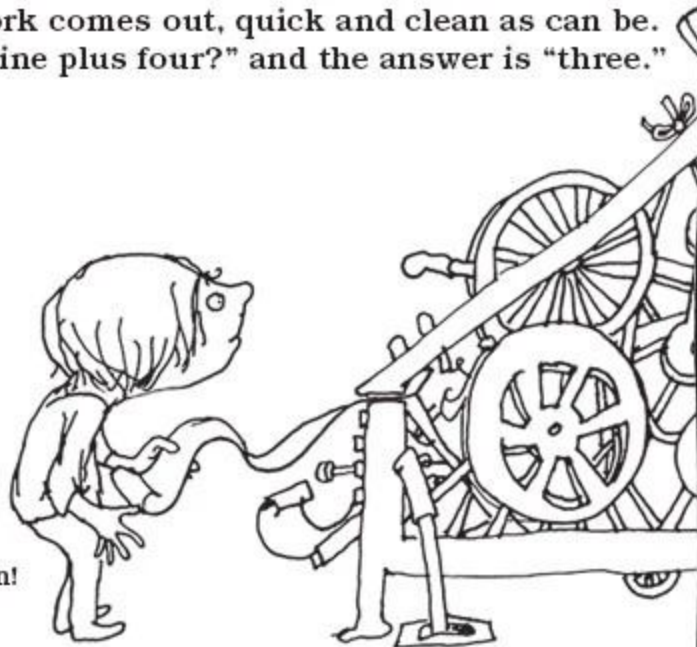
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Three?
Oh me . . .

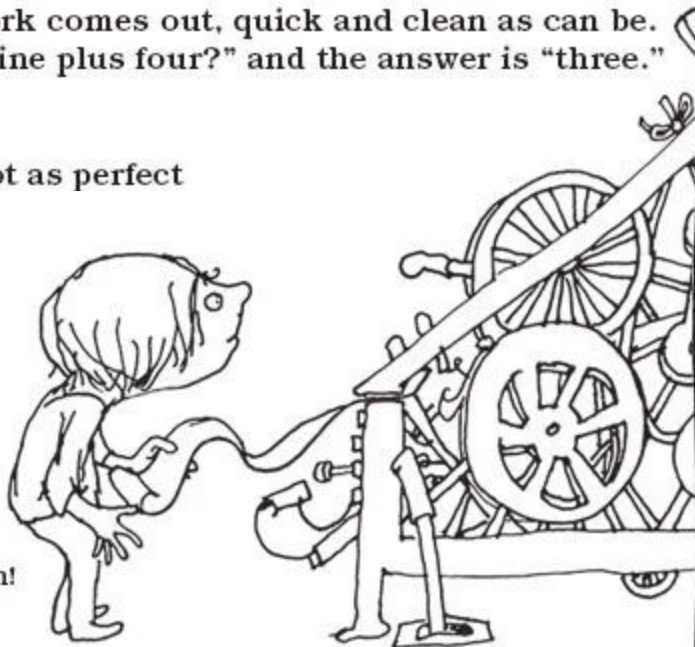
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Three?
Oh me . . .
I guess it's not as perfect

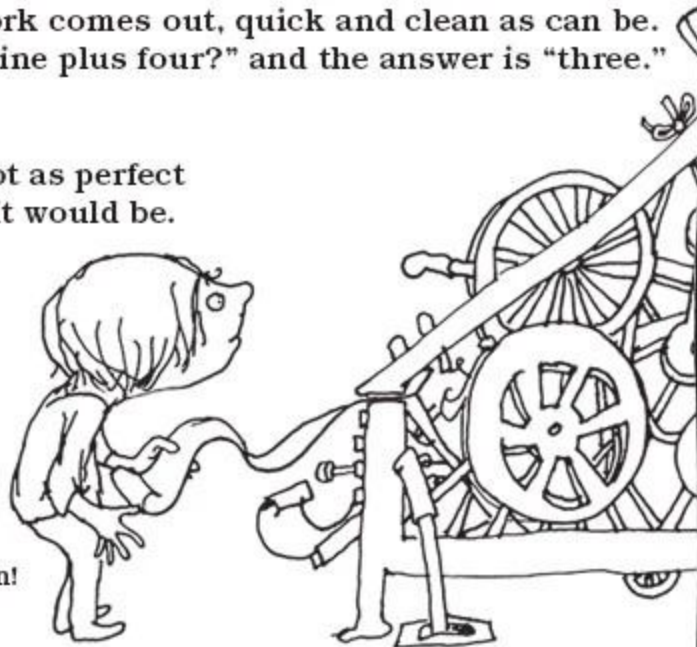
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Your homework comes out, quick and clean as can be.
Here it is—"nine plus four?" and the answer is "three."
Three?
Oh me . . .
I guess it's not as perfect
As I thought it would be.

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This is CS50

If registered for CS50's Family
Lunch after lecture, look for this
yellow duck in the lobby to follow!

